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MAT 101

Preface

The transition from Secondary School Mathematics to University Mathematics can be a nightmare for students in the first year. This is as a result of challenges in settling down in the university system for active academic work. In order to obtain a sound takeoff in Mathematics, it requires a well structured text that covers the topics which would give the student a solid foundation for successful work as well as provide a revision of their previous knowledge. Recognizing the struggle of most students in acquiring such a textbook, it became necessary to prepare one that is structured on the basic requirements for advanced Mathematics programme.

The book, Algebra and Trigonometry, is therefore written with close attention to the outline of the first Mathematics course in most Universities. The book takes into account the wide ability range of students in the choice of language, introduction of new mathematical concepts and ordering of exercises.

Algebra and Trigonometry consists of 11 chapters in all. All the topics are meant to be covered within one semester. The book is recommended to students beginning studies of Mathematics in tertiary institutions.

Prominent pedagogical features of the book include a close attention to revision of pre-university topics and their extensions; the inclusion of numerous carefully selected worked examples to illustrate the concepts being developed at that stage; and provision of several practice exercises at the end of each chapter. Answers to selected questions are attached at the end of the questions. Students are advised not to overlook the exercises, since they are meant to reinforce their understanding and confidence. One major objective that guided the choice of questions in the book was to enable students develop critical thinking ability. A student who fails to develop this ability early enough is not likely to enjoy studies in Mathematics. It is our belief that you would find Algebra and Trigonometry the book that helps in this regard.

The book is a final product of material prepared by the authors and used for teaching for several years by some lecturers of the department. We are grateful to Mrs. Henrietta Nkansah, a lecturer at the department, for helpng to typeset the manuscripts. We are also grateful to Mr. Samuel Ntim, our graphic designer, and a staff of the Department of Physics. Finally, we thank all students of Mathematics, UCC, and our families for their encouragement.

E. K. Essel and B. K. Nkansah, 2013

Chapter 1

Sets

The algebra of sets and solution to set problems are familiar. In this presentation of sets, the algebra and set problems are briefly revised. In most parts of this chapter the concept is presented in a slightly different manner from what you are used to.

1.1 Union and Intersection of Sets

Definition 1.1

Let A and B be subsets of the universal set U. The union of A and B denoted by $A \cup B$ is defined by

$$A \cup B = \{x \in U | x \in A \text{ or } x \in B\} \text{ or } \{x : x \in A \text{ or } x \in B\}.$$

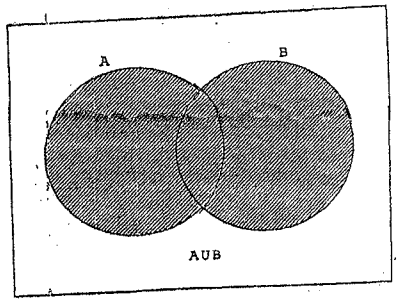
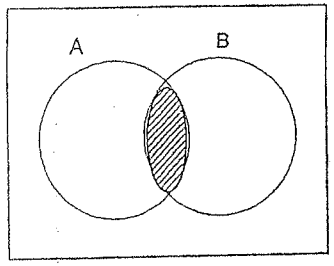


Figure 1.1

The intersection of A and B is denoted by $A \cap B$ and defined by the set

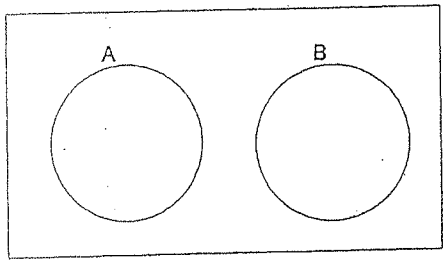
$$A \cap B = \{x : x \in A \text{ and } x \in B\}.$$



$A \cap B$

Figure 1.2

The subsets A and B are disjoint if $A \cap B = \phi$.



$A \cap B = \phi$

Figure 1.3

Theorem 1.1

Let A, B and C be sets. Then

(i) $A \cup A = A \cap A = A$

(ii)

$$A \cup B = B \cup A$$

and

$$A \cap B = B \cap A$$

i.e., the union and intersection of sets are commutative.

(iii)

$$(A \cup B) \cup C = A \cup (B \cup C)$$

and

$$(A \cap B) \cap C = A \cap (B \cap C)$$

i.e., the union and intersection of sets are associative.

(iv)

$$A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$$

i.e., union is distributive over intersection and

$$A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$$

i.e., intersection is distributive over union.

Prove Theorem 1.1 (iv)

Proof

Let $x \in A \cup (B \cap C)$.
Then

$$x \in A \text{ or } x \in B \cap C.$$

Take $x \in A$.
Then

$$x \in A \cup B \text{ and } x \in A \cup C \iff x \in (A \cup B) \cap (A \cup C).$$

Take $x \in B \cap C$. Then

$$\begin{aligned} x \in B \text{ and } x \in C &\iff x \in A \cup B \text{ and } x \in A \cup C \\ &\iff x \in (A \cup B) \cap (A \cup C) \\ &\iff x \in A \cup (B \cap C) \subseteq (A \cup B) \cap (A \cup C) \end{aligned}$$

and

$$(A \cup B) \cap (A \cup C) \subseteq A \cup (B \cap C).$$

Therefore

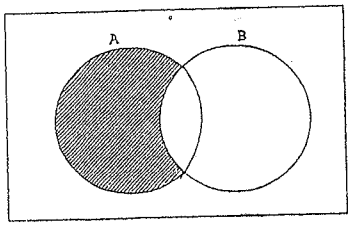
$$A \cup (B \cap C) = (A \cup B) \cap (A \cup C).$$

Definintion 1.2

Let A and B be subsets of U. The difference of A and B, written $A - B$, is the set

$$A - B = \{x : x \in A \text{ and } x \notin B\}.$$

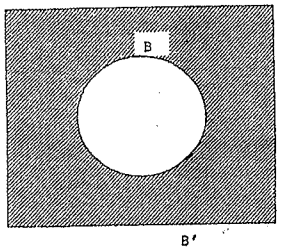
If $B \subset A$, then $A - B$ is called the complement of B in A.Note that $A - B$ is also written as $A \cap B'$.



A - B

Figure 1.4

The complement of B in the universal set U is written as $U - B$ or B', the complement of B.



B'

Figure 1.5

If

$$U = \{1,2,3,4,5,6\}, \quad A = \{1,2,3\}, \quad B = \{3,4,5\}$$

Then

$$A - B = \{1,2\} \text{ and } B - A = \{4,5\}.$$

Theorem 1.2 (DeMorgan's Laws)

(i) $(A \cap B)' = A' \cup B'$

$(A \cup B)' = A' \cap B'$

(ii) $A \cap (A \cap B')' = A \cap B$

$[A - (A - B) = A \cap B]$

(iii) $A \cap (B \cup C)' = (A \cap B') \cap (A \cap C')$

$A - (B \cup C) = (A - B) \cap (A - C)$

(iv) $A \cap (B \cap C)' = (A \cap B') \cup (A \cap C')$

$A - (B \cap C) = (A - B) \cup (A - C)$

Prove Theorem 1.2 (i)

Proof

Let $x \in (A \cap B)'$. Then

$$x \notin A \cap B \iff x \notin A \text{ and } x \notin B \iff x \in A' \text{ or } x \in B' \qquad (1.1)$$

since x cannot be in A' and B' at the same time.

Equation (1.1) implies that

$$x \in A' \cup B'.$$

Therefore

$$(A \cap B)' = A' \cup B'.$$

Next, we let $x \in (A \cup B)'$. Then

$$x \notin A \cup B \iff x \notin A \text{ or } x \notin B \iff x \in A' \text{ and } x \in B'$$

$$\Longrightarrow x \in A' \cap B'$$

Therefore

$$(A \cup B)' = A' \cap B'.$$

Prove Theorem 1.2 (iii)

Method 1

Let $x \in A \cap (B \cup C)'$. Then

$$\begin{aligned} x \in A \text{ and } x \in (B \cup C)' &\iff x \in A \text{ and } x \notin B \cup C \\ &\iff x \in A \text{ and } x \notin B \text{ and } x \notin C \text{ and } x \notin C \\ &\iff (x \in A \text{ and } x \notin B) \text{ and } (x \in A \text{ and } x \notin C) \\ &\iff (x \in A \text{ and } x \in B') \text{ and } (x \in A \text{ and } x \in C') \\ &\iff x \in A \cap B' \text{ and } x \in A \cap C' \\ &\iff x \in (A \cap B') \cap (A \cap C'). \end{aligned}$$

Therefore

$$A \cap (B \cup C)' = (A \cap B') \cap (A \cap C').$$

Method 2

Let $x \in A - (B \cup C)$. Then

$$\begin{aligned} x \in A \text{ and } x \notin (B \cup C) &\iff x \in A \text{ and } x \notin B \text{ and } x \notin C \\ &\iff (x \in A \text{ and } x \notin B) \text{ and } (x \in A \text{ and } x \notin C) \\ &\iff x \in A - B \text{ and } x \in A - C \iff x \in (A - B) \cap (A - C). \end{aligned}$$

If $B \subseteq A$ prove that

(i) $A - (A - B) = A$ Note that $A \cap (A \cap B') = B$

(ii) $(A - B) \cup B = A$

Proof

(i) Let $x \in B$. Then

$$x \in A \text{ but } x \notin (A - B) \Longrightarrow x \in A - (A - B)$$

Therefore

$$B \subseteq A - (A - B). \tag{1.2}$$

Let $x \in A - (A - B)$. Then

$$x \in A \text{ and } x \notin (A - B) \Longrightarrow x \in A \text{ and } x \in B.$$

Therefore

$$A - (A - B) \subseteq B \tag{1.3}$$

From (1.2) and (1.3) we have

$$A - (A - B) = B.$$

Let $x \in (A - B) \cup B$.Then

$$\begin{aligned} x \in A - B \text{ or } x \in B &\Longrightarrow x \in A \text{ and } x \in B \\ &\Longrightarrow x \in A \cup B \Longrightarrow x \in A. \end{aligned}$$

Therefore

$$(A - B) \cup B \subseteq A. \tag{1.4}$$

Next, let $x \in A$. Then

$$x \in B \text{ or } x \notin B.$$

If $x \in B$, then

$$x \in (A - B) \cup B$$

If $x \notin B$. Then

$$x \in A - B \Longrightarrow x \in (A - B) \cup B$$

Therefore

$$A \subseteq (A - B) \cup B \tag{1.5}$$

From (1.4) and (1.5) we have $(A - B) \cup B = A$.

1.2 Two-Set Problems

In Figure 1.6,

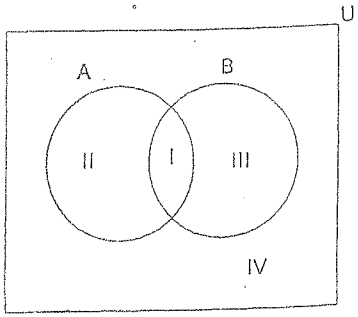


Figure 1.6

A and B are subsets of a Universal se t U. Every element $x \in U$ is a member of exactly one of the four distinct subsets of U represented by I, II, III and IV. Each of the regions are described in terms of A and B as follows:

1. Region I: The elements belong to A and B. It is therefore denoted as $A \cap B$.
2. Region II: The elements belong to A only. That is, they belong to A and not B. It is therefore denoted by $A \cap B'$.
3. Region III: The elements belong to B only. That is, they belong to B and not A. It is therefore denoted by $A' \cap B$.
4. Region IV: The elements belong to none of A and B. That is, they belong to A and not B. It is therefore denoted as $A' \cap B'$ or $(A \cup B)'$.

The regions in Figure 6 are connected by the two set equation

$$n(A \cup B) = n(A) + n(B) - n(A \cap B) \tag{1.6}$$

It should be noted that

$$n(U) = n(A \cup B) + n(A \cup B)'.$$

1.3 Three-Set Problems

In Figure 1.7, A, B and C are subsets of a universal set U. Every element $x \in U$ is a member of a exactly one of the eight distinct subsets of U. represented from I to VIII.

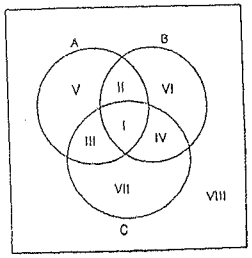


Figure 1.7

Each of the regions are described in terms of A, B and C as follows:

1. Region I : the elements belong to A, B and C. It is therefore denoted as $A \cap B \cap C$.
2. Region II: the elements belong to A and B only. That is, they belong to A, B and not C. It is therefore denoted as $A \cap B \cap C'$.
3. Region III: the elements belong to A and C only. That is, they belong to A, C and not B. It is therefore denoted as $A \cap B' \cap C$.
4. Region IV: the elements belong to B and C only. That is, they belong to B, C and not A. It is therefore denoted as $A' \cap B \cap C$.
5. Region V: the elements belong to A only. That is, they belong to A, and not to B, and not to C. It is therefore denoted as $A \cap B' \cap C'$.
6. Region VI: the elements belong to B only. That is, they belong to B, and not to A, and not to C. It is therefore denoted as $A' \cap B \cap C'$.
7. Region VII: the elements belong to C only. That is, they belong to C, and not to A, and not to B. It is therefore denoted as $A' \cap B' \cap C$.
8. Region VIII: the elements belong to none of A, B and C. That is, they belong not to A, and not to B, and not to C. It is therefore denoted as $A' \cap B' \cap C'$.

The regions in the three sets problem are connected by the equation

$$n(A \cup B \cup C) = n(A) + n(B) + n(C) - \{n(A \cap B) + n(A \cap C) + n(B \cap C)\} + n(A \cap B \cap C) \tag{1.7}$$

It should be noted that

$$n(U) = n(A \cup B \cup C) + n(A \cup B \cup C)'. \tag{1.8}$$

Example 1

A national football coach invited 18 players for a match. Out of these, 7 were attackers, 9 were defenders and 12 midfielders. The coach realized that 4 could play both in the attac and in midfield, 6 in defence and midfield, and 2 play all three roles.

(a) Use set notation to represent the information

(b) Illustrate the information in a Venn diagram

(c) Find the number of players that can play

(i) exactly two holes;

(ii) exactly one hole.

State any assumption(s) made.
Solution

(a) Let

A represent attackers;
D represent defenders;
M represent midfielders; and
U represent players invited

then

$$n(U) = 18, \quad n(A) = 7, \quad n(D) = 9, \quad n(M) = 12$$

and

$$n(A \cap M) = 4 \quad n(D \cap M) = 6 \quad n(A \cap D \cap M) = 2$$

Let $n(A \cap D \cap M') = x$. Assuming that any of the players invited, plays in, at least one of the three roles, it follows that $n(A \cup D \cup M) = 18$. Using the (1.7), we have

$$n(A \cup D \cup M) = n(A)+n(D)+n(M)-\{n(A \cap D)+n(A \cap M)+n(D \cap M)\}+n(A \cap D$$

i.e.,

$$\begin{aligned} 7+9+12-\{(2+x)+4+6\}+2 &= 18 \\ 30-(12+x) &= 18 \\ 18-x &= 18 \\ x &= 0. \end{aligned}$$

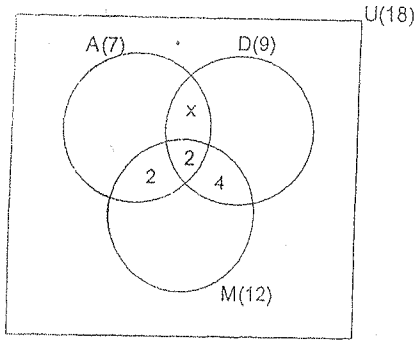


Figure 1.1: Figure 1.8

(i) The number of players that play in exactly two roles is

$$n(A \cap D \cap M')+n(A \cap M \cap D')+n(D \cap M \cap A') = 0+2+4 = 6$$

(ii) Midfield only is

$$n(M \cap A' \cap D') = 12-(2+2+4) = 12-8 = 4$$

Attack only is

$$n(A \cap M' \cap D') = 7-(2+2) = 7-4 = 3.$$

Defence only is

$$n(D \cap A' \cap M') = 9-(2+4) = 9-6 = 3$$

Therefore, the number of players that play in exactly one role is

$$= 4+3+3 = 10$$

Exercise

1. Prove that $A \cup (A \cap B) = A \cap (A \cup B) = A$.
2. Prove that $(A \cap B') \cup (A' \cap B) = (A \cup B) \cap (A \cap B)'$.
   Illustrate the results in Venn diagrams.

3. Verify that the following results are true

   (a) $(A \cup B \cup C)' = A' \cap B' \cap C'$

   (b) $(A \cap B \cap C)' = A' \cup B' \cup C'$

   In each case illustrate the result using Venn diagrams.

4. Prove that

$$A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$$

   Illustrate the result using Venn diagrams.

5. Describe the region in a three-set problem represented by

$$(A - B) \cap (A - C).$$

6. In an Arts class of 100 students, 37 students study English Literature, 32 study French and 31 study Economics. 9 students study English Literature and French, 12 English Literature and Economics, 11 study French and Economics, 28 study none of the three subjects.

   Illustrate the information in a Venn diagram.

   Find

   (a) the numbe of students who study all three subjects;

   (b) the number of students who study only one subject;

   (c) the number of students who study exactly two subjects.

7. Let $U = \{-3 \leq x \leq 12\}$. Let P and Q two subsets of U given by $P = \{0 \leq x \leq 12\}$ and $Q = \{-3 < x < 10\}$.

   Find $P' \cap Q'$.

   Ans. $\{-3\}$

8. Consider the following sets.

   $U = \{\text{triangles}\}$ $E = \{\text{equilateral triangles}\}$

   $I = \{\text{isosceles triangles}\}$ $R = \{\text{right-angled triangles}$

   $A = \{$ triangles of area $32\text{cm}^2\}$

   (a) Illustrate the sets in a Venn diagram

   (b) Find

      i. the ratio of the sides of triangle $I \cap R$;

      ii. the length of the sides of triangle $E \cap A$.

   Answer: (a) $1 : 1 : \sqrt{2}$ (b) 8.6 cm.

1.4 Cartesian Product of Sets

We know that points in the Euclidean Plane are defined by the set $\mathbb{R} \times \mathbb{R}$ or

$$\mathbb{R}^2 = \{P : P = (x, y);\ x,\ y \in \mathbb{R}\}.$$

Using a similar notion, the Cartesian product of two sets S and T is defined as

$$S \times T = \{P : P = (s, t),\ s \in S,\ t \in T\}$$

That is, $S \times T$ is the set of all ordered pairs of elements (s, t) with the first member of each pair always from the first set, S and the second member always from the second set T. For example,

$$\{2, 4\} \times \{2, 4, 7\} = \{(2, 2), (2, 4), (2, 7), (4, 2), (4, 4), (4, 7)\}.$$

1.4.1 Characteristics of Cartesian Product

1. Two elements of the product $S \times T$ are equal if and only if they are identical i.e.,

$$(s, t) = (a, b) \Longrightarrow s = a \text{ and } t = b.$$

2. If

$$S = \phi \text{ or } T = \phi, \text{ then } S \times T = \phi.$$

3. Clearly,

$$S \times T \neq T \times S$$

   since if

$$(s, t) \in S \times T \not\Rightarrow (s, t) \in T \times S.$$

4. The Cartesian product of n sets $S_1, S_2, S_3, ..., S_n$ is the set of n-tuples

$$S_1 \times S_2 \times S_3 \times ... \times S_n$$
$$= \{(x_1, x_2, x_3, \cdots, x_n) : x_1 \in S_1, x_2 \in S_2, \cdots, x_n \in S_n\}.$$

If

$$S_i = S, \quad i = 1, 2, \cdots, n$$

then,

$$S_1 \times S_2 \times S_3 \times ... \times S_n = S^n.$$

In particular,

$$S_1 \times S_2 = S^2 \text{ and } S_1 \times S_2 \times S_3 = S^3.$$

Example 2

Prove that

$$A \times B = B \times A$$

if and only if either $A = B$ or at least one is empty.

Solution

Let $A = B$

$\Longrightarrow$

$$A \times B = B^2 = B \times A$$

On the other hand, if

$$A = \phi \text{ or } B = \phi \Longrightarrow A \times B = \phi = B \times A$$

Therefore,

$$A \times B = B \times A \text{ when } A = B, \text{ or } A = \phi \text{ or } B = \phi.$$

Let

$$A \times B = B \times A \text{ and let } A \neq \phi \text{ and } B \neq \phi.$$

Let $a \in A$, $b \in B$ (Since $A \neq \phi$ and $B \neq \phi$ by assumption). Then

$(a, b) \in A \times B \Longrightarrow (a, b) \in B \times A$, since $A \times B = B \times A$ (assumed)
$\Longrightarrow a \in B$ and $b \in A \Longrightarrow B \subseteq A$ and $A \subseteq B$
$\Longrightarrow A = B.$

Example 3

Let the sets A, B and C be as defined in each of the following.
(a) $A = \{x_1, x_2\}$, $B = \{y_1, y_2\}$ and $C = \{z_1, z_2\}$.
(b) $A = \{x_1, x_2\}$, $B = \{y_1, y_2, y_3\}$
(c) $A = \{x_1, x_2\}$, $B = \{y_1, y_2\}$ and $C = \{z_1, z_2, z_3\}$.
Find the cartesian product in each case.

Solution

(a) Constructing all possible triples (x, y, z) such that $x \in A$, $y \in B$ and $z \in C$, we obtain

$$\begin{aligned} A \times B \times C &= \{x_1, x_2\} \times \{y_1, y_2\} \times \{z_1, z_2\} \\ &= \{(x_1, y_1, z_1), (x_1, y_1, z_2), (x_1, y_2, z_1), \\ &\quad (x_1, y_2, z_2), (x_2, y_1, z_1), (x_2, y_1, z_2), \\ &\quad (x_2, y_2, z_1), (x_2, y_2, z_2)\}. \end{aligned}$$

(b) Constructing all possible pairs $(x \times y)$ such that $x \in A$, and $y \in B$, we obtain

$$\begin{aligned} A \times B &= \{x_1, x_2\} \times \{y_1, y_2, y_3\} \\ &= \{(x_1, y_1), (x_1, y_2), (x_1, y_3), (x_2, y_1), (x_2, y_2), (x_2, y_3)\} \end{aligned}$$

(c) Constructing all possible triples $\{x, y, z\}$ such that $x \in A$, $y \in B$ and $z \in C$, we obtain

$$\begin{aligned} A \times B \times C &= \{x_1, x_2\} \times \{y_1, y_2\} \times \{z_1, z_2, z_3\} \\ &= \{(x_1, y_1, z_1), (x_1, y_1, z_2), (x_1, y_1, z_3) \\ &\quad (x_1, y_2, z_1), (x_1, y_2, z_2), (x_1, y_2, z_3), \\ &\quad (x_2, y_1, z_1), (x_2, y_1, z_2), (x_2, y_1, z_3), \\ &\quad (x_2, y_2, z_1), (x_2, y_2, z_2), (x_2, y_2, z_3)\} \end{aligned}$$

In the above example, we see that in (a), $n(A) = 2$, $n(B) = 2$ and $n(C) = 2$. For the resulting product, $n(A \times B \times C) = 8$. That is,

$$n(A \times B \times C) = n(A) \times n(B) \times n(C).$$

In (b), $n(A) = 2$, $n(B) = 3$. For the resulting product, $n(A \times B) = 6$. That is,

$$n(A \times B) = n(A) \times n(B).$$

In (c), $n(A) = 2$, $n(B) = 2$ and $n(C) = 3$. For the resulting product, $n(A \times B \times C) = 12$. That is,

$$n(A \times B \times C) = n(A) \times n(B) \times n(C).$$

The results in this example shows that the elements of the cartesian product of any number of sets is the product of the number of elements in the respective sets.

In the next example, we consider a condition under which cartesian product is associative.

Example 4

Prove that

$$(A \times B) \times C = A \times (B \times C)$$

if and only if at least one of the sets A, B, C is empty.

Solution

Let A, B and C be empty. Then,

$$(A \times B) \times C = \phi = A \times (B \times C).$$

Suppose

$$(A \times B) \times C = A \times (B \times C) \text{ and } A \neq \phi,\ B \neq \phi \text{ and } C \neq \phi$$

then there exists

$$(x, y) \in (A \times B) \times C \Longrightarrow x \in A \times B \text{ and } y \in C.$$

But

$$(x, y) \in A \times (B \times C) \Longrightarrow x \in A \text{ and } y \in B \times C.$$

Now,

$x \in A$, and $x \in A \times B$ is., a contradiction.

The assumption $A \neq \phi$, $B \neq \phi$ and $C \neq \phi$ is false.

Therefore, A, B and C are empty.

Example 5

Let

$$A = \{1 \leq x \leq 2\}, \quad B = \{-2 \leq x \leq 3\}.$$

Sketch and describe the graph of $A \times B$ in the xy-plane.

Solution

The points that constitute the set $A \times B$ are given as

$$\begin{aligned} A \times B = \{&(1, -2), \cdots, (1, 0), \cdots, (1, 3), \\ &\cdots, (1.5, -2), \cdots, (1.5, 0), \cdots, (1.5, 3), \\ &\cdots, (2, -2), \cdots, (2, 0), \cdots, (2, 3)\}. \end{aligned}$$

The graph of $A \times B$ is given as shown.

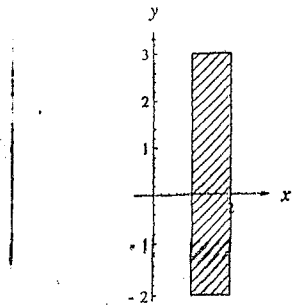


Figure 1.9

The graph is the set of all points in the rectangular region in the plane with vertices $(1, -2)$, $(1, 3)$, $(2, -2)$ and $(2, 3)$.

Example 6

Let

$$A = \{x \leq 2\}, \quad B = \{-1 \leq x \leq 3\}.$$

Sketch and describe the graph of $A \times B$ in the xy- plane.

Solution

The points that constitute the set $A \times B$ are given as

$$A \times B = (2,-1), \cdots, (2,3), \cdots, (1,-1), \cdots, (1,3), \cdots\}.$$

The graph of $A \times B$ is given as shown.

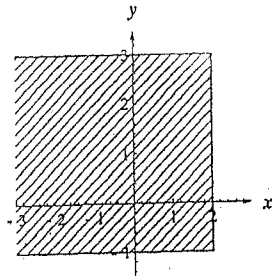


Figure 1.10

Exercise

1. Let $A = \{1 \leq x \leq 3\}, \quad B = \{2 \leq x \leq 3\}.$
   Sketch the graphs $A \times B$ and $B \times A$ in the plane. Describe your sketch in each case.
2. Let $A = \{1 \leq x \leq 2\}, \quad B = \{0 \leq x \leq 1\}$ and $C = \{-1 \leq x \leq 2\}$
   Sketch in the xyz space the graph of $A \times B \times C$. Describe your sketch.

Chapter 2

The Real Number System

Aspects of the real number system are already familiar to you. These include the Natural numbers ($\mathbb{N}$), the Whole numbers ($\mathbb{W}$), Integers ($\mathbb{Z}$), Rational numberss ($\mathbb{Q}$) and Irrational numbers ($\mathbb{Q}'$). By definition

1. The Counting (or Natural) numbers

   $$\mathbb{N} = \{1, 2, 3, ...\}.$$

2. The Whole numbers

   $$\mathbb{W} = \{0, 1, 2, 3, ...\} = \mathbb{N} \cup \{0\}$$

   thus, $\mathbb{N} \subset \mathbb{W}$.

3. The Integers

   $$\mathbb{I} \text{ or } \mathbb{Z} = \{..., -3, -2, -1, 0, 1, 2, 3, ...\}$$

   thus, $\mathbb{W} \subset \mathbb{Z}$.

4. The Rational numbers

   $$\mathbb{Q} = \left\{q : q = \frac{a}{b}, \quad a, b \in Z, \quad b \neq 0\right\}.$$

   and $\mathbb{Z} \subset \mathbb{Q}$. These are numbers whose decimal representation either terminate or recur.

5. The Irrational numbers

$\mathbb{Q}' = \{\text{numbers whose decimal representation do not terminate or recur}\}$.

6. The Real numbers

$$\mathbb{R} = \mathbb{Q} \cup \mathbb{Q}'.$$

Numbers which are not $\mathbb{Q}$ or $\mathbb{Q}'$ are called *complex* numbers denoted by $\mathbb{C}$.

In the following sections, we examine in some detail the sets of rational, irrational, and real numbers.

2.1 Properties of Real Numbers

1. Let $a, b, c \in \mathbb{R}$

(a) Reflexive property

$$a = a$$

1. (b) Symmetric/Substitution property

$$\text{If } a = b, \text{ then } b = a$$

(c) Transitive property

$$\text{If } a = b \text{ and } b = c \text{ then } a = c.$$

2. Axioms of Addition and Multiplication

(a) Closure property for:

i. Addition
For any $a, b \in \mathbb{R}$

$$a + b \in \mathbb{R}$$

ii. Multiplication
For any $a, b \in \mathbb{R}$

$$ab \in \mathbb{R}$$

(b) Commutative property
$\forall\, a, b \in \mathbb{R}$

$$a + b = b + a \text{ and } ab = ba$$

(c) Associative property
$\forall\, a, b, c \in \mathbb{R}$

i.

$$(a + b) + c = a + (b + c) \text{ and } (ab)\, c = a\,(bc)$$

(d) Distributive property
$\forall\, a, b, c \in \mathbb{R}$

$$a\,(b + c) = ab + ac.$$

(e) Identity property

i. Additive identity
There exist one and only one real number z such that

$$a = a + z \quad \forall a \in \mathbb{R} \tag{2.1}$$

The number $z = 0$ satisfies (2.1) and is called the additive identity

ii. Multiplicative identity
There exist one and only one real number v such that

$$a = av \quad \forall a \in \mathbb{R} \tag{2.2}$$

The number $v = 1$ satisfies (2.2) and is called the multiplicative identity

(f) Inverse property

i. Additive inverse
Corresponding to all $a \in \mathbb{R}$,

$$a + a' = 0$$

The number a' is called the additive inverse of a and we write $a' = -a$.

ii. Multiplicative inverse
For a non-zero number $a \in \mathbb{R}$, there exists $u \in \mathbb{R}$ such that

$$au = 1$$

The number u is called the multiplicative inverse of a and we write $u = \frac{1}{a}$.

Further Properties

3. (a) Corresponding to the pair $a, b \in \mathbb{R}$, there exists a real number c such that

$$a = b + c$$
$$a = bc$$

(b) If $b > c$, then $\forall a \in \mathbb{R}$,

$$a + b > a + c$$

(c) $\forall a, b \in \mathbb{R}$, if

$$a > 0 \text{ and } b > 0 \text{ then } ab > 0$$

(d) Uniqueness: If

$$a = a' \text{ and } b = b' \text{ then } a + b = a' + b'.$$

2. Zero-Multiplication Theorem

$$\begin{aligned} ab &= ab \quad \text{(reflexive property)} \\ ab &= a(b+0) \\ ab &= ab + a0 \\ ab + 0 &= ab + a0 \end{aligned}$$

By the cancellation law we have

$$a0 = 0$$

3. The zero-product Theorem

If $ab = 0$, then $a = 0$ or $b = 0$ (or possibly both).

Proof

Let $a \neq 0$, and suppose $ab = 0$. Then,

$$\begin{aligned} \frac{1}{a}(ab) &= \frac{1}{a}(0) \\ 1 \times b &= 0 \\ b &= 0. \end{aligned}$$

Let $b \neq 0$. Then from $ab = 0$ we have

$$\begin{aligned} (ab)\frac{1}{b} &= (0)\frac{1}{b} \\ a \times 1 &= 0 \\ a &= 0. \end{aligned}$$

4. If $a > b$ and $a, b > 0$ then $\frac{1}{a} < \frac{1}{b}$.

The real numbers together with the relation of equality and operations of $+$ and $\times$ satisfy the field properties.

The set $\mathbb{R}$ is closed for $+$, $\times$, $-$, non-zero division and taking of roots of non-negative numbers.

2.2 Rational Numbers

The set of rational numbers denoted by $\mathbb{Q}$ is defined as

$$\begin{aligned} \mathbb{Q} &= \left\{ q : q = \frac{a}{b}, \quad a, b \in Z, \quad b \neq 0 \right\} \\ &= \{\text{numbers whose decimal representation terminate or recur}\}. \end{aligned}$$

Properties of Rational Numbers

1. Let $a, b, c, d, e \in \mathbb{Z}$ and $b, d \neq 0$. Then

$$\frac{a}{b} = \frac{c}{d} \quad \text{iff } ad = bc.$$

Proof

$$\begin{aligned}
\frac{a}{b} &= \frac{c}{d} \\
a \times \frac{1}{b} &= c \times \frac{1}{d} \\
a \times \frac{1}{b} \times bd &= c \times \frac{1}{d} \times bd \\
a \times d \times \frac{1}{b} \times b &= b \times c \times \frac{1}{d} \times d \\
a \times d \times 1 &= b \times c \times 1 \\
ad &= bc.
\end{aligned}$$

2. Fundamental property of $\mathbb{Q}$: For any non-zero number x and rational number $\frac{a}{b}$ we have

$$\frac{a \cdot x}{b \cdot x} = \frac{a}{b}.$$

It allows for the reduction of fractions with the least common factor (L.C.F) for both numerator and denominator.

For example,

$$\frac{12}{18} = \frac{4 \times 3}{6 \times 3} = \frac{4}{6} = \frac{2 \times 2}{3 \times 2} = \frac{2}{3}$$

3. Theorem 2.1

There is no rational number whose square root is $\sqrt{2}$. (in other words $\sqrt{2}$ is not a rational number.)

Proof

Let $a, b \in \mathbb{Z}$, Assume that $(a, b) = 1$ (i.e, the H.C.F of a and b is 1) and that $\sqrt{2}$ is rational number. Then

$$\begin{aligned}
\sqrt{2} &= \frac{a}{b} \quad b \neq 0, \\
a &= b\sqrt{2} \\
a^2 &= 2b^2.
\end{aligned}$$

The coefficient of b^2 being 2 implies that a^2 is an even number and hence a is even. Let $a = 2k$, $k \in \mathbb{N}$. Substituting for a^2 gives

$$\begin{aligned}
2b^2 &= (2k)^2 \\
2b^2 &= 4k^2 \\
b^2 &= 2k^2.
\end{aligned}$$

The coefficient of k^2 being 2 implies that b^2 is an even number and hence b is even. Now a, b being both even means they have 2 as a factor and so $\frac{a}{b}$ is not in its reduced form This contradicts the earlier assumption that $(a, b) = 1$. Therefore $\sqrt{2}$ is not rational.

2.3 Irrational Numbers

Irrational numbers are defined as

$$\begin{aligned}
Q' &= \{\text{numbers whose decimal equivalent do not terminate or recur}\} \\
&= \{\text{numbers that do not have fractional representation}\}.
\end{aligned}$$

Examples of irrational numbers are $\sqrt{2}$, $\sqrt{3}$, $\sqrt{6}$ etc. The expression $\sqrt{b}$ is called a radical expression and b the radicand.

Properties of Irrational Numbers

1. If $\sqrt{b}$ is irrational, then $a + c\sqrt{b} \in Q' \ \forall\ a, b \in Q$.
2. $\left(a + c\sqrt{b}\right)\left(a - c\sqrt{b}\right) = a^2 - bc^2 \in Q$. That is, the product of two irrational numbers can be rational. The number $a - c\sqrt{b}$ is called the conjugate of $a + c\sqrt{b}$.
3. Axiom of Order
   (a) Trichotomy law: Given any two numbers $a, b \in \mathbb{R}$, one of the following three statements hold:
   $$a = b, \quad a > b, \quad \text{or} \quad a < b.$$
   (b) Transitive law: If
   $$a > b, \quad b > c, \quad \text{then} \quad a > c.$$
4. If $a = a'$, $b = b'$ and $a > b$, then $a' > b'$.
5. If $a, b > 0$ then
   $$\begin{aligned}
   a < b &\implies ka > kb, \quad k > 0 \\
   a < b &\implies ka > kb, \quad k, < 0
   \end{aligned}$$

ii) Open right infinity interval

$$(a, \infty) = \{x : a < x < \infty\}$$

iii) Closed left infinity interval

$$(-\infty, a] = \{x : -\infty < x \leq a\}$$

iv) Open left infinity interval

$$(-\infty, a) = \{x : -\infty < x < a\}$$

Note that

$$a < x < \infty \text{ means } x > a, \text{ and}$$

the whole number line is considered to be an interval expressed as

$$(-\infty, \infty) = \{x : -\infty < x < \infty\}.$$

Example 3

Solve the following inequalities and illustrate your answer on the number line.

(a) $|x-1| \leq 2$

(b) $|x-2| \geq 4$

(c) $x^2 - 4 \leq -3$

Solution

(a)

$$\begin{aligned} |x-1| \leq 2 &\implies -2 \leq x-1 \leq 2 \\ &\implies -2+1 \leq x \leq 2+1 \\ &\implies -1 \leq x \leq 3 \\ &\text{or} \quad x \in [-1, 3]. \end{aligned}$$

(b)

$$\begin{aligned} |x-2| \geq 4 &\implies x-2 \geq 4 \quad \text{or} \quad x-2 \leq -4 \\ &\implies x \geq 6 \quad \text{or} \quad x \leq -2 \end{aligned}$$

i.e., $\{x : x \leq -2 \cup x \geq 6\}$

or $x \in \{(-\infty, -2] \cup [6, \infty)\}$

(c)

$$\begin{aligned} x^2 - 4 \leq -3 &\implies x^2 - 1 \leq 0 \\ &\implies (x-1)(x+1) \leq 0 \end{aligned}$$

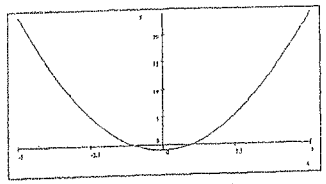


Figure 2.2

From the sketch we see that the region between the x-axis and the curve is satisfies $(x-1)(x+1) \leq 0$. This region is the interval $-1 \leq x \leq 1$ i.e., $x \in [-1, 1]$.

Example 4

Find the interval of values that satisfies the inequality

$$\frac{1}{2x-1} > \frac{1}{x-3}$$

Solution

First, we multiply both sides by the square of the denominators to obtain

$$(2x-1)^2 (x-3)^2 \frac{1}{2x-1} > \frac{1}{x-3} (2x-1)^2 (x-3)^2,$$

noting that multiplying an inequality by a positive number does not change the direction of the inequality. Simplifying we obtain

$$\begin{aligned}(2x-1)(x-3)^2 &> (2x-1)^2(x-3)\\ (2x-1)(x-3)^2-(2x-1)^2(x-3) &> 0\\ (2x-1)(x-3)\{(x-3)-(2x-1)\} &> 0\\ (2x-1)(x-3)(-x-2) &> 0\\ (2x-1)(3-x)(x+2) &> 0 \qquad (2.3)\end{aligned}$$

The sketch of the polynomial $f(x)=(2x-1)(3-x)(x+2)$ is as shown.

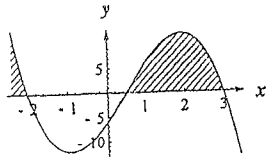


Figure 2.3

The shaded portions in the sketch corresponds to the values of x for which the inequality (2.3) is satisfied. Thus, the range of values of x for which the given inequality is satisfied is $(-\infty,-2)\cup(\frac{1}{2},3)$.

Exercise

1. Solve $x^2-4x+4\geq 0$.

2. By illustrating appropriate sets on the number line, find the values of such that
$$(2x-1)(2x+5)>0.$$

3. Find the range of values of x for which the the following inequalities are satisfied

(a)
$$|x+1|\geq|x+2|+|x+3|$$

(b)
$$|x+2|\geq|x+1|+|x-3|$$

Answer: $2\leq x\leq 4$

(c)
$$-1\leq\frac{x^2}{x-2}\leq 8$$

Answer: $-2\leq x\leq 1$.

4. For any two real numbers x and y, prove that

(a)
$$|x+y|\leq|x|+|y|$$

(b)
$$x>y \Leftrightarrow x^2>y^2 \text{ if } x,\ y>0$$

(a) For what values of t is the inquality
$$-3<\frac{x^2+tx-2}{x^2-x+1}<2$$
satisfied for all values of x ?

(b) Solve the inequality
$$x+2\left|x-\frac{3}{2}\right|>|x+1|-1$$

5. Show that for any number $x\neq 1$, $x^2+1>2x$.

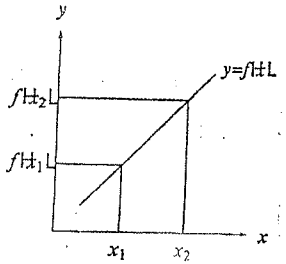


Figure 3.1

4. If $m < 0$, then $x_1 < x_2 \Rightarrow f(x_1) > f(x_2)$. The function $f(x)$ is referred to as a decreasing function. This is illustrated as shown.

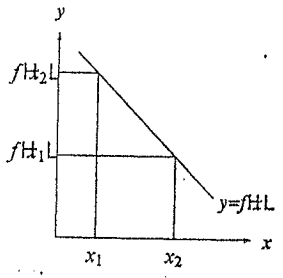


Figure 3.2

5. If $m = 0$, then $x_1 < x_2 \Rightarrow f(x_1) = f(x_2)$. The function $f(x)$ is said to be constant as shown.

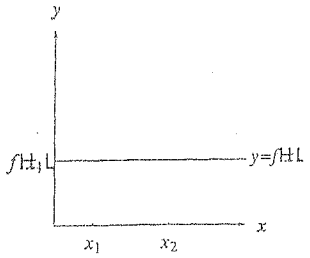


Figure 3.3

Remark

Let l_1 and l_2 be the graphs of the functions $f(x)$ and $g(x)$, respectively, with gradients m_1 and m_2.

1. l_1 and l_2 are parallel if and only if $m_1 = m_2$.
2. l_1 and l_2 are perpendicular if and only if $m_1 m_2 = -1$.
3. A non-constant linear function defines a one-to-one mapping in the set of $\mathbb{R}$.

Example 1

Given that $\sin 11 = 0.1908$, $\sin 12 = 0.2079$, $\sin 13 = 0.225$.
Find a linear approximation for (a) $\sin 11.3$ (b) $\sin 12.6$

Solution

Given that $\sin 11 = 0.1908$ and $\sin 12 = 0.2079$. Suppose that in $11 \leq x \leq 12$, $f(x) = \sin x$ is approximately linear, then points (11, 0.1908) (12, 0.2079) are on the line.

Gradient of f

$$f = \frac{0.2079 - 0.1908}{12 - 11} = 0.0171$$

Then

$$\begin{aligned} \sin x &= 0.0171(x - 11) + 0.1908 \\ &= 0.0171x - 0.1881 + 0.1908 \\ &\Rightarrow \sin x = 0.0171x + 0.0027. \end{aligned}$$

When $x = 11.3$

$$\begin{aligned}\sin 11.3 &= 0.0171(11.3) + 0.0027 \\ &= 0.19323 + 0.0027 \\ &= 0.19593\end{aligned}$$

The above approach is referred to as linear interpolation.

3.2 Quadratic Functions

A quadratic function is given by the rule

$$f : x \longrightarrow ax^2 + bx + c,$$

where a, b and c are constants and $a \neq 0$.

We examine the graph of this function by two principles: translation and completing of squares.

3.2.1 Characteristics of the Quadratic Function

1. By translation principle, $f(x) = ax^2 + bx + c \iff$ the graph of $f(x)$ is the image of the parabola $y = ax^2$ under a suitable translation. This suitable translation will be clear after completing the squares.

2. By Square Completion

   Let

   $$\begin{aligned} y &= ax^2 + bx + c \\ &= a\left[x^2 + \tfrac{b}{a}x + \tfrac{c}{a}\right], && \text{Factorise the leading coefficient} \\ &= a\left[x^2 + \tfrac{b}{a}x + \left(\tfrac{b}{2a}\right)^2 - \left(\tfrac{b}{2a}\right)^2 + \tfrac{c}{a}\right], && \text{Add and subtract square of } \tfrac{b}{2a} \\ &= a\left[\left(x + \tfrac{b}{2a}\right)^2 + \tfrac{4ac-b^2}{4a^2}\right], && \text{Express first three terms as perfect square} \\ &= a\left(x + \tfrac{b}{2a}\right)^2 + \tfrac{4ac-b^2}{4a}, && \text{Multiply through by leading coefficient} \end{aligned}$$

   The result is in the form

   $$y = a(x-p)^2 + q$$

where $p = -\frac{b}{2a}$ and $q = \frac{4ac-b^2}{4a}$.

The equation may be rewritten as $y - q = a(x-p)^2$. In this form, the graph of $f(x)$ is the image of the graph of $y = ax^2$ under a translation by the point $(p,\ q) = \left(-\frac{b}{2a},\ \frac{4ac-b^2}{4a}\right)$.

3. Turning point of the curve

   If $a > 0$, the point $(p,\ q)$ is a minimum turning point of the curve of $f(x)$. Thus, $f(x)$ has a minimum value of $y = q\ \left(= \frac{4ac-b^2}{4a}\right)$ at $x = p\ \left(= -\frac{b}{2a}\right)$. If $a < 0$, the point $(p,\ q)$ is a maximum turning point of the curve of $f(x)$. Thus, $f(x)$ has a maximum value of $y = q\ \left(= \frac{4ac-b^2}{4a}\right)$ at $x = p\ \left(= -\frac{b}{2a}\right)$. The equation $x = -\frac{b}{2a}$ represents the line of symmetry of the curve.

4. The zeros of $f(x)$

   The zeros are the values of x at which $f(x) = 0$. That is,

   $$a\left(x + \frac{b}{2a}\right)^2 + \frac{4ac - b^2}{4a} = 0.$$

   Simplifying gives

   $$\begin{aligned} a\left(x + \frac{b}{2a}\right)^2 &= \frac{b^2 - 4ac}{4a} \\ \left(x + \frac{b}{2a}\right)^2 &= \frac{b^2 - 4ac}{4a^2} \\ x + \frac{b}{2a} &= \frac{\pm\sqrt{b^2 - 4ac}}{2a} \\ x &= -\frac{b}{2a} \pm \frac{\sqrt{b^2 - 4ac}}{2a} \\ &= \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} \end{aligned}$$

   Therefore, the zeros of $f(x)$ are given as

   $$x_1 = \frac{-b + \sqrt{b^2 - 4ac}}{2a}, \qquad x_2 = \frac{-b - \sqrt{b^2 - 4ac}}{2a}$$

   The zeros are also referred to as the *roots* of the function.

5. Type of roots

(a) The roots x_1 and x_2 are *real* if the discriminant is non-negative, i.e., $b^2 - 4ac \geq 0$.

(b) The roots x_1 and x_2 are *repeated* if $b^2 - 4ac = 0$. In this case, the repeated root is $x = -\frac{b}{2a}$, and $q = 0$.
For example, the function $y = x^2 - 2x + 1$ can be expressed as $y = (x-1)^2$. Equating to zero gives $(x-1)^2 = 0, \implies x = 1$ repeated. This means that the curve touches the x-axis at $x = 1$, and its value at this point is $y = 0$. The graph of this function is given in Figure.3.4

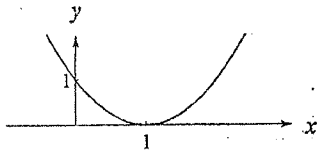


Figure 3.4

(c) The roots x_1 and x_2 are *complex* if the discriminant is negative, i.e., $b^2 - 4ac < 0$. In this case, either of the following two cases is implied.

i. $f(x)$ is strictly positive, i.e., $f(x) > 0$, if $a > 0$. This is illustrated as shown.

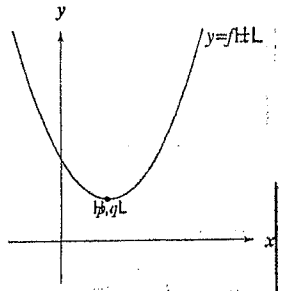


Figure 3.5

ii. $f(x)$ is strictly negative, i.e., $f(x) < 0$, if $a < 0$.

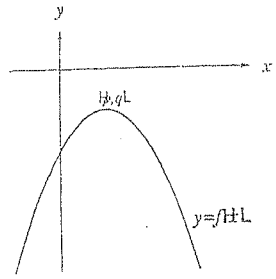


Figure 3.6

Example 2

Given the function

$$f(x) = 1 - 2x - 2x^2,$$

(a) show that $\forall x \in \mathbb{R}, \quad f(x) \leq \frac{3}{2}$;

(b) find the equation of the axis of symmetry, the turning point and the roots of $f(x)$.

Solution

(i) By completing squares, we have

$$\begin{aligned} f(x) &= 1 - 2x - 2x^2 \\ &= -2x^2 - 2x + 1 \\ &= -2\left[x^2 + x - \frac{1}{2}\right] \\ &= -2\left[x^2 + x + \left(\frac{1}{2}\right)^2 - \left(\frac{1}{2}\right)^2 - \frac{1}{2}\right] \\ &= -2\left[\left(x + \frac{1}{2}\right)^2 - \frac{3}{4}\right] \\ &= -2\left(x + \frac{1}{2}\right)^2 + \frac{3}{2} \end{aligned}$$

Since

$$x_1 + x_2 = -\frac{b}{a} = -26, \quad \Longrightarrow \quad a = 1$$

Substituting into $ax^2 - a(x_1 + x_2)x + ax_1x_2 = 0$, we obtain

$$\begin{aligned} x^2 - (-26)x + (-27) &= 0 \\ x^2 + 26x - 27 &= 0. \end{aligned}$$

Exercises

1. Show that the curve $y = ax^2 + bx + c$ is a translation of the curve $y = ax^2$, and determine the translation vector in terms of a, b and c.

2. Let α and β be the roots of the equation $ax^2 + bx + c = 0$. Show that the equation whose roots are $(\alpha+\beta)^2$ and $(\alpha-\beta)^2$ is $a^4x^2 + -a^2(2b^2 - 4ac)x + b^2(b^2 - 4ac) = 0$

Chapter 4

Polynomial Functions

A polynomial function is generally defined by

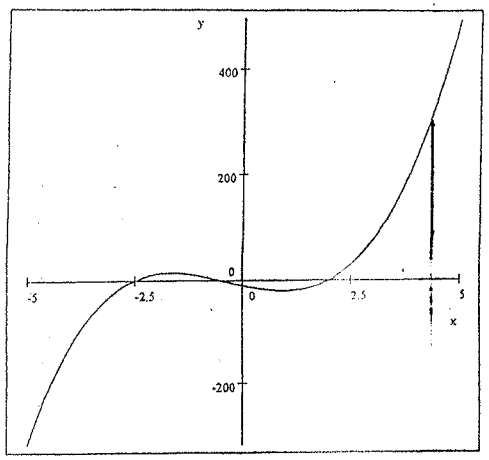
$$P(x) = a_nx^n + a_{n-1}x^{n-1} + a_{n-2}x^{n-2} + \cdots + a_1x + a_o$$

where $a_i \in \mathbb{R}$, $i = 0, 1, 2, \cdots, n$, $n \geq 0$, $n \in \mathbb{Z}^+$. *(That is, n is a non-negative integer.)* $P(x)$ is said to be a polynomial function of degree n, written as $\deg P(x) = n$, and a_n is called the leading coefficient.

Some Operations and Characteristics of a Polynomial Function

1. *$P(x)$ has a smooth curve*

2. If $P(x)$ is a polynomial of degree $n > 1$, then the graph of $P(x)$ intersects the $x-$axis (or any line in the plane) in at most n points. Figures 4.1 and 4.2 show the graphs of polynomial of degree 3. Figure 4.3 shows the graph of a polynomial of degree 4. Note the distinction between the shapes of the curves in Figures 4.1 and 4.2. Particularly, in Figure4.1, the leading coefficient a_n is positive, whilst in Figure 4.2, a_n is negative. In both graphs, the curves cuts the $x-$axis in three points, which is the degree of the function. In Figure 4.3, the curve cuts any suitably drawn line (e.g, $x = 2$) in the plane in at most four points, which is the degree of the function.



.Figure 4.1

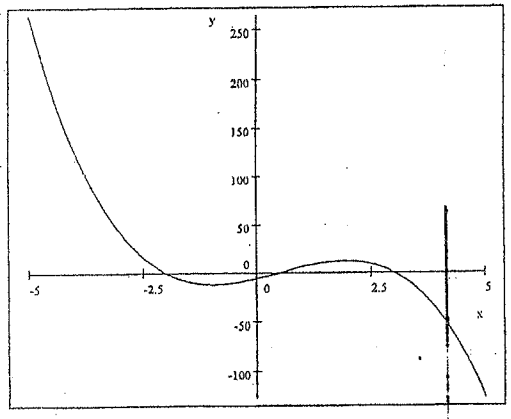


Figure 4.2

Figures 4.1 and 4.2 show the graphs of polynomials of degree 3 given respectively, by $y = 4x^3 + 4x^2 - 19x - 10$ and $y = -2x^3 + 3x^2 + 11x - 6$.

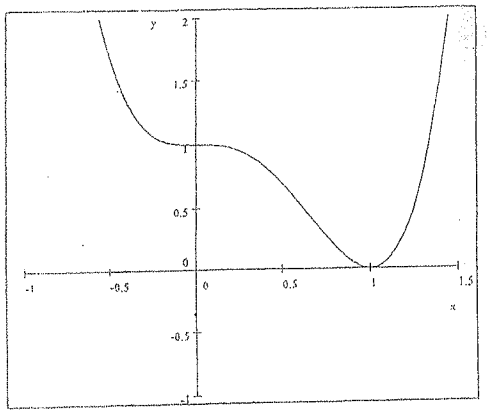


Figure 4.3

Figure 4.3 shows the graph of a polynomial of degree 4 given by $y = 3x^4 - 4x^3 + 1$.

3. For large values of x, the graph of $P(x)$ looks more and more like the graph of its leading term. For example, we observed in the previous chapter that the second degree polynomial $y = ax^2 + bx + c$ is just a translation of $y = ax^2$.

4. A zero of $P(x)$ is an x value x_0 for which $P(x_o) = 0$.

5. $P(x) + Q(x) = (P + Q)(x)$. That is, evaluating the polynomials and adding results is the same as adding the polynomials and evaluating the resulting polynomial. Suppose that $\deg P(x) = m$, $\deg Q(x) = n$ such that $m > n$, then $\deg(P + Q)(x) = m$.

6. $P(x)Q(x) = PQ(x)$. That is, evaluating the polynomials and multiplying results is the same as multiplying the polynomials and evaluating the resulting polynomial. Suppose that $\deg P(x) = m$, $\deg Q(x) = n$, then $\deg PQ(x) = m + n$.

7. Suppose $x = a$ is a zero of $P(x)$. Then $x = 0$ is a zero of $P(x + a)$. This point will be explained in a subsequent section.

8. By a process called 'Long Division', if a polynomial $P(x)$ is divided by another $D(x)$, then it yields the polynomials $Q(x)$ and $R(x)$ such that

$$P(x) = D(x)Q(x) + R(x) \tag{4.1}$$

where $\deg D(x) > 0$ and $\deg R(x) < \deg D(x)$.

The polynomial $Q(x)$ is called the quotient, $D(x)$ is the divisor, and $R(x)$ is the remainder polynomial.

Exanple 1

Find the remainder when $P(x) = x^3 + 4x^2 - 2x + 3$ is divided by $D(x) = 2x^2 - 1$.

Solution

By the long division process, we divide $P(x)$ by $D(x)$ as follows.

$$
\begin{array}{r|l}
 & \frac{1}{2}x + \quad 2 \\
\hline
2x^2 - 1 & x^3 + \quad 4x^2 - \quad 2x + \quad 3 \\
 & x^3 + \quad 0x^2 - \quad \frac{1}{2}x \\
\cline{2-2}
 & \qquad 4x^2 - \quad \frac{3}{2}x + \quad 3 \\
 & \qquad 4x^2 + \quad 0x - \quad 2 \\
\cline{2-2}
 & \qquad -\frac{3}{2}x + \quad 5
\end{array}
$$

Therefore, $P(x)$ is written such that

$$
\begin{aligned}
P(x) &= D(x)Q(x) + R(x) \\
&= (2x^2 - 1)\left(\frac{1}{2}x + 2\right) - \frac{3}{2}x + 5
\end{aligned}
$$

Note from the example that the 'long division' amounts to a series of successive multiplication and subtraction and terminates when the condition that $\deg R(x) < \deg D(x)$ is met. Equation(4.1) is what we refer to as the Remainder Theorem. In this equation, if $R(x) = 0$, then $P(x) = D(x)Q(x)$. In this case, $P(x)$ is said to be divisible by $D(x)$ or $D(x)$ is a factor of $P(x)$, and $Q(x)$ is also a factor of $P(x)$. $P(x)$ is then said to be factorised into its factors. If $D(x)$ is linear, then this case is stated in another theorem that follows.

4.1 The Factor Theorem

Let $P(x)$ be a polynomial. $P(a) = 0 \Leftrightarrow P(x) = (x - a)Q(x)$ for some polynomial $Q(x)$.

Proof

Let

$$P(x) = (x - a)Q(x).$$

Then at

$$
\begin{aligned}
x &= a, \quad P(a) = (a - a)Q(a) = 0 \\
&\Rightarrow P(a) = 0.
\end{aligned}
$$

Suppose now that $P(a) = 0$. That is, $x = a$ is a zero of $P(x)$. Then we prove that $(x - a)$ is a factor of $P(x)$.

Express $P(x)$ as

$$
\begin{aligned}
P(x) &= D(x)Q(x) + R(x) \\
&= (x - a)Q(x) + R(x)
\end{aligned}
$$

But $R(x)$ is a polynomial of degree 0, i.e., a constant. Since it is assumed that $P(a) = 0$, we have

$$
\begin{aligned}
(x - a)Q(x) + R(x) &= 0, \\
(a - a)Q(a) + R(a) &= 0, \\
R(a) &= 0
\end{aligned}
$$

This implies that

$$P(x) = (x - a)Q(x).$$

which means $(x - a)$ is a factor of $p(x)$. Hence, the result.

Note

In the Factor Theorem, if $P(x) = (x - a)Q(x)$, then $(x - a)$ and $Q(x)$ are both factors of $P(x)$.

Example 2

Find a factor of the polynomial $P(x) = x^3 - x^2 - 5x - 3$. Hence, find the other factor(s) of $P(x)$.

Solution

If $x = 3$, then

$$\begin{aligned} P(3) &= 3^3 - 3^2 - 5(3) - 3 \\ &= 27 - 9 - 15 - 3 \\ &= 27 - 27 = 0 \end{aligned}$$

Since $P(3) = 0, \Rightarrow x - 3$ is a factor of $P(x)$.

Thus,

$$P(x) = (x - 3)Q(x) .$$

By long division, we obtain

$$\begin{aligned} P(x) &= (x - 3)(x^2 + 2x + 1) \\ &= (x - 3)(x + 1)^2 \end{aligned}$$

Therefore, the other factors of $P(x)$ are $(x + 1)$ and $(x + 1)^2$

4.1.1 Extended Factor Theorem

Let $a_1, a_2, a_3, \cdots, a_m$ be m distinct zeros of the polynomial $P(x)$. Then, for some polynomial $Q(x)$,

$$P(x) = (x - a_1)(x - a_2)(x - a_3) \cdots (x - a_m)Q(x)$$

In the above theorem, $P(x)$ is factorised into m linear factors and another polynomial factor, $Q(x)$. Example 2 is actually an illustration of an extended factor theorem. As another example, we express the polynomial $P(x) = x^3 - 6x^2 + 11x - 6$ as

$$\begin{aligned} P(x) &= (x - 1)(x - 2)Q(x) \\ &= (x - 1)(x - 2)(x - 3) \end{aligned}$$

The polynomial $P(x)$ is factorized into two linear factors, $(x-1)$ and $(x-2)$, and another polynomial $Q(x) = (x - 3)$, which is also linear.

4.1.2 Extended Factor Theorem with Multiplicities

Let $a_1, a_2, a_3, \cdots, a_m$ be m distinct zeros of the polynomial $P(x)$. Then, $P(x)$ can be factorised as

$$P(x) = (x - a_1)^{r_1}(x - a_2)^{r_2}(x - a_3)^{r_3} \cdots (x - a_m)^{r_m}Q(x)$$

where $Q(x)$ is a polynomial having no real zeros or is a constant..

The $r_i \in Z^+$ is called the multiplicity of the zero a_i for $P(x)$, $i = 1, 2, 3, \cdots, m$.

Example 3

Factorise $P(x) = 2x^4 - 5x^3 + 3x^2 + x - 1$.

Solution

Clearly, $x = 1$ is a zero of $P(x)$.

By repeated long division, we obtain

$$\begin{aligned} P(x) &= 2x^4 - 5x^3 + 3x^2 + x - 1 \\ &= (x - 1)(2x^3 - 3x^2 + 1) \\ &= (x - 1)(x - 1)(2x^2 - x - 1) \\ &= (x - 1)(x - 1)(x - 1)(2x + 1) \\ &= (x - 1)^3(2x + 1). \end{aligned}$$

Therefore, the zero $x = 1$, has a multiplicity of 3, and the zero $x = -\frac{1}{2}$ has a multiplicity of 1.

The extended factor theorem has some two consequences which are stated as follows:

1. The n zeros Theorem: A non-zero polynomial of degree n has at most n distinct zeros.

2. Let $\deg P(x) > 1$. Then the graph of $P(x)$ intersects any line in the plane in at most n points.

This remark is illustrated in the second of the characteristics of polynomials presented earlier in this section.

Example 4

A polynomial has $(x+1)$ as a factor if and only if the alternating sum of its coefficients is 0.

Solution

Let

$$P(x) = a_n x^n + a_{n-1}x^{n-1} + a_{n-2}x^{n-2} + \cdots + a_1 x + a_0$$

Suppose $(x+1)$ is a factor of $P(x)$, $\Rightarrow$ $x = -1$ is a zero, $\Rightarrow$ $P(-1) = 0$.That is,

$$a_n(-1)^n + a_{n-1}(-1)^{n-1} + a_{n-2}(-1)^{n-2} + \cdots + a_1(-1) + a_0 = 0$$
$$a_o - a_1 + a_2 - \cdots + (-1)^{n-1}a_{n-1} + (-1)^n a_n = 0 .$$

Now, suppose that

$$a_o - a_1 + a_2 - \cdots + (-1)^{n-1}a_{n-1} + (-1)^n a_n = 0$$

or

$$a_n(-1)^n + a_{n-1}(-1)^{n-1} + a_{n-2}(-1)^{n-2} + \cdots + a_1(-1) + a_0 = 0.$$

Put $x = -1$. This implies that

$$a_n x^n + a_{n-1}x^{n-1} + a_{n-2}x^{n-2} + \cdots + a_1 x + a_o = 0$$

That is, $P(x) = 0$, when $x = -1$, $\Rightarrow$ $(x+1)$ is a factor of $P(x)$.

For example, for $P(x) = 3x^2 + 5x + 2$, the alternating sum of coefficients is $2 - 5 + 3 = 0$.

This implies that $(x+1)$ is a factor. Therefore, $P(x) = (x+1)(3x+2)$.

Exercises

1. Factorize
$$P(x) = x^5 - 4x^4 + 7x^3 - 10x^2 + 10x - 4$$
and state the multiplicities of the zeros of $P(x)$.

2. A polynomial is given as
$$Q(x) = 2x^4 + 15x^3 + 27x^2 - 27x - 81$$
Verify that $x = -3$ is a zero of $Q(x)$ and find its multiplicity.

3. Divide $P(x) = x^4 + 9x^2 + 2x - 1$ by $D(x) = x^2 + 4x + 1$.

4.2 The Replacement Operation

Suppose that a polynomial $P(x)$ is in terms of x. If we replace x by another polynomial $t(x)$, in $P(x)$, we obtain a polynomial that is in terms of $t(x)$. Assume, without loss of generality, that $t(x) = x + a$, is linear. If $t(x)$ is actually a factor of $P(x)$, then $P(t(x) - a)$ is a polynomial whose zero is zero (0). It will then be easier to determine the other factors of $P(x)$. This approach to finding the factors of a polynomial is what is referred to as the *Replacement Operation*. We will illutrate this operation with the following example.

Example 5

A polynomial is given as

$$P(x) = 2x^3 - x^2 - 8x + 4.$$

Find a polynomial whose zero is 0. Hence, factorise $P(x)$ completely.

Solution

Clearly, $x = 2$ is a zero.
Let

$$t = x - 2, \quad \Rightarrow \quad x = t + 2.$$

Substituting for x in $P(x) = 2x^3 - x^2 - 8x + 4$, we obtain

$$\begin{aligned} P(t+2) &= 2(t+2)^3 - (t+2)^2 - 8(t+2) + 4 \\ &= 2(t^3 + 6t^2 + 12t + 8) - (t^2 + 4t + 4) - 8t - 16 + 4 \\ &= 2t^3 + 12t^2 + 24t + 16 - t^2 - 4t - 4 - 8t - 16 + 4 \\ &= 2t^3 + 11t^2 + 12t \\ G(t) &= t(2t^2 + 11t + 12) \end{aligned}$$

which is a polynomial whose zero is 0.

Now factorising $G(t)$, gives

$$G(t) = t(t+4)(2t+3).$$

Substituting back $(x-2)$ for t and simplifying gives

$$\begin{aligned} P(x) &= (x-2)\,[(x-2)+4]\,[2(x-2)+3] \\ &= (x-2)(x+2)(2x-1). \end{aligned}$$

4.3 Factorization of the Polynomial $x^n - a^n$

It is clear that a factor of the polynomial $P(x) = x^n - a^n$ is $(x-a)$. We are now interested in finding the other factors of $P(x)$.

Long division method is used to find the other factor as follows.

$$\begin{array}{r|l} & x^{n-1} + ax^{n-2} + a^2x^{n-3} + \cdots + a^{n-2}x + a^{n-1} \\ \hline x-a & x^n - a^n \\ & x^n - ax^{n-1} \\ \hline & ax^{n-1} + 0x^{n-2} \\ & ax^{n-1} - a^2x^{n-2} \\ \hline & a^2x^{n-2} + 0x^{n-3} \\ & a^2x^{n-2} - a^3x^{n-3} \\ \hline & a^3x^{n-3} + 0x^{n-4} \\ & \cdots \quad \cdots \\ \hline & a^{n-2}x^2 + 0x \\ & a^{n-2}x^2 - a^{n-1}x \\ \hline & a^{n-1}x - a^n \\ & a^{n-1}x - a^n \\ \hline & 0 \end{array}$$

$P(x)$ is then written as

$$P(x) = (x-a)\left(x^{n-1} + ax^{n-2} + a^2x^{n-3} + \cdots + a^{n-2}x + a^{n-1}\right)$$

Observe that for the other factor, all the coefficients are in terms of the constant, a.

Example 6

Factorise $P(x) = x^4 - 16$.

Solution

We write $P(x)$ as $P(x) = x^4 - 2^4$.
Clearly, $x = 2$ is a zero, $\Rightarrow x - 2$ is a factor.
Then factorsing, we have

$$\begin{aligned} P(x) &= (x-2)\left(x^{4-1} + 2x^{4-2} + 2^2x^{4-3} + 2^{4-1}\right) \\ &= (x-2)\left(x^3 + 2x^2 + 4x + 8\right) \\ &= (x-2)(x+2)\left(x^2+4\right). \end{aligned}$$

Factorization of $(x^n + a^n)$ is another interesting exercise you should try.

4.4 The Rational Zero Theorem

Theorem

Let

$$f(x) = a_nx^n + a_{n-1}x^{n-1} + a_{n-2}x^{n-2} + \cdots + a_1x + a_0, \qquad a_i \in Z$$

If $\frac{p}{q}$ is a rational number (in lowest terms) is a zero of $f(x)$, then p is a factor of a_o and q of a_n.

Proof

Suppose that $x = \frac{p}{q}$ is a zero of $f(x)$ such that the highest common factor of p and q is 1, denoted as $(p,q) = 1$. Then $p\left(\frac{p}{q}\right) = 0$. That is,

$$\begin{aligned} a_n\left(\frac{p}{q}\right)^n + a_{n-1}\left(\frac{p}{q}\right)^{n-1} + a_{n-2}\left(\frac{p}{q}\right)^{n-2} + \cdots + a_1\left(\frac{p}{q}\right) + a_o &= 0 \\ a_np^n + a_{n-1}p^{n-1}q + a_{n-2}p^{n-2}q^2 + \cdots + a_1pq^{n-1} + a_oq^n &= 0 \quad (4.2) \end{aligned}$$

Equation (4.2) may be written as

$$\begin{aligned} a_n p^n + a_{n-1}p^{n-1}q + a_{n-2}p^{n-2}q^2 + \cdots + a_1 pq^{n-1} &= -a_o q^n \\ p\left(a_n p^{n-1} + a_{n-1}p^{n-2}q + a_{n-2}p^{n-3}q^2 + \cdots + a_1 q^{n-1}\right) &= -a_o q^n \end{aligned}$$

That is, p is a factor of the LHS $\Rightarrow p$ is a factor of the RHS.

But

$$(p, q) = 1 \Rightarrow (p, q^n) = 1.$$

Then p divides a_0.

Now, writing (4.2) as

$$\begin{aligned} -a_n p^n &= a_{n-1}p^{n-1}q + a_{n-2}p^{n-2}q^2 + \cdots + a_1 pq^{n-1} + a_o q^n \\ &= q\left(a_{n-1}p^{n-1} + a_{n-2}p^{n-2}q + \cdots + a_1 pq^{n-2} + a_o q^{n-1}\right) \end{aligned}$$

That is, q is a factor of the RHS $\Rightarrow q$ is a factor of the LHS.

But

$$(p, q) = 1 \Rightarrow (p^n, q) = 1.$$

Then q divides a_n.

Hence, the result.

The theorem shows that to determine the factors of a polynomial, one needs to first consider the two sets of factors of both the leading coefficient and the constant of the polynomial and the pairwise combinations of values from both sets.

Example 7

Factorise completely the polynomials

(a) $P(x) = 12x^3 - 44x^2 + 39x - 10$

(b) $Q(x) = 12x^4 + 4x^3 - 21x^2 - 16x - 3$

Solution

(a) Given $P(x) = 12x^3 - 44x^2 + 39x - 10$. Let $a_n = 12$, and $a_0 = 10$. The set of factors of 12 is given by

$$q = \{1,\ 2,\ 3,\ 4,\ 12\}$$

and the set of factors of 10 is

$$p = \{1,\ 2,\ 5,\ 10\}.$$

Consider the set of ratios,

$$\frac{p}{q} = \left\{\frac{1}{2},\ \frac{2}{3},\ \frac{5}{2}\right\}$$

Take, for example, $\frac{p}{q} = \frac{1}{2}$.

$$\begin{aligned} P\left(\frac{1}{2}\right) &= 12\left(\frac{1}{8}\right) - 44\left(\frac{1}{4}\right) + 39\left(\frac{1}{2}\right) - 10 \\ &= \frac{3}{2} - \frac{22}{2} + \frac{39}{2} - 10 \\ &= \frac{1}{2}[3 - 22 + 39 - 20] \\ &= \frac{1}{2}[42 - 42] = 0 \end{aligned}$$

Therefore, $x = \frac{1}{2}$ is a zero, $\Rightarrow 2x - 1$ is a factor.

Taking $\frac{p}{q} = \frac{2}{3}$.

$$\begin{aligned} P\left(\frac{2}{3}\right) &= 12\left(\frac{8}{27}\right) - 44\left(\frac{4}{9}\right) + 39\left(\frac{2}{3}\right) - 10 \\ &= \frac{32}{9} - \frac{176}{9} + \frac{78}{3} - 10 \\ &= \frac{1}{9}[32 - 176 + 234 - 90] \\ &= \frac{1}{2}[266 - 266] = 0. \end{aligned}$$

Therefore, $x = \frac{2}{3}$ is a zero, $\Rightarrow 3x - 2$ is a factor.

Taking $\frac{p}{q} = \frac{5}{2}$.

$$\begin{aligned} P\left(\frac{5}{2}\right) &= 12\left(\frac{125}{8}\right) - 44\left(\frac{25}{4}\right) + 39\left(\frac{5}{2}\right) - 10 \\ &= \frac{375}{2} - \frac{550}{2} + \frac{195}{2} - 10 \\ &= \frac{1}{2}[375 - 550 + 195 - 20] \\ &= \frac{1}{2}[570 - 570] = 0 \end{aligned}$$

Therefore, $x = \frac{5}{2}$ is a zero, $\Rightarrow 2x - 5$ is a factor.

Therefore, $P(x)$ is factorised as

$$P(x) = (2x - 1)(3x - 2)(2x - 5).$$

(b) Given $Q(x) = 12x^4 + 4x^3 - 21x^2 - 16x - 3$.

Let $a_n = 12$, and $a_0 = 3$.

The set of factors of 12,

$$q = \{1,\ 2,\ 3,\ 4,\ 12\}$$

and the set of factors of 3 is

$$p = \{1,\ 3\}$$

Consider the ratio, $\frac{p}{q} = \frac{3}{2}$. Since

$$Q(\frac{3}{2}) = 0,$$

it implies $(2x - 3)$ is a factor of $Q(x)$. By long division, we have

$$Q(x) = (2x - 3)(6x^3 + 11x^2 + 6x + 1).$$

By inspection, $x = -1$ is a zero of the other polynomial factor. $Q(x)$ is further written as

$$Q(x) = (2x - 3)(x + 1)(6x^2 + 5x + 1)$$

Using the rational zero theorem again, observe that since 2 and 3 are the factors of 6 and 1 is its own factor, the zeros of the quadratic factor will be $x = -\frac{1}{2}$ and $x = -\frac{1}{3}$.

Therefore, $Q(x)$ is further factorised as

$$Q(x) = (2x - 3)(x + 1)(2x + 1)(3x + 1).$$

4.5 Relation Between Roots and Coefficients of Polynomials

We have already shown how to derive the quadratic equation from its roots. We know that if x_1 and x_2 are the roots of the quadratic equation

$$ax^2 + bx + c = 0$$

then, it can be written as

$$a\left[x^2 - (x_1 + x_2)x + x_1x_2\right] = 0 \tag{4.3}$$

Expanding and simplifying gives

$$\begin{aligned} a\left[x^2 - x_1x - x_2x + x_1x_2\right] &= 0 \\ a\left[x(x - x_1) - x_2(x - x_1)\right] &= 0 \\ a(x - x_1)(x - x_2) &= 0 \end{aligned} \tag{4.4}$$

It was therefore realized that

$$x_1 + x_2 = -\frac{b}{a}, \quad \text{and} \quad x_1x_2 = \frac{c}{a}. \tag{4.5}$$

Consider the cubic polynomial

$$ax^3 + bx^2 + cx + d = 0.$$

Factorizing out a gives

$$a\left(x^3 + \frac{b}{a}x^2 + \frac{c}{a}x + \frac{d}{a}\right) = 0. \tag{4.6}$$

Suppose the roots are $r_1,\ r_2,\ r_3$. Then, we have

$$(x-r_1)(x-r_2)(x-r_3)=0.$$

Expanding and simplifying gives

$$\begin{aligned}(x-r_1)(x-r_2)(x-r_3) &= 0\\ (x^2-r_2x-r_1x+r_1r_2)(x-r_3) &= 0\\ \left[x^2-(r_1+r_2)x+r_1r_2\right](x-r_3) &= 0\\ x^3-r_3x^2-(r_1+r_2)x^2+r_3(r_1+r_2)x+r_1r_2x-r_1r_2r_3 &= 0\\ x^3-(r_1+r_2+r_3)x^2+(r_1r_2+r_1r_3+r_2r_3)x-r_1r_2r_3 &= 0 \qquad (4.7)\end{aligned}$$

Comparing Equation (4.6) and (4.7), we have

$$r_1+r_2+r_3=-\frac{b}{a};\qquad r_1r_2+r_1r_3+r_2r_3=\frac{c}{a}\ \text{ and }\ r_1r_2r_3=-\frac{d}{a} \qquad (4.8)$$

Therefore, in terms of its roots r_1, r_2, and r_3, the cubic equation is given as

$$a\left[x^3-(r_1+r_2+r_3)x^2+(r_1r_2+r_1r_3+r_2r_3)x-r_1r_2r_3\right]=0 \qquad (4.9)$$

The result in Equation (4.9) can be generalized for the general polynomial

$$a_nx^n+a_{n-1}x^{n-1}+a_{n-2}x^{n-2}+\cdots a_2x^2+a_1x+a_o=0 \qquad (4.10)$$

Suppose the roots are

$$r_1,\ r_2,\ r_3,\ \cdots,\ r_n$$

From Equation (4.9), Equation (4.10) can be written as

$$a_n\left[x^n-\left(\sum_{i=1}^{n}r_i\right)x^{n-1}+\left(\sum_{i<j}r_ir_j\right)x^{n-2}+\cdots+(-1)^n\prod_{i=1}^{n}r_i\right]=0 \qquad (4.11)$$

Thus, the zeros are related to the coefficients as follows:

$$\begin{aligned}\sum_{i=1}^{n}r_i &= -\frac{a_{n-1}}{a_n}\\ \sum_{i<j}r_ir_j &= \frac{a_{n-2}}{a_n}\\ \sum_{i<j<k}r_ir_jr_k &= -\frac{a_{n-3}}{a_n}\\ &\vdots\\ \prod_{i=1}^{n}r_i &= (-1)^n\frac{a_o}{a_n}\end{aligned}$$

Exercises

1. Factorise the following completely
   (a) $P(x)=24x^3-94x^2+117x-45$
   (b) $P(x)=4x^3-10x^2+6x-1.$
2. Factorise completely the polynomial
   (a) $P(x)=12x^3-44x^2+39x-10$
   (b) Hence, use the replacement operation to factorise completely the polynomial
   $$Q(x)=12x^4+4x^3-21x^2-16x-3.$$
3. Suppose
   $$P(x)=tx^4+96x^3+216x^2+mx+n$$
   is a perfect square. Determine the values of t, m and n.
   Answer: $t=16,\ m=216,\ \ n=81.$
   (a) Let
   $$P(x)=x^n+a^n,\quad n\in\mathbb{Z}^+.$$
   Show that $(x+a)$ is a factor of $P(x)\ \Leftrightarrow\ n$ is odd. Find the other factor.
   (b) Use your result above to factorize x^3+1.

4. Factorize the polynomial

$$P(x) = x^4 + 1$$

in the form

$$P(x) = (x^2 + mx + n)(x^2 + px + q)$$

where $m,\ n,\ p,\ q \in \mathbb{R}$.

Answer; $m = \sqrt{2},\ \ n = 1,\ \ p = -\sqrt{2},\ \ q = 1$.

5. If α and β are the roots of the equation $px^2 + qx + r = 0$, form without solving the equation, the equation whose roots are $(\alpha + 1)$ and $(\beta + 1)$.

6. The equation $3x^2 + 3kx + k = 0$ where k is a non-zero constant has roots α and β. Find the quadratic equation whose roots are

$$\frac{\alpha^3}{\beta\left(\alpha^3 + \beta^3 + k^3\right)} \quad \text{and} \quad \frac{\beta^3}{\alpha\left(\alpha^3 + \beta^3 + k^3\right)}.$$

Chapter 5

FURTHER FUNCTIONS

You are already familiar with polynomial functions. In this chapter, we will examine other functions that are not polynomials. These may be a function that is a ratio of two polynomial functions, called a rational function; it may be absolute function; and a function that involve radicals. For rational functions, we will examine their graphs, operations, and other forms of their presentation. For absolute functions, we will learn to sketch their graphs and examine the relationship between two given functions. Relationship between functions would be examined using the concept of transformations. In a similar manner, radical functions will then be studied.

5.1 Rational Functions

Definition

A function $f(x)$ which can be expressed as a quotient $\frac{p(x)}{q(x)}$, of two polynomials is called a rational function.

5.1.1 Domain and Range of a Function

The domain of $f(x)$ is given by $\{x : x \in \mathbb{R},\ q(x) \neq 0\}$.

In the above definition, it can be observed that if $\deg q(x) = n$, then there are at most n points where $f(x)$ is not defined.

Consider the rational function $y = \frac{p(x)}{q(x)},\ q(x) \neq 0$. The range of f are the values of y corresponding to the domain of f. The range is obtained by finding an expression for x from the given function. Such an expression is in

terms of y, of the form $x = h(y)$. The range is therefore the values of y for which x is defined. The value of y for which x is undefined is what is referred to as the *horizontal asymptote* to the curve of the function. The curve can also have an *oblique* asymptote. In this case, you will observe that the value of y for which x is undefined is also in terms of x. The *vertical asymptote* is the value of x for which the function is undefined. The curve of a function only approaches the asymptote but does not cross or touch it. However there could be a few exceptions.

Example 1

Determine the domain and range of the function

$$f(x) = \frac{1}{x^2+1}.$$

Examine the behaviour of the function, and hence, sketch its graph.

Solution

Let

$$y = \frac{1}{x^2+1}.$$

Clearly, for all values of $x \in \mathbb{R}$, $x^2 + 1 \neq 0$. Thus, y is defined for all values of x. Therefore, the domain of f is $\{x : x \in \mathbb{R}\}$.

Finding an expression for x, we obtain

$$\begin{aligned} y(x^2+1) &= 1 \\ x^2y &= 1-y \\ x^2 &= \frac{1-y}{y}. \end{aligned}$$

Therefore, x is given by

$$x = \pm\sqrt{\frac{1-y}{y}}$$

For real values of x, we must have

$$\frac{1-y}{y} \geq 0$$

Multiplying through by y^2 gives the inequality

$$y(1-y) \geq 0$$

which is satisfied by the values, $0 \leq y \leq 1$. However, if $y = 0$, x is not defined. Therefore, the range of the function is $\{y : 0 < y \leq 1\}$.

The line $y = 0$ is thus the horizontal asymptote. Since $f(-x) = f(x)$, the function is even. That is, the graph is symmetrical about the $y-$axis.

As

$$\begin{aligned} y &\longrightarrow 0^+, \quad x \longrightarrow +\infty; \\ y &\longrightarrow 0^-, \quad x \longrightarrow -\infty; \\ x &\longrightarrow 0^+, \quad y \longrightarrow 1; \\ x &\longrightarrow 0^-, \quad y \longrightarrow 1. \end{aligned}$$

The graph of $f(x)$ is given as shown in Figure 5.1.

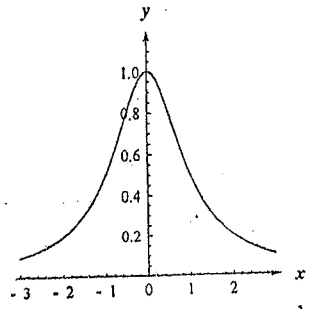


Figure 5.1: Graph of $y = \frac{1}{x^2+1}$

Example 2

Determine the domain and range of the function

$$f(x) = \frac{x^3}{x^2-4}.$$

Examine the behaviour of the function, and hence, sketch its graph.

Solution

Putting the denominator to zero, we have $x^2 - 4 = 0, \implies x = \pm 2$. Thus, y is defined for all values of x, except for $x = \pm 2$. Therefore, the domain of f is $\{x : x \in \mathbb{R}, x \neq \pm 2\}$. Finding an expression for x, we obtain

$$\begin{aligned} y(x^2 - 4) &= x^3 \\ x^2 y - x^3 &= 4y \\ x^2 &= \frac{4y}{y - x} \end{aligned}$$

Therefore, x is given by

$$x = \pm\sqrt{\frac{4y}{y - x}}.$$

For real values of x, we must have

$$\frac{4y}{y - x} \geq 0.$$

Multiplying through by $(y - x)^2$ and simplifying gives the inequality

$$y(y - x) \geq 0$$

which is satisfied by the values, $y \leq 0 \cup\ y \leq x$. However, if $y = x$, x is not defined. Therefore, the range of the function is $\{y : y \leq 0 \cup\ y < x\}$.

The line $y = x$ is thus an oblique asymptote. Since for $x = \pm 2$, the function is undefined, the line $x = 2$ and $x = -2$ are vertical asymptotes. We also note that

$$f(-x) = \frac{(-x)^3}{(-x)^2 - 4} = \frac{-x^3}{x^2 - 4} = -f(x)$$

This implies that the function is odd. That is, the graph is symmetrical about the origin.

Dividing both the numerator and denominator by the leading term of the denominator with unit coefficient.

$$y = \frac{\frac{x^3}{x^2}}{\frac{x^2}{x^2} - \frac{4}{x^2}} = \frac{x}{1 - \frac{4}{x^2}}.$$

In this form, we see that as

$$\begin{aligned} &x \longrightarrow \infty, \quad y \longrightarrow x; \\ &x \longrightarrow -2^-, \quad y \longrightarrow -\infty; \\ &x \longrightarrow -2^+, \quad y \longrightarrow \infty; \\ &x \longrightarrow 2^-, \quad y \longrightarrow -\infty; \\ &x \longrightarrow 2^+, \quad y \longrightarrow \infty. \end{aligned}$$

When $x = 0$, $y = 0$, $\implies (0, 0)$ is on the curve. The graph of $f(x)$ is given in Figure 5.2

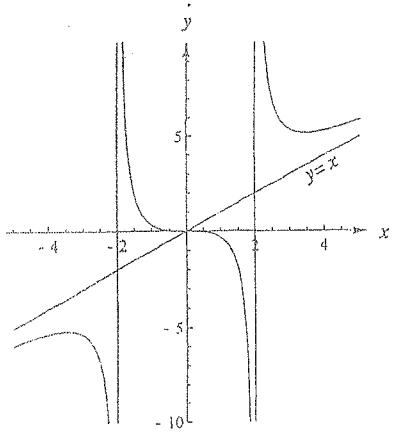


Figure 5.2: Graph of $y = \frac{x^3}{x^2-4}$

5.1.2 Decomposition of Rational Functions into Partial Fractions

Two rational functions can be added, multiplied, divided and subtracted to obtain another rational function. For example,

$$\frac{2}{(x-1)^2} + \frac{3}{x(x-1)} = \frac{5x - 3}{x(x-1)^2}.$$

In this section, we consider an important operation in which a rational function is decomposed into its partial fractions. In decomposition, we express

complicated rational functions as the sum of simple rational functions. Thus, it is a process that reverses the operation in the given example. We shall consider four cases of decomposition.

Case I

Let $f(x) = \frac{p(x)}{q(x)}$, where $\deg p(x) < \deg q(x)$. In this case, suppose that the denominator $q(x)$ can be factorised into *non-repeating linear factors*. That is, if $\deg q(x) = n$, then

$$f(x) = \frac{p(x)}{q(x)} = \frac{p(x)}{(x-a_1)(x-a_2)\cdots(x-a_n)}.$$

The function is decomposed as

$$\frac{p(x)}{(x-a_1)(x-a_2)\cdots(x-a_n)} = \frac{A_1}{x-a_1} + \frac{A_2}{x-a_2} + \cdots \frac{A_n}{x-a_n}$$

where A_i, $i = 1,\ 2,\ \cdots,\ n$, are constants.

Example 3

Express $f(x) = \frac{7x+11}{x^2+3x+2}$ as partial fractions.

Solution

We first factorize the denominator of the rational function. This gives

$$\frac{7x+11}{x^2+3x+2} = \frac{7x+11}{(x+1)(x+2)}$$

The numerator corresponding to each of the denominator linear factors is a constant. Thus,

$$\frac{7x+11}{(x+1)(x+2)} = \frac{A}{x+1} + \frac{B}{x+2} \tag{5.1}$$

Multiplying through Equation (5.1) by $(x+1)(x+2)$, we obtain

$$7x+11 = A(x+2) + B(x+1) \tag{5.2}$$

When

$$x = -1, \implies -7+11 = A \implies A = 4.$$

When

$$x = -2, \implies -14+11 = -B \implies B = 3.$$

Therefore,

$$f(x) = \frac{4}{x+1} + \frac{3}{x+2}.$$

Note that in the example above, the values of x chosen to obtain the values of A and B were not determined arbitrarily. They were chosen to make sore of the terms in the RHS of (5.2) equal to zero.

Alternative Approach

An alternative method to determining the values of A and B in the above example is to use what we call the method of undetermined coefficients. In this method, we expand and equate the corresponding coefficients in (5.2) to obtain

$$\begin{aligned} 7x+11 &= A(x+2) + B(x+1) \\ &= (A+B)x + 2A + B \end{aligned}$$

Thus,

$$\begin{aligned} A+B &= 7 \\ 2A+B &= 11 \end{aligned}$$

Solving the two equations simultaneously gives $A = 4$ and $B = 3$ as before.

In some problems, there will be the need to use the two methods to obtain the desired partial fractions for the rational function.

Case II

Let $f(x) = \frac{p(x)}{q(x)}$, where $\deg p(x) < \deg q(x)$. In this case, suppose that the denominator $q(x)$ can be factorised into *repeating linear factors*. That is, if $\deg q(x) = n$, then

$$f(x) = \frac{p(x)}{q(x)} = \frac{p(x)}{(x-a)^n}.$$

The function is decomposed as

$$\frac{p(x)}{(x-a)^n} = \frac{A_1}{x-a} + \frac{A_2}{(x-a)^2} + \cdots \frac{A_{n-1}}{(x-a)^{n-1}} + \frac{A_n}{(x-a)^n}$$

where A_i, $i = 1, 2, \cdots, n$, are constants.

Example 4

Express $f(x) = \frac{3x+1}{(x-3)(x+2)^2}$ as partial fractions.

Solution

We note that the denominator is already factorized into linear factors including a repeating factor, $(x+2)^2$. Thus,

$$\frac{3x+1}{(x-3)(x+2)^2} = \frac{A}{x-3} + \frac{B}{x+2} + \frac{C}{(x+2)^2} \tag{5.3}$$

Multiplying through Equation (5.3) by $(x-3)(x+2)^2$, we obtain

$$3x+1 = A(x+2)^2 + B(x-3)(x+2) + C(x-3) \tag{5.4}$$

When

$$x = -2, \implies -6+1 = C(-2-3) \implies C = 1.$$

When

$$x = 3, \implies 9+1 = 25A \implies A = \frac{10}{25} = \frac{2}{5}.$$

Thus,

$$\begin{aligned} 3x+1 &= A(x^2+4x+4) + B(x^2-x-6) + C(x-3) \\ &= (A+B)x^2 + (4A-B+C)x + 4A - 6B - 3C \end{aligned}$$

Comparing coefficients we have

$$A + B = 0 \implies B = -\frac{2}{5}$$

Therefore,

$$f(x) = \frac{2}{5(x-3)} - \frac{2}{5(x+2)} + \frac{1}{(x+2)^2}.$$

Exercises

1. Express $f(x) = \frac{x^3+2x-1}{(x^2-4)(x^2-1)}$ as partial fractions.
2. Decompose the following into partial fractions
   (a) $f(x) = \frac{1}{x(x+1)^2}$
   (b) $g(x) = \frac{x^2+11x+5}{x^2(x+5)}$

Case III

Let $q(x)$ has a non-repeating quadratic factor, i.e.,

$$q(x) = (a_1x^2 + b_1x + c_1)(a_2x^2 + b_2x + c_2)\cdots(a_nx^2 + b_nx + c_n)$$

Write

$$f(x) = \frac{A_1x + B_1}{a_1x^2 + b_1x + c_1} + \frac{A_2x + B_2}{a_2x^2 + b_2x + c_2} + \cdots + \frac{A_nx + B_n}{a_nx^2 + b_nx + c_n}.$$

Example 5

Express $f(x) = \frac{3-x}{x^3+1}$ as partial fractions.

Solution

$$\frac{3-x}{(x+1)(x^2-x+1)} = \frac{A}{x+1} + \frac{Bx+C}{x^2-x+1} \tag{5.5}$$

Multiplying through Equation (5.5) by $(x+1)(x^2-x+1)$, we obtain

$$3-x = A(x^2-x+1) + (Bx+C)(x+1) \tag{5.6}$$

When

$$x = -1, \implies 4 = A(1+1+1) \implies 3A = 4 \implies A = \frac{4}{3}.$$

That is,

$$\begin{aligned} 3-x &= A(x^2-x+1) + Bx^2 + x + Cx + C \\ &= (A+B)x^2 + (B+C-A)x + A + C \end{aligned}$$

$$\implies A+B=0 \implies B=-\frac{4}{3}$$
$$\implies A+C=3 \implies C=3-\frac{4}{3}=\frac{5}{3}$$

Therefore,

$$f(x)=\frac{\frac{4}{3}}{x+1}+\frac{-\frac{4}{3}x+\frac{5}{3}}{x^2-x+1}=\frac{4}{3(x+1)}+\frac{5-4x}{3(x^2-x+1)}.$$

Example 6

Express $f(x)=\frac{x^2}{(x+1)^2(x^2+1)^2}$ in partial fractions.

Solution

$$\frac{x^2}{(x+1)^2(x^2+1)^2}=\frac{A}{x+1}+\frac{B}{(x+1)^2}+\frac{Cx+D}{x^2+1}+\frac{Ex+F}{(x^2+1)^2} \tag{5.7}$$

Multiplying through Equation (5.7) by $(x+1)^2(x^2+1)^2$, we obtain

$$x^2=A(x+1)(x^2+1)^2+B(x^2+1)^2+(Cx+D)(x+1)^2(x^2+1)+(Ex+F) \tag{5.8}$$

Substituting $x=-1$ into (5.8), we obtain

$$1=4B \implies B=-\frac{1}{4}$$

Expanding (5.8) and simplifying gives

$$\begin{aligned}x^2 = {} & (A+C)x^5+(A+2C+B+D)x^4+(2A+2C+2D+E)x^3\\ & +(2A+2B+2C+2D+2E+F)x^2+(A+C+2D+E+2F)x\\ & +(A+B+D+F)\end{aligned}$$

Equating like terms on either sides of the above equation gives

$$\begin{aligned}A+C &= 0 && (1)\\ A+B+2C+D &= 0 && (2)\\ 2A+2C+2D+E &= 0 && (3)\\ 2A+2B+2C+2D+2E+F &= 1 && (4)\\ A+C+2D+E+2F &= 0 && (5)\\ A+B+D+F &= 0 && (6)\end{aligned}$$

From Equation (1), $C=-A$. Substituting for C in Equation (2) and simplifying gives

$$\begin{aligned}A-2A+\frac{1}{4}+D &= 0\\ -A+D &= -\frac{1}{4}\\ D &= A-\frac{1}{4}\end{aligned}$$

Substituting for C and D in Equation (3) gives

$$\begin{aligned}2A-2A+2\left(A-\frac{1}{4}\right)+E &= 0\\ -\frac{1}{2}+2A+E &= 0\\ E &= \frac{1}{2}-2A\end{aligned}$$

From Equation (6)

$$\begin{aligned}A+\frac{1}{4}+A-\frac{1}{4}+F &= 0\\ 2A+F &= 0\\ F &= -2A\end{aligned}$$

Substituting for C, D, E and F in Equation (5),

$$A - A + 2\left(A - \frac{1}{4}\right) + \left(\frac{1}{2} - 2A\right) + 4A = 0$$

Simplifying, we have $A = 0$. This implies that $C = 0$, $D = -\frac{1}{4}$, $E = \frac{1}{2}$, $F = 0$.

Therefore,

$$\begin{aligned} f(x) &= \frac{\frac{1}{4}}{(x+1)^2} + \frac{-\frac{1}{4}}{x^2+1} + \frac{\frac{1}{2}x}{(x^2+1)^2} \\ &= \frac{1}{4(x+1)^2} - \frac{1}{4(x^2+1)} + \frac{x}{2(x^2+1)^2}. \end{aligned}$$

5.2 Translations of a Function

Let $f(x)$ be a function and $a > 0$ and $c > 1$. The following translations of $f(x)$ can be considered.

1. $f(x-a)$ is a translation of $f(x)$ 'a' units to the right
2. $f(x+a)$ is a translation of $f(x)$ 'a' units to the left
3. $f(x)+a$ is a translation of $f(x)$ 'a' units up
4. $f(x)-a$ is a translation of $f(x)$ 'a' units down
5. $f(\frac{x}{c})$ is a shrink in x-direction by factor c.
6. $f(cx)$ is a stretch in x-direction by factor c.
7. $cf(x)$ is a stretch in y-direction by factor c.
8. $\frac{1}{c}f(x)$ is a shrink in y-direction by factor c.
9. $f(-x)$ is a reflection of $f(x)$ in y-axis.
10. $-f(x)$ is a reflection of $f(x)$ in x-axis.

We will demonstrate the use of translation with some examples.

Example 7

(a) Sketch the graph of

$$f(x) = \frac{1}{x}$$

(b) Deduce from the function in (a), the function

$$g(x) = \frac{1}{x+2} - 3$$

Hence, or otherwise sketch the graph of $g(x)$.

Solution

(a) Given the function $f(x) = \frac{1}{x}$

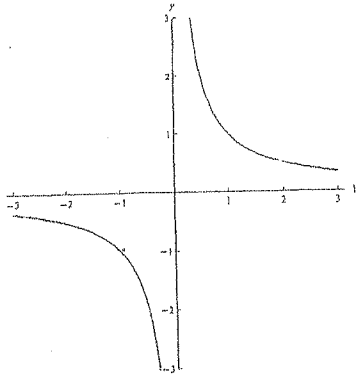


Figure 5.3: Graph of $y = \frac{1}{x}$

$x = 0$ is a vertical asymptote

As $x \to \infty$, $y \to 0$. Therefore, $y = 0$ is a horizontal asymptote.

As $x \to 0^-$, $y \to -\infty$ and $x \to 0^+$ $y \to +\infty$

The graph of $g(x)$ is shown in the Figure 5.3

(b) The graph of $g(x)$ is shown in the Figure 5.4.

$$\implies A+B=0 \implies B=-\frac{4}{3}$$
$$\implies A+C=3 \implies C=3-\frac{4}{3}=\frac{5}{3}$$

Therefore,

$$f(x)=\frac{\frac{4}{3}}{x+1}+\frac{-\frac{4}{3}x+\frac{5}{3}}{x^2-x+1}=\frac{4}{3(x+1)}+\frac{5-4x}{3(x^2-x+1)}.$$

Example 6

Express $f(x)=\frac{x^2}{(x+1)^2(x^2+1)^2}$ in partial fractions.

Solution

$$\frac{x^2}{(x+1)^2(x^2+1)^2}=\frac{A}{x+1}+\frac{B}{(x+1)^2}+\frac{Cx+D}{x^2+1}+\frac{Ex+F}{(x^2+1)^2} \tag{5.7}$$

Multiplying through Equation (5.7) by $(x+1)^2(x^2+1)^2$, we obtain

$$x^2=A(x+1)(x^2+1)^2+B(x^2+1)^2+(Cx+D)(x+1)^2(x^2+1)+(Ex+F) \tag{5.8}$$

Substituting $x=-1$ into (5.8), we obtain

$$1=4B \implies B=-\frac{1}{4}$$

Expanding (5.8) and simplifying gives

$$\begin{aligned}x^2 &= (A+C)x^5+(A+2C+B+D)x^4+(2A+2C+2D+E)x^3\\ &\quad+(2A+2B+2C+2D+2E+F)x^2+(A+C+2D+E+2F)x\\ &\quad+(A+B+D+F)\end{aligned}$$

Equating like terms on either sides of the above equation gives

$$\begin{aligned}A+C &= 0 &(1)\\ A+B+2C+D &= 0 &(2)\\ 2A+2C+2D+E &= 0 &(3)\\ 2A+2B+2C+2D+2E+F &= 1 &(4)\\ A+C+2D+E+2F &= 0 &(5)\\ A+B+D+F &= 0 &(6)\end{aligned}$$

From Equation (1), $C=-A$. Substituting for C in Equation (2) and simplifying gives

$$\begin{aligned}A-2A+\frac{1}{4}+D &= 0\\ -A+D &= -\frac{1}{4}\\ D &= A-\frac{1}{4}\end{aligned}$$

Substituting for C and D in Equation (3) gives

$$\begin{aligned}2A-2A+2\left(A-\frac{1}{4}\right)+E &= 0\\ -\frac{1}{2}+2A+E &= 0\\ E &= \frac{1}{2}-2A\end{aligned}$$

From Equation (6)

$$\begin{aligned}A+\frac{1}{4}+A-\frac{1}{4}+F &= 0\\ 2A+F &= 0\\ F &= -2A\end{aligned}$$

Substituting for C, D, E and F in Equation (5),

$$A - A + 2\left(A - \frac{1}{4}\right) + \left(\frac{1}{2} - 2A\right) + 4A = 0$$

Simplifying, we have $A = 0$. This implies that $C = 0$, $D = -\frac{1}{4}$, $E = \frac{1}{2}$, $F = 0$.

Therefore,

$$\begin{aligned} f(x) &= \frac{\frac{1}{4}}{(x+1)^2} + \frac{-\frac{1}{4}}{x^2+1} + \frac{\frac{1}{2}x}{(x^2+1)^2} \\ &= \frac{1}{4(x+1)^2} - \frac{1}{4(x^2+1)} + \frac{x}{2(x^2+1)^2}. \end{aligned}$$

5.2 Translations of a Function

Let $f(x)$ be a function and $a > 0$ and $c > 1$. The following translations of $f(x)$ can be considered.

1. $f(x-a)$ is a translation of $f(x)$ 'a' units to the right
2. $f(x+a)$ is a translation of $f(x)$ 'a' units to the left
3. $f(x)+a$ is a translation of $f(x)$ 'a' units up
4. $f(x)-a$ is a translation of $f(x)$ 'a' units down
5. $f(\frac{x}{c})$ is a shrink in x-direction by factor c.
6. $f(cx)$ is a stretch in x-direction by factor c.
7. $cf(x)$ is a stretch in y-direction by factor c.
8. $\frac{1}{c}f(x)$ is a shrink in y-direction by factor c.
9. $f(-x)$ is a reflection of $f(x)$ in y-axis.
10. $-f(x)$ is a reflection of $f(x)$ in x-axis.

We will demonstrate the use of translation with some examples.

Example 7

(a) Sketch the graph of

$$f(x) = \frac{1}{x}$$

(b) Deduce from the function in (a), the function

$$g(x) = \frac{1}{x+2} - 3$$

Hence, or otherwise sketch the graph of $g(x)$.

Solution

(a) Given the function $f(x) = \frac{1}{x}$

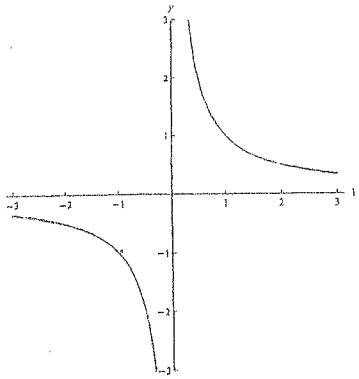


Figure 5.3: Graph of $y = \frac{1}{x}$

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As $x \to \infty$, $y \to 0$. Therefore, $y = 0$ is a horizontal asymptote.

As $x \to 0^-$, $y \to -\infty$ and $x \to 0^+$ $y \to +\infty$

The graph of $g(x)$ is shown in the Figure 5.3

(b) The graph of $g(x)$ is shown in the Figure 5.4.

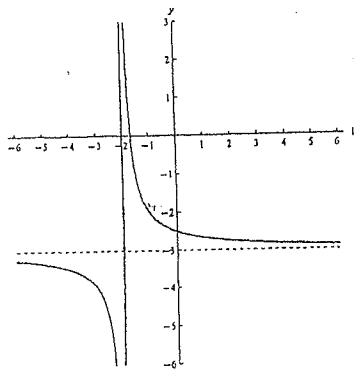


Figure 5.4: Graph of $y = \frac{1}{x+2} - 3$

Given $f(x) = \frac{1}{x}$ we obtain $f(x+2) = \frac{1}{x+2}$ and $f(x+2) - 3 = \frac{1}{x+2} - 3$.

Therefore, $g(x) = f(x+2) - 3$. Thus, $g(x)$ is a translation of $f(x)$ 2 units to the left and 3 units down, i.e., a translation by the vector $\begin{pmatrix} -2 \\ -3 \end{pmatrix}$.

Alternative Method

For $g(x) = \frac{1}{x+2} - 3$, when $x = -2$, $g(x)$ is undefined. Thus, $x = -2$ is a vertical asymptote.

Dividing numerator and denominator by x gives

$$y = \frac{\frac{1}{x}}{1 + \frac{2}{x}} - 3.$$

As $x \to \infty$, $y \to -3$. This means $y = -3$ is a horizontal asymptote.

As $x \to -2^-$, $y \to -\infty$ and $x \to -2^+$, $y \to +\infty$.

The points of intercepts are $(0, -2.5)$ and $(-1.67, 0)$. The graph of $g(x)$ is shown in Figure 5.4.

The method of translation of functions will be used very often in subsequent sections.

5.3 Absolute Functions and Functions involving Radicals

5.3.1 Absolute Functions

Recall that

$$y = |x| = \begin{cases} x, & x \geq 0 \\ -x, & x < 0 \end{cases} \quad \text{and } \sqrt{x^2} = |x|$$

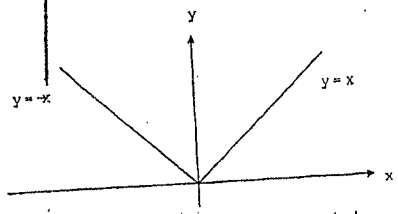


Figure 5.5: Graph of $y = |x|$

Example 8

Sketch the graph of the following functions

(a) $f(x) = |x| + 1$

(b) $f(x) = |x+1|$

Solution

(a) Since $|x| = x$, $x \geq 0$, but $|x| = -x$, $x < 0$, we obtain

$$|x| + 1 = \begin{cases} x+1, & x \geq 0 \\ -x+1, & x < 0 \end{cases}$$

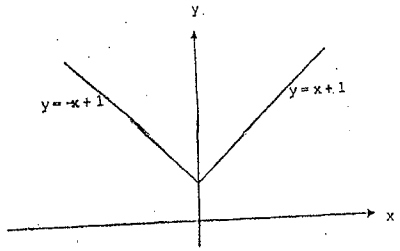


Figure 5.6: Graph of $y = |x| + 1$

(b) $|x+1| = \begin{cases} x+1, & x+1 \geq 0 \quad \text{or} \quad x \geq -1 \\ -x-1, & x+1 < 0 \quad \text{or} \; x < -1 \end{cases}$

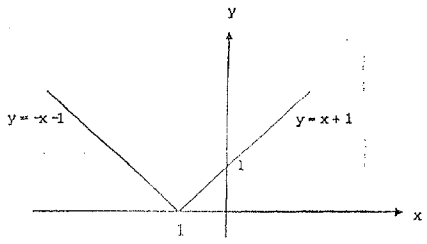


Figure 5.7: Graph of $y = |x+1|$

Example 9

Sketch the graph of the function $f(x) = |3 - 2x - x^2|$

Solution

$$\begin{aligned} f(x) &= |3 - 2x - x^2| \\ &= \begin{cases} 3 - 2x - x^2, & 3 - 2x - x^2 \geq 0 \\ -(3 - 2x - x^2), & 3 - 2x - x^2 < 0 \end{cases} \\ &= \begin{cases} (3+x)(1-x), & (3+x)(1-x) \geq 0 \\ (3+x)(x-1), & (3+x)(1-x) < 0 \end{cases} \end{aligned}$$

We sketch the3 curves $f(x) = (3+x)(1-x)$ and $g(x) = (3+x)(x-1)$ and highlighted within regions subject to conditions.

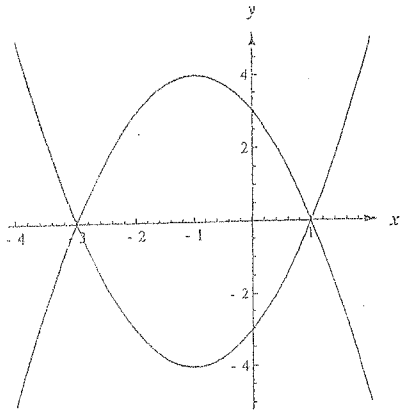


Figure 5.8: Graph of $y = |3 - 2x - x^2|$

5.3.2 Functions involving Radicals

Consider equations in terms of x and y in which y depends on the value of x, but not the other way round.

That is x is the independent variable and y is the dependent variable. For instance, for $y = \sqrt{x}$, we consider only positive values of y. Dom$f$ = $\{x : x \geq 0\}$ and Rg= $\{f(x) \geq 0 : x \geq 0\}$. Figure 5.9 is the graph of $y = \sqrt{x}$.

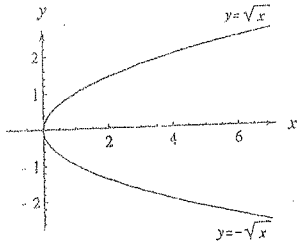


Figure 5.9: Graph of $y = \sqrt{x}$

The graphs of the expressions $\sqrt{x-2}$, $\sqrt{x-3}$, $\sqrt{\frac{1}{2}x - 2}$, $\sqrt{x+3}$ etc. are translation of $\sqrt{x}$.

Let $f(x) = \sqrt{x}$ and $g(x) = \sqrt{x-2}$. Thus, $f(x-2) = g(x)$, which is a translation of $f(x)$ by (2, 0). Dom$g = \{x : x \geq 2\}$ and $Rg = \{g(x) \geq 0 : x \geq 2\}$ Let $h(x) = \sqrt{\frac{1}{2}x - 2}$. Then $f(\frac{1}{2}x - 2) = h(x)$ is a shrink in $x-$direction by 2 and then a shift to the right by 2. D$g = \{x : x \geq 4\}$.. The graphs of $\sqrt{x}$ and its translations are given in the figure shown.

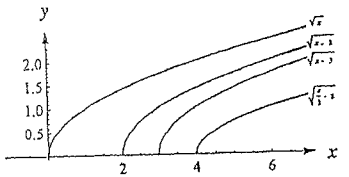


Figure 5.10: Translations of $\sqrt{x}$

Example 10

By considering the graph of $f(x) = \sqrt{1-x^2}$, or otherwise, sketch the graph of the following in the same plane.

(i) $g(x) = \sqrt{1-(x+2)^2}$

(ii) $h(x) = -\sqrt{1-(x-2)^2}$

(iii) $t(x) = 1 - \sqrt{1-(x+2)^2}$

Solution

(i) $f(x) = \sqrt{1-x^2}$, i.e $y^2 = 1 - x^2 \implies x^2 + y^2 = 1 \implies f(x)$ is upper half of circle with radius 1 and centre $(0, 0)$. $g(x) = \sqrt{1-(x+2)^2} = f(x+2) \implies g(x)$ is a translation of $f(x)$, 2 units to the left.

Alternative approach

$y^2 = 1-(x+2)^2 \implies (x+2)^2 + y^2 = 1$, i.e., a circle of radius 1unit with centre $(-2, 0)$. Therefore, $g(x)$ is the upper half of the circle.

(ii) $h(x) = -\sqrt{1-(x-2)^2} = -f(x-2)$, i.e., a translation of $f(x)$ 2 units to the right and reflected in the $x-$axis.

(iii) $t(x) = 1 - \sqrt{1-(x+2)^2} = 1 - f(x+2)$, i.e., a translation of $f(x)$ 2 units to the left and reflected in the $x-$axis, followed by shift up on the y-axis by 1 unit.

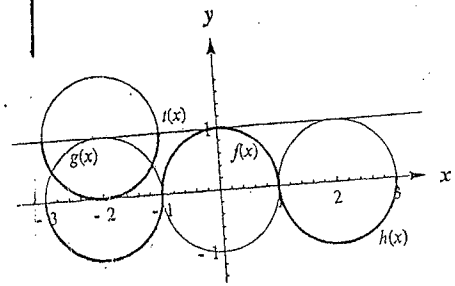


Figure 5.11: Translations of some semi-circles

Exercise

1. Sketch the curve of the function $f : x \to \sqrt{3-2x-x^2}$.

   Deduce also the graph of $g : x \to 2 - \sqrt{3-2x-x^2}$ in the same plane as f and describe this curve completely.

2. In the diagram, write down the equation of each of the semi-circles. In each case, describe the circle in relation to the basic semi-circle centred at the origin.

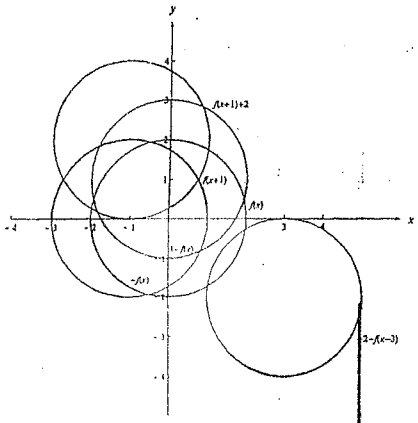


Figure 5.12

3. For each of the following functions determine the domain and range. Examine the behaviour of the function, and hence, sketch its graph.

   a. $f(x) = \frac{1}{x^2-1}$ b. $t(x) = \frac{x^3}{x^2-1}$

   c. $f(x) = \frac{x+2}{x-3}$ d. $g(x) = \frac{x^2+1}{4^2-3}$

   e. $h(x) = \frac{x^2+2x+3}{x+2}$ f. $g(x) = \frac{x}{x^2-1}$

Chapter 6

Linear Inequalities

The line $l : ax + by + c = 0$, as we know is the linear equation in terms of x and y.

It may be written in a set form as

$$l = \{(x, y) : ax + by + c = 0\}.$$

That is, the line l consists of all points that satisfy the equation $ax+by+c = 0$. Similarly, an inequality such as $l : ax + by + c \geq 0$ is the set of all points for which $ax + by + c > 0$ and $ax + by + c = 0$ are satisfied. A simple way of determining the set of points that satisfy the inequality $ax + by + c > 0$ is to conduct what is called the *origin* test. To conduct this test, we first note that the entire plane is divided into two regions by the line $ax+by+c = 0$ which is the boundary of the solution set of the inequality. The test simply examines whether or not the origin $(0, 0)$ satisfies the inequality by substituting it into the inequality. If it satisfies it, it implies that the entire region that contains the origin satisifes the inequality. This region is then shaded accordingly.

Example 1

Find the region in the plane which satisfies the inequality

$$3x - 2y - 6 < 0$$
$$3x + 4y - 12 \geq 0$$

Solution

Consider

$$\begin{aligned} 3x - 2y - 6 &= 0 \\ 3x - 2y &= 6 \\ \frac{x}{2} - \frac{y}{3} &= 1 \end{aligned}$$

This implies that the x-intercept $= 2$, and y-intercept $= -3$.

At $(0, 0)$,

$$3(0) - 2(0) - 6 = -6 < 0$$

The origin therefore satisfies the inequality. It implies that the region containing the origin satisfies the inequlaity. The region is as shaded in Figure 6.1.

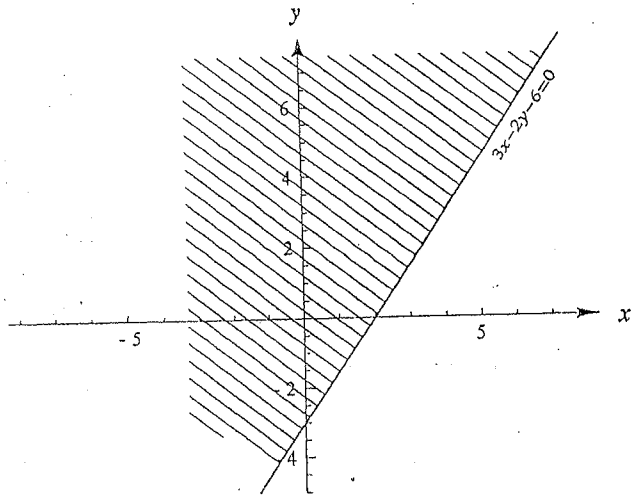


Figure 6.1

Example 2

Find the region which satisfies simultaneously the inequalities

$$\begin{aligned} 3x - 2y - 6 &< 0 \\ 3x + 4y - 12 &\geq 0 \end{aligned}$$

Solution

The region satisfied by $3x - 2y - 6 < 0$ is as shown in the previous example. Now we consider $3x + 4y - 12 \geq 0$. Taking the equality,

$$\begin{aligned} 3x + 4y - 12 &= 0 \\ 3x + 4y &= 12 \\ \frac{x}{4} + \frac{y}{3} &= 1 \end{aligned}$$

Thus, x-intercept $= 4$ and y-intercept $= 3$.

At $(0, 0)$,

$$3(0) + 4(0) - 12 = -12 \not> 0$$

The origin therefore does not satisfy the inequality. It implies that the region that does not contain $(0, 0)$ is the solution region. Shading this region in the same plane as $3x - 2y - 6 < 0$, the simultaneous solution region is as shaded in Figure 6.2..

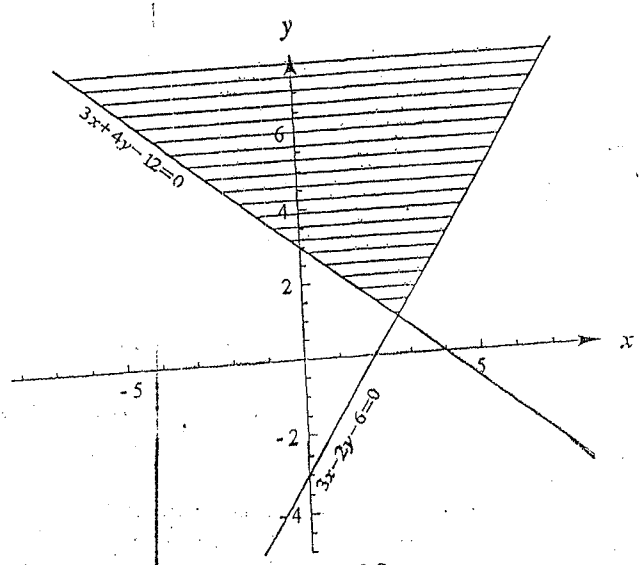


Figure 6.2

Exercise

Show by shading the region that satisfies simultaneously the following sets of linear inequalities.

1.

$$\begin{aligned} 4y+3x-12 &\geq 0 \\ y-2x &\geq -2 \\ x-y+3 &\geq 0 \end{aligned}$$

2.

$$\begin{aligned} 2x-3y &\geq -6 \\ 2y-x+2 &\geq 0 \\ y-2x-2 &\leq 0 \\ x &\leq 3 \end{aligned}$$

3.

$$\begin{aligned} y+2x &\leq 2x \\ y &\leq 2x \\ 3y &\geq 2. \end{aligned}$$

6.1 Maximizing or Minimizing a Linear Expression

Let $ax+by$ $(=c)$ be a given linear expression. It is usually of interest to find the particular members of the solution set of a system of linear inequalities that maximizes or minimizes c. That is, we want to determine the pair (x,y) for which c is greatest or smallest and also satisfies the given system of inequalities.

It should be noted that the farther away one moves from the origin, the greater c becomes. Therefore, the maximizing or minimizing pair (x,y) of c occurs at the extremeties of the solution set. That is, c is optimized at a vertex of the figure formed by the solution set of the given system of inequalities. Graphs of $ax+by$ are parallel for all values of c.

Example 3

Shade in the $x-y$ plane, the region that satisfies simultaneously the linear inequalities

$$\begin{aligned} x+2y &\leq 14; \quad 3x+2y \leq 26; \quad 2x+y \geq 10 \\ x+4y &\geq 12; \quad x \geq 0; \quad y \geq 0. \end{aligned}$$

Find the members of the solution set which maximize and minimize the expression $5x+9y$. Hence, find the maximum and minimum values of the expression.

Solution

Consider $x+2y=14$
When $x=0$, $2y=14 \implies y=7 \implies (0,7)$ is on the line.
When $y=0$, $x=14 \implies (14,0)$ is on the line.
At $(0,0)$,

$$0+2(0)=0 \leq 14$$

$\implies$ region containing $(0,0)$ is the solution set of $x+2y \leq 14$.
For $3x+2y=26$
When $x=0$, $2y=26 \implies y=13 \implies (0,13)$ is on the line.
When $y=\frac{5}{2}$, $3x=21 \implies x=7 \implies (7,\frac{5}{2})$.
At $(0,0)$,

$$3(0)+2(0)=0 \leq 26.$$

$\implies$ region containing $(0,0)$ is the solution set of $3x+2y \leq 26$.
For $2x+y=10$
When $x=0$, $y=10 \implies (0,10)$ is on the line.
When $y=0$, $2x=10 \implies x=5 \implies (5,0)$ is on the line.
At $(0,0)$,

$$2(0)+0=0 \not\geq 10.$$

$\implies$ region not containing $(0,0)$ is the solution set of $2x+y \geq 10$.
For $x+4y=12$
When $x=0$, $4y=12 \implies y=3 \implies (0,3)$ is on the line.
When $y=0$, $x=12 \implies (12,0)$ is on the line.
At $(0,0)$,

$$0+4(0)=0 \not\geq 12.$$

$\Longrightarrow$ region not containing $(0,0)$ is the solution set of $x + 4y \geq 12$.

Solution set of $x \geq 0$ and $y \geq 0$ means that the solution set is located in the right upper part of the plane. The solution set is as shaded in Figure 6.3.

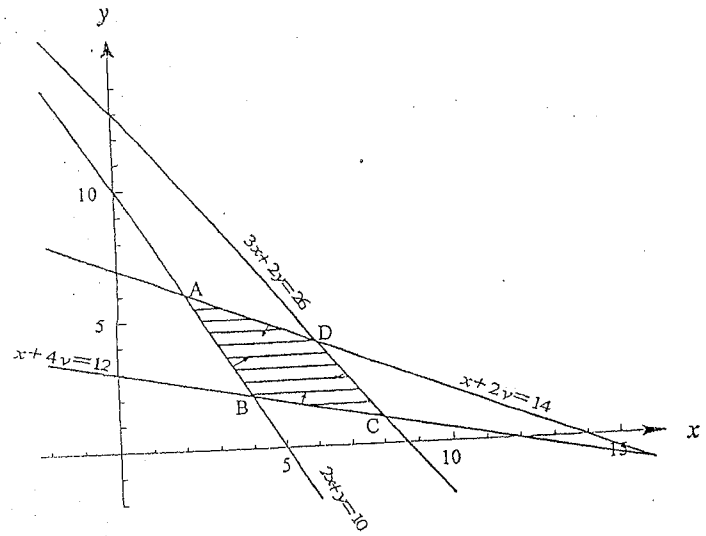


Figure 6.3

In order to determine the coordinates of the points of intersection A, B, C and D, we solve the system of equations pairwisely. Taking

$$x + 2y = 14 \quad (6.1)$$
$$2x + y = 10 \quad (6.2)$$

The solution to the equation is $A(2,6)$.
Taking the pair

$$x + 4y = 12 \quad (1)$$
$$3x + 2y = 26 \quad (2)$$

The solution to the equation is $C(8,1)$.

Solving

$$x + 2y = 14 \quad (1)$$
$$3x + 2y = 26 \quad (2)$$

The solution to the equation is $D(6,4)$.
Solving the pair

$$2x + y = 10$$
$$x + 4y = 12$$

we obtain $B(4,2)$.
Let $5x + 9y = c$.
At

$$(2,6) \Longrightarrow c = 5(2) + 9(6) = 64$$
$$(4,2) \Longrightarrow c = 5(4) + 9(2) = 38$$
$$(8,1) \Longrightarrow c = 5(8) + 9(1) = 49$$
$$(6,4) \Longrightarrow c = 5(6) + 9(4) = 66$$

Therefore, maximum value of $5x + 9y$ is 66 and occurs at $(6,4)$.
Minimum value of $5x + 9y$ is 38 and occurs at $(4,2)$.

6.2 Application of Linear Programming

In practice, we may be interested in finding the values of the unknown quantities x and y, for which a linear expression $ax + by$ is either a maximum or minimum. The expression may represent profit, cost, combination of materials for producing an item, etc. The process of determing the values of x and y maximizes or minimizes a linear expression is called *linear programming*.

Example 4

A manufacturing company makes GH¢ 2000.00 profit on each radio set it produces and GH¢ 3000.00 on each colour television. The production of radio set requires 1hour of section A, 3hours of section B, 2hours of section C and 5hours of section D of the company. The production of colour television requires 4hrs of A, 4hrs of B, 5hrs of C and 4hrs of D. In any given

week, section A,B,C and D work up to a maximum of 44, 52, 51 and 85 hrs respectively. How many radio sets and televisions should the company produce per week to maximize profit.

Solution

Let r be the number of radio set and t the number of television set produced per week. Then the set of inequalities are

$$\begin{aligned} r + 4t &\leq 44 \\ 3r + 4t &\leq 52 \\ 2r + 5t &\leq 51 \\ 5r + 4t &\leq 85 \\ r \geq 0, \quad t &\geq 0. \end{aligned}$$

Noting that r and t are non-negative.

$$\text{Total profit} = 2000r + 3000t.$$

Taking $r + 4t = 44$,
When

$$r = 0 \implies t = 11 \implies (0, 11)$$

is on the line.
When

$$t = 10 \implies r = 4 \implies (4, 10)$$

is on the line.
When

$$t = 0 \implies r = 44 \implies (44, 0)$$

is on the line.
At $(0, 0)$,

$$0 + 4(0) = 0 \leq 44$$

Therefore, region containing $(0, 0)$ is the truthset of $r + 4t \leq 44$.
Taking $3r + 4t = 52$,
When

$$r = 0 \implies 4t = 52 \implies t = 13 \implies (0, 13)$$

is on the line.
When

$$t = \frac{5}{2} \implies 3r = 42 \implies r = 14 \implies \left(14, \frac{5}{2}\right)$$

is on the line.
At $(0, 0)$,

$$3(0) + 4(0) = 0 \leq 52$$

Therefore, region containing $(0, 0)$ is the truthset of $3r + 4t \leq 52$.
Taking $2r + 5t = 51$,
When

$$t = 3 \implies 2r = 36 \implies r = 18 \implies (18, 3)$$

is on the line.
When

$$t = 5 \implies 2r = 26 \implies r = 13 \implies (13, 5)$$

is on the line.
At $(0, 0)$,

$$2(0) + 5(0) = 0 \leq 51$$

Therefore, region containing $(0, 0)$ is the truthset of $2r + 5t \leq 51$.
Taking $5r + 4t = 85$,
When

$$t = 0 \implies 5r = 85 \implies r = 17 \implies (17, 0)$$

is on the line.
When

$$t = 5 \implies 5r = 65 \implies r = 13 \implies (13, 5)$$

is on the line. Note that $r \geq 0, \quad t \geq 0$ implies first quadrant.

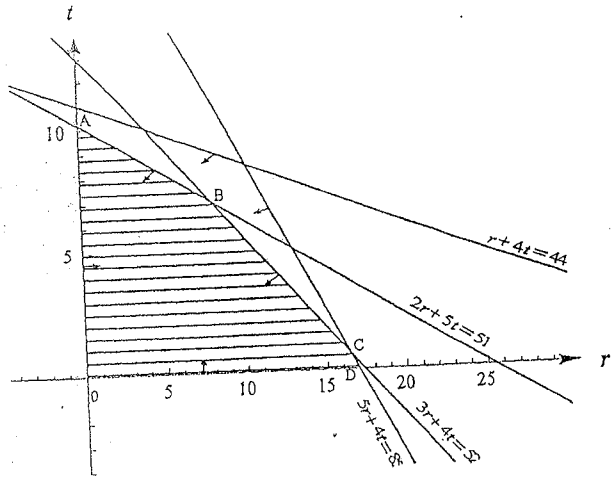


Figure 6.4

The vertices of the solution region is obtained by solving the system of equations pairwisely simultaneously.

Taking

$$\begin{aligned} 3r + 4t &= 52 \\ 2r + 5t &= 51 \end{aligned}$$

The solution is the point $B\,(8,7)$.

Taking

$$\begin{aligned} 3r + 4t &\leq 52 \\ 5r + 4t &\leq 85 \end{aligned}$$

The solution is the point $C\left(16\frac{1}{2}, \frac{5}{8}\right)$.

From Figure 6.4, the other vertices are $A\left(0, \frac{51}{5}\right)$ and $D\,(17,0)$.

The profit function is given by

$$P = 200r + 300t.$$

At $A\left(0, \frac{51}{5}\right)$

$$P = 200\,(0) + 300\left(\frac{51}{5}\right) = 3060$$

At $B\,(8,7)$

$$P = 200\,(8) + 300\,(7) = 3700$$

At $C\left(16\frac{1}{2}, \frac{5}{8}\right)$

$$P = 200\left(16\frac{1}{2}\right) + 300\left(\frac{5}{8}\right) = 3487.5$$

At $D\,(17,0)$

$$P = 200\,(17) + 300\,(0) = 3400.$$

Therefore, the maximum profit that can be made is GH¢ 3,700.00. For this maximum profit, 8 radio sets and 7 colour television sets should be produced.

Exercise

Shade in the $x-y$ plane, the region that satisfies simultaneously the linear inequalities

$$\begin{aligned} x + y &\leq 6 \\ 3x - y &\geq 2 \\ 3y - x &\leq 2 \end{aligned}$$

Find the members of the solution set which make (i) $2y + x$ max./min. and hence find the max./min. value of $2y + x$.

Chapter 7

Indices and Logarithm

7.1 Indices

Suppose a given quantity a multiplies itself n times. This is expressed as

$$a \times a \times a \times \cdots \times a = a^n$$

where a^n is the power of n factors. The number n expressing the number of factors is called the *index* or *exponent*. The number a is called the base of the indicial expression a^n.

7.1.1 Fundamental law of Combining Indices

Suppose $m,\ n > 0\ \ \forall m,\ n \in \mathbb{Z}$, the following are the three fundamental laws of Indices.

(a) $a^m.a^n = a^{m+n}$

(b) $a^m \div a^n = a^{m-n} = \frac{a^m}{a^n}$

(c) $\left(a^m\right)^n = a^{mn}$

Fractional Indices

Suppose $p, q > 0$, then, $\forall p, q \in \mathbb{Z}$, since

$$a^m.a^n = a^{m+n},$$

we have,

$$a^{\frac{p}{q}}.a^{\frac{p}{q}} = a^{\frac{2p}{q}}.$$

Similarly,

$$a^{\frac{p}{q}}.a^{\frac{p}{q}}.a^{\frac{p}{q}} = a^{\frac{3p}{q}}.$$

Therefore, for any factor q,

$$a^{(\frac{p}{q})q} = a^p.$$

Note that

$$a^{\frac{p}{q}} = (a^p)^{\frac{1}{q}}$$

which is the qth root of a^p. That is,

$$a^{\frac{p}{q}} = \sqrt[q]{a^p}.$$

Example 1

Find the values of the following

(i) $\frac{16^{\frac{1}{3}} \times 4^{\frac{1}{3}}}{8}$ (ii) $(1.331)^{\frac{1}{3}}$ (iii) $\frac{32^{\frac{2}{3}} \times 16 \times 8^{\frac{5}{4}}}{(128)^{\frac{3}{2}}}$

Solution

(i)

$$\frac{16^{\frac{1}{3}}.4^{\frac{1}{3}}}{8} = \frac{2^{\frac{4}{3}}.2^{\frac{2}{3}}}{2^3} = \frac{2^2}{2^3} = 2^{-1} = \frac{1}{2}$$

(ii)

$$(1.331)^{\frac{1}{3}} = \left(\frac{1331}{1000}\right)^{\frac{1}{3}} = \left(\frac{11^3}{10^3}\right)^{\frac{1}{3}} = \frac{11}{10} = 1.1$$

(iii)

$$\frac{32^{\frac{2}{3}} \times 16 \times 8^{\frac{5}{4}}}{(128)^{\frac{3}{2}}} = \frac{(2^5)^{\frac{2}{3}} \times 2^4 \times (2^3)^{\frac{5}{4}}}{(2^7)^{\frac{3}{2}}}$$

Therefore,

$$\begin{aligned}\frac{32^{\frac{2}{3}} \times 16 \times 8^{\frac{5}{4}}}{(128)^{\frac{3}{2}}} &= \frac{2^{\frac{10}{3}} \times 2^4 \times 2^{\frac{15}{4}}}{(2^7)^{\frac{3}{2}}} \\ &= \frac{2^{\frac{133}{12}}}{2^{\frac{21}{2}}} \\ &= 2^{\frac{133}{12} - \frac{21}{2}} = 2^{\frac{7}{12}}.\end{aligned}$$

7.1.2 The Zero Index

We know that for all $m, n > 0$, $m, n \in \mathbb{Z}$,

$$a^m \times a^n = a^{m+n}, \tag{7.1}$$

putting $m = 0$

$$\begin{aligned}\implies a^0.a^n &= a \\ a^0 &= \frac{a^n}{a^n} = 1\end{aligned}$$

Therefore

$$a^0 = 1.$$

This result means that any number raised to the power 0 is equal to 1.

Interpretation of a^{-n}

From (7.1), putting $m = -n$

$$a^{-n}.a^n = a^{-n+n} = a^0 = 1$$

Therefore

$$a^{-n} = \frac{1}{a^n}.$$

Example 2

Simplify the following

(i) $16^{\frac{3}{4}n} \div 8^{\frac{5}{3}n} \times 4^{n+1}$ (ii) $\frac{x^{3n+1}}{x^{2n+\frac{5}{2}} \times \sqrt{(x^{2n-3})}}$ (iii) $\frac{(1+x)^{\frac{1}{3}} - \frac{1}{3}x(1+x)^{-\frac{2}{3}}}{(1+x)^{\frac{2}{3}}}$

Solution

(i)

$$\begin{aligned}16^{\frac{3}{4}n} \div 8^{\frac{5}{3}n} \times 4^{n+1} &= \left(2^{3n} \div 2^{5n}\right) \times 2^{2n+2} \\ &= 2^{-2n} \times 2^{2n+2} \\ &= 2^2 = 4\end{aligned}$$

(ii)

$$\frac{x^{3n+1}}{x^{2n+\frac{5}{2}} \times \sqrt{(x^{2n-3})}} = \frac{x^{3n+1}}{x^{2n+\frac{5}{2}} \times (x^{2n-3})^{\frac{1}{2}}}$$
$$= \frac{x^{3n+1}}{x^{2n+\frac{5}{2}+n-\frac{3}{2}}} = 1$$

$$\frac{(1+x)^{\frac{1}{3}} - \frac{1}{3}x(1+x)^{-\frac{2}{3}}}{(1+x)^{\frac{2}{3}}} = \frac{(1+x)^{\frac{1}{3}} \cdot \left[1 - \frac{1}{3}x(1+x)^{-1}\right]}{(1+x)^{\frac{1}{3}}}$$
$$= \frac{1 - \frac{1}{3}x(1+x)^{-1}}{(1+x)^{\frac{1}{3}}} = \frac{1+x-\frac{1}{3}x}{(1+x)^{\frac{4}{3}}}$$
$$= \frac{1+\frac{2}{3}x}{(1+x)^{\frac{4}{3}}}$$

7.2 Equations Involving Indices

Solution to equations involving indices makes use of two basic rules. These rules are that for any positive numbers a, b, if

1.
$$a^x = b^x \implies a = b$$
since the powers are the same, bases must be the same.

2.
$$a^x = a^y \implies x = y$$
since the bases are the same, powers must be the same.

When solving equations involving indices, the process frequently may involve solving polynomial equations or application of logarithm.

Example 3

1. Find the values of x in the following

(a) $10^x = 0.0001$ (b) $2^{2x} - 5 \times 2^x + 4 = 0$ (c) $2^{x+3} - 15 = 2^{1-x}$

Solution

(a) Expressing both sides to the same base, we have

$$10^x = 10^{-3}$$

Since bases are the same, we have $x = -3$.

(b) Given

$$2^{2x} - 5 \times 2^x + 4 = 0$$
$$(2^x)^2 - 5 \times 2^x + 4 = 0$$

Let $y = 2^x$, then

$$y^2 - 5y + 4 = 0$$
$$(y-1)(y-4) = 0$$

which implies that

$$y - 1 = 0 \implies y = 1$$
$$y - 4 = 0 \implies y = 4.$$

When $y = 1$, we have

$$2^x = 1 \implies 2^x = 2^0$$
$$x = 0$$

When $y = 4$, we have

$$2^x = 4 \implies 2^x = 2^2$$
$$x = 2.$$

(c)

$$2^{x+3} - 15 = 2^{1-x}$$
$$2^x \times 2^3 - 15 = \frac{2}{2^x}$$

Let $y = 2^x$. Then

$$8y - 15 = \frac{2}{y}.$$

Multiplying through by y gives

$$\begin{aligned} 8y^2 - 15y &= 2 \\ 8y^2 - 15y - 2 &= 0 \\ (8y+1)(y-2) &= 0 \end{aligned}$$

This implies that

$$\begin{aligned} 8y + 1 = 0 &\implies y = -\frac{1}{8} \\ y - 2 = 0 &\implies y = 2. \end{aligned}$$

Since $y = 2^x > 0 \ \forall\, x$, we reject $y = -\frac{1}{8}$.

When $y = 2$, we have

$$\begin{aligned} 2^x &= 2 \implies 2^x = 2^1 \\ x &= 1. \end{aligned}$$

7.3 Logarithm

7.3.1 Definition

The log of a positive quantity b to a given base a is the index of the power to which a must be raised to be equal to the given quantity. That is, if

$$a,\ b \in \mathbb{R}^+,\ \ a \neq 1 \ \text{ and } b = a^x, \ \text{ then } \log_a b = x$$

From this definition, a number of deduction can be made. Notable among these are as follows:

1. $a = a^1$ by symmetry $\implies$

$$\log_a a = 1.$$

2. $a^0 = 1 \implies$

$$\log_a 1 = 0, \quad \forall a \in \mathbb{R}^+.$$

3.

$$b = a^{\log_a b}.$$

Example 4

Find $\log_{\frac{1}{2}} 8$.

Solution

Let

$$\log_{\frac{1}{2}} 8 = x \implies 8 = \left(\frac{1}{2}\right)^x \implies 8 = \left(2^{-1}\right)^x \implies 2^3 = 2^{-x} \implies x = -3$$

The log of a number b to base 10 is called the common logarithm of b and is usually written as $\log b$. The logarithm of a number b to base e $(= 2.7132)$ is called the natural logarithm of b and is written as $\log_e b$.

7.3.2 Fundamental Laws of Logarithm

For any two numbers $a,\ b \in \mathbb{R}^+$, the following hold:

1.

$$\log_c ab = \log_c a + \log_c b$$

2.

$$\log_c \frac{a}{b} = \log_c a - \log_c b$$

3.

$$\log_a b^n = n \log_a b$$

We will prove Laws 1 and 3. Law 2 can then be easily derived.

Proof of (1) and (3)

Suppose that $a > 0$, and $b > 0$. Then

$$\begin{aligned} a &= c^{\log_c a} \\ b &= c^{\log_c b} \end{aligned}$$

Multiplying the two gives

$$\begin{aligned} ab &= c^{\log_c a} \times c^{\log_c b} \\ &= c^{\log_c a + \log_c b} \end{aligned}$$

Taking log of both sides to base c gives

$$\begin{aligned} \log_c ab &= \log_c \left(c^{\log_c a + \log_c b}\right) \\ &= \log_c a + \log_c b \end{aligned}$$

$$\begin{aligned} \log_a b^n &= \log_a (b \cdot b \cdot b \cdots b) \text{ multiply } b \text{ } n\text{times} \\ &= \log b + \log b + \log b + \cdots + \log b, \text{ using law 1} \\ &= n \log_a b \end{aligned}$$

Note from the above results that

$$\log_a b^n \neq (\log_a b)^n \text{ and } \log_a b \log_a c \neq \log_a bc$$

Example 5

If $\log 2 = 0.3010$, $\log 3 = 0.477$, find without calculator or tables, $\log \frac{1}{7.5}$.

Solution

$$\begin{aligned} \log \frac{1}{7.5} &= \log (7.5)^{-1} = \log \left(\frac{75}{10}\right)^{-1} \\ &= \log \frac{10}{75} = \log 10 - \log 75 \\ &= 1 - \log \left(5^2 \times 3\right) \end{aligned}$$

Thus,

$$\begin{aligned} \log \frac{1}{7.5} &= 1 - 2\log 5 - \log 3 \\ &= 1 - 2\log \left(\frac{10}{2}\right) - \log 3 \\ &= 1 - 2(\log 10 - \log 2) - \log 3 \\ &= 1 - 2 + 2\log 2 - \log 3 \\ &= -1 + 2(0.3010) - 0.4770 \\ &= -0.875. \end{aligned}$$

Example 6

Find, without using calculator, $\log 2.1 + \log 25 - \log 15 - \log 35$.

Solution

$$\begin{aligned} &\log \frac{21}{10} + \log 25 - \log 15 - \log 35 \\ &= \log 21 - \log 10 + \log 5^2 - \log(3 \times 5) - \log (5 \times 7) \\ &= \log (3 \times 7) - 1 + 2\log 5 - (\log 3 + \log 5) - (\log 5 + \log 7) \\ &= \log 3 + \log 7 - 1 + 2\log 5 - \log 3 - \log 5 - \log 5 - \log 7 \\ &= -1. \end{aligned}$$

Example 7

Given that $\log 2 = 0.3010$, $\log 3 = 0.4771$, solve for x in

$$3.5^{3x} = 2^{7x+4}$$

to four decimal places.

Solution

$$\begin{aligned} 3.5^{3x} &= 2^{7x+4} \\ \log 3.5^{3x} &= \log 2^{7x+4} \\ \log 3 + \log 5^{3x} &= (7x+4)\log 2 \\ \log 3 + 3x \log 5 &= 7x \log 2 + 4\log 2 \\ 3x \log 5 - 7x \log 2 &= 4\log 2 - \log 3 \\ x(3\log 5 - 7\log 2) &= 4\log 2 - \log 3 \end{aligned}$$

Therefore,

$$\begin{aligned} x &= \frac{4\log 2 - \log 3}{3\log\left(\frac{10}{2}\right) - 7\log 2} = \frac{4\log 2 - \log 3}{3(\log 10 - \log 2) - 7\log 2} \\ &= \frac{4\log 2 - \log 3}{3 - 10\log 2} = \frac{4(0.3010) - 0.4771}{3 - 10(0.3010)} \\ &= -72.69. \end{aligned}$$

7.3.3 Some further laws of Logarithm

1. Change of Base

We know that

$$\log_a b = x \implies b = a^x.$$

Taking log to both sides to base c, we have

$$\begin{aligned} \log_c b &= x \log_c a \\ \implies x &= \frac{\log_c b}{\log_c a} \end{aligned}$$

Therefore,

$$\log_a b = \frac{\log_c b}{\log_c a}$$

This rule enables us to change the log in a given base to a different base. For example,

$$\log_3 5 = \frac{\log 5}{\log 3} = \frac{0.6990}{0.4771} = 1.4651$$

Example 8

Solve $\log_3 y + \log_9 y = 6$.

Solution

Converting base 9 to 3, we obtain

$$\begin{aligned} \log_3 y + \log_9 y &= 6 \\ \log_3 y + \frac{\log_3 y}{\log_3 9} &= 6 \end{aligned}$$

Since, $\log_3 9 = \log_3 3^2 = 2\log_3 3 = 2$, we have

$$\begin{aligned} \log_3 y + \frac{1}{2}\log_3 y &= 6 \\ 2\log_3 y + \log_3 y &= 12 \\ 3\log_3 y &= 12 \\ \log_3 y &= 4 \end{aligned}$$

This implies that

$$y = 3^4 = 81.$$

2. Reciprocal Law

By definition,

$$\log_a b = x \implies b = a^x.$$

Taking log to base b of both sides, we have

$$\begin{aligned} \log_b b &= x \log_b a \\ x &= \frac{1}{\log_b a} \end{aligned}$$

Therefore,

$$\log_a b = \frac{1}{\log_b a}$$

which implies that

$$\log_a b . \log_b a = 1$$

Example 9

Solve the equation

$$(\log_3 x)^2 + 11 = 6\log_3 x + 6\log_x 3$$

Solution

Given

$$(\log_3 x)^2 + 11 = 6\log_3 x + 6\log_x 3$$

Since $\log_x 3 = \frac{1}{\log_3 x}$, we see that

$$(\log_3 x)^2 + 11 = 6\log_3 x + \frac{6}{\log_3 x}$$

Let $\log_3 x = t$. Then

$$t^2 + 11 = 6t + \frac{6}{t}$$
$$t^3 + 11t = 6t^2 + 6$$
$$t^3 - 6t^2 + 11t - 6 = 0$$

Since $f(1) = 0 \implies t - 1$ is a factor, we have

$$(t-1)(t^2 - 5t + 6) = 0$$
$$(t-1)(t-2)(t-3) = 0$$

$t - 1 = 0 \implies t = 1$; $t - 2 = 0 \implies t = 2$; $t - 3 = 0 \implies t = 3$.

When $t = 1$

$$\log_3 x = 1 \implies x = 3.$$

When $t = 2$

$$\log_3 x = 2 \implies x = 9.$$

When $t = 3$

$$\log_3 x = 3 \implies x = 27.$$

Therefore, the truthset is $\{x : x = 3,\ 9,\ 27\}$.

7.4 Application of Indices and Logarithm

7.4.1 Antilogarithm

Consider the common logarithm of a number b. Let b be expressed in standard form as

$$b = k \times 10^n, \quad 1 \leq k < 10$$

Taking the logarithm of both sides, we get

$$\begin{aligned} \log b &= \log(k \times 10^n) \\ &= \log k + \log 10^n \\ &= n \log 10 + \log k \\ &= n + \log k, \quad 0 \leq \log k < 1,\ n \in \mathbb{Z} \end{aligned} \tag{7.2}$$

The number n is called the *characteristic* of $\log b$ and $\log k$ is called the *mantissa* of $\log b$, which are compiled in tables.

Example 10

If $\log 7.045 = 0.8479$, find the log of

(a) 70450 (b) 0.007045

Solution

(a) Express 70450 in standard form as 7.045×10^4. Then,

$$\begin{aligned} \log 70450 &= \log(7.045 \times 10^4) \\ &= \log 7.045 + 4 \log 10 \\ &= 4 + 0.8479 \\ &= 4.8479 \end{aligned}$$

(b) Express 0.007045 in standard form as 7.045×10^{-3}. Then,

$$\begin{aligned} \log 0.007045 &= \log(7.045 \times 10^{-3}) \\ &= \log 7.045 - 3 \log 10 \\ &= -3 + 0.8479 \\ &= \overline{3}.8479 \end{aligned}$$

Note that in (b) of the example above, $\overline{3}.8479$ is the same as -2.1521. In the form presented, it is possible to identify the characteristic of the logarithm of the number as -3 and mantissa as 0.8479.

Now, suppose $\log b$ is given but b unknown. That is,

$$\log b = x, \quad x \text{ given.}$$

Then, from (7.2)

$$x = n + m, \quad n \in \mathbb{Z}, \quad 0 \leq m < 1$$

Therefore,

$$\begin{aligned} b &= 10^x = 10^{n+m} = 10^n \times 10^m \\ &= 10^n \operatorname{antilog}_{10} m, \quad 1 \leq \operatorname{antilog} m < 10 \end{aligned}$$

where $\operatorname{antilog}_{10} m = 10^m \quad 0 \leq m \leq 1$.

Antilog of m values have been compiled in Tables.

Example 11

If $\log b = \bar{3}.8913$, find b.

Solution

$$\log b = \bar{3}.8913 = \bar{3} + 0.8913$$

This implies that

$$\begin{aligned} b &= 10^{(\bar{3}+0.8913)} = 10^{-3} \times 10^{0.8913} \\ &= 10^{-3} \text{ anti}\log 0.8913 \\ &= \frac{1}{1000} \times 7.7857 \\ &= 0.007786 \end{aligned}$$

Example 12

Use log to find the values of the following to three significant figures.
(a) $\sqrt[3]{0.1247}$ (b) $\frac{35.67\times0.861}{22.55\times0.0057}$,
given that $\log 4.996 = 0.6986$, and $\log 2.3895 = 0.3783$

Solution

(a) Let $y = \sqrt[3]{0.1247} = (0.1247)^{\frac{1}{3}}$.

$$\begin{aligned} \log y &= \frac{1}{3}\log 0.1247 = \frac{1}{3}\left(\bar{1}.09587\right) = \frac{1}{3}\left(\bar{1} + 0.09587\right) \\ &= \frac{1}{3}\left(\bar{3} + 2 + 0.09587\right) = -1 + 0.6986 = \bar{1}.6986 \end{aligned}$$

Therefore,

$$\begin{aligned} y &= 10^{(-1+0.6986)} = 10^{-1} \text{ anti}\log 0.6986 \\ &= 10^{-1} \times 4.9960 = 0.4996 \\ &= 0.5 \quad (3 \text{ s.f.}) \end{aligned}$$

(b) Let

$$\begin{aligned} y &= \frac{35.67 \times 0.861}{22.55 \times 0.0057} \\ \log y &= \log\left(35.67 \times 0.861\right) - \log\left(22.55 \times 0.0057\right) \\ &= \log 35.67 + \log 0.861 - \log 22.55 - \log 0.0057 \\ &= 1.5523 + \bar{1}.935 - \left(1.3531 + \bar{3}.7559\right) \\ &= 1.4873 - \bar{1}.1090 \\ &= 2.3783 \end{aligned}$$

Therefore,

$$\begin{aligned} y &= 10^{(2+0.3783)} \\ y &= 10^{2} \text{ anti}\log 0.3783 \\ &= 10^{2} \times 2.3895 = 238.95 \\ &\simeq 239. \end{aligned}$$

Example 13

Two varaibles x and y are related by a law of the form $y = ab^x$. It is assumed that x and y are also approximately linearly related and the graph of its function passes through the points $(-2.16,\ 0)$ and $(0,\ 0.38)$.

Find, to three significant figures, estimates of the values of a and b.

Solution

Given

$$\begin{aligned} y &= ab^x \\ \log y &= \log ab^x \\ &= \log a + \log b^x \\ \log y &= x\log b + \log a \end{aligned}$$

of the form

$$y = mx + c$$

Approximate straight line passes through $(-2.16,\ 0)$ and $(0,\ 0.38)$.

$$\text{Gradient, } m = \frac{0.38 - 0}{0 + 2.16} = 0.1759$$

Therefore,

$$\begin{aligned} \log b &= 0.1759 \\ b &= \textit{anti}\log 0.1759 = 1.4993 \end{aligned}$$

but $c = 0.38$

$\Longrightarrow$

$$\begin{aligned} \log a &= 0.38 \\ a &= \text{anti}\log 0.38 = 2.3988 \end{aligned}$$

Therefore,

$$\begin{aligned} a &= 2.4 \text{ and } b = 1.5 \\ y &= 2.4(1.5^x) \end{aligned}$$

Exercises

1. Without using tables, simplify

$$\log_{10} 2\,(2\log_{10} 5) + \log_{10} 2$$

2. Prove that

(a)

$$\log_a x = \log_{a^k} x^k \quad x > 0, k \in \mathbb{N}$$

(b)

$$\log_{a^k} x^m = \frac{m}{k}\log_a x \quad x > 0, k \in \mathbb{N}$$

3. If p, q and a are positive numbers, prove that

$$(\log_q p)\,(\log_a q) = \log_a p.$$

Hence, deduce that if m, n and x are positive numbers

$$\log_{mn} x = \frac{\log_n x}{1 + \log_n m}.$$

4. Find the truthset of the equation

$$5^{3x+1} - 5^{3x} - 2^{x+4} + 2^{x+2}$$

given that $\log 2 = 0.3010$ and log3=0.4771.

5. If p and q are positive numbers, prove that

$$\log_q p \times \log_p q = 1.$$

Hence, express y in terms of x if

$$3\left(\log_x y - \log_y x\right) = 8.$$

6. If $\log_2 x = \mu$, show that

$$\log_{2x} 2 = \frac{1}{1+\mu}.$$

Hence or otherwise, solve the equation

$$\log_{2x} 8 - 3\log_2 2x + 6 = 0$$

7. Solve the equation

$$(x+1)^{\log(x+1)} = 100\,(x+1).$$

8. Solve the simultaneous equations

$$\begin{aligned} \log_y x - 4\log_x y &= 0 \\ \log_{10} x + \log_{10} y &= 1. \end{aligned}$$

9. Find x from the equation

(a)

$$3^{2x} = 5^{x+1}$$

(b)

$$\ln x - \ln\sqrt{3} = \ln\sqrt{x+16}.$$

10. If a, b and c are positive real numbers

(a) show that

$$\log_b a = \frac{\log_c a}{\log_c b}.$$

Hence or otherwise find

i. the values of x for which

$$2\log_4 x + 9\log_x 4 = 9$$

ii. the limit as n tends to infinity of

$$\log_8 (12n - 5) - \log_2 (3n + 4)^{\frac{1}{3}}.$$

11. Solve the equations

(a)

$$\log_3 x + \log_x 3 = \frac{10}{3}$$

(b)

$$2\log_4 x + 9\log_x 4 = 9$$

12. If

$$10^x + 10^{-x} = x,$$

prove that

$$x = \pm\log_{10}\left(2 + \sqrt{3}\right).$$

13. If

$$\log_{10} 2 = t,$$

show that

(a)

$$\log_{10}\frac{125}{128} = 3 - 10t$$

(b)

$$\log_{10}\frac{625}{572} = 4 - 13t$$

Deduce that $\frac{3}{10} < t < \frac{4}{13}$.

14. An amount Q, invested at $r\%$ compoud interest, payable annually accumulates to an amount $Q\left(1 + \frac{r}{100}\right)^n$ at the end of n years.

Find how long it will take for a sum of money invested at 9% per annum compound interest to

(a) double its value

(b) triple it value.

15. Given that

$$\log_b a + \log_c a = 2\log_b a \log_c a,$$

express a in terms of b and c.

16. Find the values of x which satisfy $4^{\frac{1}{2}+x} + 4^{\frac{3}{2}-x} \leq 17$.

17. Find the value of x in following equation $3^{8x+1} - 28(3^{x-1}) + 1 = 0$.

18. Show that

$$\sqrt[q]{a^p} = \left(\sqrt[q]{a}\right)^p.$$

Chapter 8

BINOMIAL THEOREM

Consider the expansion of the expression $(a+b)^n, n \in Z^+$. For

$$
\begin{aligned}
n = 0, &\quad (a+b)^0 = 1 \\
n = 1, &\quad (a+b) = a+b \\
n = 2, &\quad (a+b)^2 = a^2 + 2ab + b^2 \\
n = 3, &\quad (a+b)^3 = a^3 + 3a^2b + 3ab^2 + b^3 \\
n = 4, &\quad (a+b)^4 = a^4 + 4a^3b + 6a^2b^2 + 4ab^3 + b^4
\end{aligned}
$$

The coefficients of the expansions are given by the triangular pattern below.

$n = 0,$ 1

$n = 1,$ 1 1

$n = 2,$ 1 2 1

$n = 3,$ 1 3 3 1

$n = 4,$ 1 4 6 4 1

$n = 5,$ 1 5 10 10 5 1

$n = 6,$ 1 6 15 20 15 6 1

$n = 7,$ — — — — — — —

The triangular shape is called the Pascal Triangle.
There are two observations that are obvious from the table:

1. Each expansion begins and ends with 1;
2. For any value of n, the coefficients of the expansion can be generated from those of the previous value, $(n-1)$. For example, for $n=7$, the coefficients are

$$\begin{array}{ll} 1, & 7\ (=1+6), 21(=6+15), \\ 35\ (=15+20), & 35\ (=20+15), \\ 21\ (=15+6), & 7\ (=6+1),\ 1. \end{array}$$

Write down the coefficients for $n=8$ and $n=9$.

The two observations above can be generalised as follows:

1. The coefficients of the expansion can be generalised for any value of n. Notice that for any value $n=r$, the general coefficients are

$$^nC_0,\ ^nC_1,\ ^nC_2,\ ^nC_3,\ \cdots,\ ^nC_r,\ \cdots,\ ^nC_{n-1},\ ^nC_n$$

where

$$^nC_r = \frac{n!}{r!(n-r)!},$$

and

$$n! = n(n-1)(n-2)(n-3)\cdots 3.2.1$$

Verify your coefficients for $n=7$ using these general forms.

2. The triangle is symmetrical about the vertical line that can be drawn through the peak of the triangle. This property means that

$$^nC_r = {}^nC_{n-r}.$$

Thus,

$$\begin{aligned} ^nC_0 &= {}^nC_n = 1,\ ^nC_1 = {}^nC_{n-1} = n, \\ ^nC_2 &= {}^nC_{n-2} = \frac{n!}{2!(n-2)!}, \end{aligned}$$

etc.

The two properties can further be exapanded into four characteristics as follows.

Characteristics of Expansion of $(a+b)^n$

1. There are $(n+1)$ terms.
2. Coefficients in the expansion are the same from either end.
3. Each term is of degree n.
4. As the power of first monomial decreases (from n to 0), that of the second monomial increases (from 0 to n).

The general expansion of $(a+b)^n$ is therefore given as

$$\begin{aligned} (a+b)^n = a^n &+ {}^nC_1a^{n-1}b + {}^nC_2a^{n-2}b^2 \\ &+ {}^nC_3a^{n-3}b^3 + \cdots + {}^nC_ra^{n-r}b^r + \cdots + b^n \end{aligned} \quad (8.1)$$

The expansion in (8.1) is called the Binomial Thereom for $n \in Z^+$. The term

$$^nC_ra^{n-r}b^r$$

is called the *general term* and is the $(r+1)^{th}$ term in the expansion. The general term is written as

$$U_{r+1} = {}^n C_ra^{n-r}b^r.$$

Example 1

Expand and simplify $(x-3y)^8$.

Solution

From the expansion (8.1), comparing $(x-3y)$ to $(a+b)$, let $a=x$, and $b=-3y$. Substituting these and the coefficients into the above, gives

$$\begin{aligned} (x-3y)^8 &= x^8 + 8x^7(-3y) + 28x^6(-3y)^2 + 56x^5(-3y)^3 + 70x^4(-3y)^4 \\ &\quad +56x^3(-3y)^5 + 28x^2(-3y)^6 + 8x(-3y)^7 + (-3y)^8 \\ &= x^8 - 24x^7y + 252x^6y^2 - 1512x^5y^3 + 5670x^4y^4 \\ &\quad -13608x^3y^5 + 20412x^2y^6 - 17496xy^7 + 6561y^8 \end{aligned}$$

The expansion in Equation (8.1) can be made simple if the first monomial, a is equal to 1. In this case, $a^r = 1$, for all values of r, and making $b = x$, the equation becomes

$$(1+x)^n = 1 + {}^nC_1x + {}^nC_2x^2 + {}^nC_3x^3 + \cdots + {}^nC_rx^r + \cdots + x^n \quad (8.2)$$

We will demonstrate the use of Equation (8.2) in the next example.

Example 2

(a) Write down the first four terms of the binomial exapansion of $(2+\frac{x}{4})^{10}$, simplifying all terms.

(b) Use your result above to evaluate $(2.025)^{10}$, correct to two decimal places.

Solution

Write $(2+\frac{x}{4})^{10} = 2^{10}\left(1+\frac{1}{8}x\right)^{10}$

(a) Substituting $\frac{1}{8}x$ for x in (8.2) and simplifying the corresponding coefficients gives

$$\begin{aligned}(2+\tfrac{x}{4})^{10} &= 2^{10}\left(1+\tfrac{1}{8}x\right)^{10}\\ &= 2^{10}\left[1+10(\tfrac{1}{8}x)+45(\tfrac{1}{8}x)^2+120(\tfrac{1}{8}x)^3+\cdots\right]\\ &= 2^{10}+2^7(10)x+2^4(45)x^2+2(120)x^3+\cdots\\ &= 1024+1280x+720x^2+240x^3+\cdots\end{aligned}$$

(b) Writing 2.025^{10} as

$$(2+0.025)^{10} \equiv (2+\frac{x}{4})^{10}$$

This implies that $\frac{x}{4} = 0.025 \Rightarrow x = 4\times 0.025 = 0.1$. Substituting 0.1 for x into the expansion in (a) gives

$$\begin{aligned}2.025^{10} &= 1024+1280(0.1)+720(0.1)^2+240(0.1)^3+\cdots\\ &= 1024+128+7.2+0.24+\cdots\\ &= 1159.44\end{aligned}$$

8.1 The *rth* Term of a Binomial Expansion

In the previos section, the general term of a binomial expansion of $(a+b)^n$ was given as

$$U_{r+1} = {}^nC_r\, a^{n-r}\, b^r, \quad r = 0,\ 1,\ 2,\ \cdots \quad (8.3)$$

The *rth* term of the expansion is obtained by putting $r' = r+1, \Rightarrow r = r'-1$. Substituting for r in Equation (8.3) gives

$$U_r = {}^nC_{r-1}\, a^{n-r+1}\, b^{r-1}, \quad r = 1,\ 2,\ \cdots \quad (8.4)$$

This term is called the *rth* term of the binomial expansion. We can use this expression to find a particular term in a binomial expansion without necessarily carrying out the expansion.

Example 3

(a) Show that the general term in the expansion of $\left(\frac{1}{2}x-\frac{1}{x^2}\right)^{12}$ can be expressed in the form

$$U_{r+1} = K\, x^{12-3r},$$

where K is in terms of r only. Hence, find

(b) (a) the term independent of x, to one significant figure;

(c) the coefficient of the term in x^{-3}.

Solution

(a) The general term of $(a+b)^n$ is given by

$$U_{r+1} = {}^nC_r\, a^{n-r}b^r.$$

Given $(\frac{1}{2}x-\frac{1}{x^2})^{12}$, we take $a = \frac{1}{2}x$, $b = -\frac{1}{x^2}$, $n = 12$, and substituting into U_{r+1} gives

$$\begin{aligned}U_{r+1} &= {}^{12}C_r(\tfrac{1}{2}x)^{12-r}(-\tfrac{1}{x^2})^r\\ &= (-1)^r\ {}^{12}C_r\cdot\tfrac{1}{2^{12-r}}x^{12-r}\cdot\tfrac{1}{x^{2r}}\\ &= (-1)^r\ {}^{12}C_r\cdot\tfrac{1}{2^{12-r}}\cdot x^{12-3r}\end{aligned}$$

which is of the form

$$U_{r+1} = K\, x^{12-3r},$$

where $K = (-1)^r\ {}^{12}C_r \cdot \frac{1}{2^{12-r}}$

(b) For the term to be a constant,

$$x^{12-3r} = x^0 \Rightarrow 12 - 3r = 0 \Rightarrow r = 4$$

Substituting 4 for r in the expression for U_{r+1}, we have

$$U_5 = (-1)^4\ {}^{12}C_4 \frac{1}{2^8} = \frac{495}{256} = 1.9336$$

Therefore, the term independent of x is the 5th term of the expansion and is equal to 2.0. (1 s.f.).

(c) From $U_{r+1} = (-1)^r\ {}^{12}C_r \cdot \frac{1}{2^{12-r}} . x^{12-3r}$, putting

$$x^{12-3r} = x^{-3} \Rightarrow 12 - 3r = -3 \quad \Rightarrow 3r = 15 \ \Rightarrow r = 5$$

Substituting 5 for r in U_{r+1} gives

$$U_6 = (-1)^5\ {}^{12}C_5 \frac{1}{2^7} . x^{-3} = -\frac{792}{128} x^{-3} = -\frac{99}{16} x^{-3}.$$

Therefore, the coefficient of x^{-3} is $-\frac{99}{16}$.

8.2 Binomial Theorem for Rational powers

An observation in Equation (8.1) is that the expansion terminates. This is the case when the power n is a positive integer.However, if n is a rationa number and $n \notin \mathbb{Z}^+$ the expansion in (8.1) and (8.2) does not terminate. In this case, the coefficients ${}^nC_r$ is generally given by

$${}^nC_r = \frac{n(n-1)(n-2)\cdots(n-r+1)}{r!} \tag{8.5}$$

where $n \in \mathbb{Q}$. Equation (8.2) is valid for $-1 < x < 1$.

Example 4

(a) Decompose $\frac{3+x}{(1+x)^2}$ in partial fractions.

(b) Hence,or otherwise, find it expansion as far as the term in x^4.

(c) Deduce the rth term of the expansion.

Solution

$$\begin{aligned} \frac{3+x}{(1+x)^2} &= \frac{A}{x+1} + \frac{B}{(1+x)^2} \\ x+3 &= A(x+1) + B \\ &= Ax + A + B \end{aligned}$$

Equating corresponding coefficients, we have

$$\begin{aligned} A &= 1 \\ A + B &= 3, \Rightarrow B = 2. \end{aligned}$$

Therefore,

$$\frac{3+x}{(1+x)^2} = \frac{1}{x+1} + \frac{2}{(1+x)^2}$$

Taking the first fraction on the RHS and using (8.5), we have

$$\begin{aligned} \frac{1}{1+x} &= (1+x)^{-1} = 1 + (-1)x + \frac{(-1)(-2)}{2!}x^2 \\ &\quad + \frac{(-1)(-2)(-3)}{3!}x^3 + \frac{(-1)(-2)(-3)(-4)}{4!}x^4 + \cdots \\ &= 1 - x + x^2 - x^3 + x^4 + \cdots \end{aligned}$$

Similarly, we consider

$$\begin{aligned} \frac{1}{(1+x)^2} &= (1+x)^{-2} = 1 + (-2)x + \frac{(-2)(-3)}{2!}x^2 \\ &\quad + \frac{(-2)(-3)(-4)}{3!}x^3 + \frac{(-2)(-3)(-4)(-5)}{4!}x^4 + \cdots \\ &= 1 - 2x + 3x^2 - 4x^3 + 5x^4 + \cdots \end{aligned}$$

Therefore,

$$\begin{aligned}\frac{x+3}{(1+x)^2} &= (1-x+x^2-x^3+x^4+\cdots)\\ &\quad +2(1-2x+3x^2-4x^3+5x^4+\cdots)\\ &= (1-x+x^2-x^3+x^4+\cdots)\\ &\quad +(2-4x+6x^2-8x^3+10x^4+\cdots)\\ &= 3-5x+7x^2-9x^3+11x^4+\cdots\\ &= \sum_{r=0}^{\infty}(3+2r)(-x)^r.\end{aligned}$$

The general term of the expansion of $\frac{x+3}{(1+x)^2}$ is

$$U_{r+1} = (3+2r)(-x)^r.$$

Therefore the rth term is

$$U_r = (-1)^{r-1}(1+2r)\,x^{r-1}.$$

Example 5

Show that if x is small enough to neglect its cube and higher powers.

$$\sqrt{\frac{1-x}{1+x}} = 1-x+\frac{x^2}{2}.$$

Deduce that

$$\sqrt{7} = 2\frac{83}{128}$$

by putting $x = \frac{1}{8}$.

Solution

$$\sqrt{\frac{1-x}{1+x}} = \frac{\sqrt{1-x}}{\sqrt{1+x}} = (1-x)^{\frac{1}{2}}(1+x)^{-\frac{1}{2}}$$

Take

$$\begin{aligned}(1-x)^{\frac{1}{2}} &= (1+(-x))^{\frac{1}{2}} = 1+\frac{1}{2}(-x)+\frac{\frac{1}{2}(-\frac{1}{2})}{2!}(-x)^2\\ &\quad +\frac{\frac{1}{2}(-\frac{1}{2})(-\frac{3}{2})}{3!}(-x)^3+\cdots\\ &= 1-\frac{1}{2}x-\frac{1}{8}x^2-\frac{1}{16}x^3+\cdots\end{aligned}$$

Also

$$\begin{aligned}(1+x)^{-\frac{1}{2}} &= 1+(-\frac{1}{2})x+\frac{(-\frac{1}{2})(-\frac{3}{2})}{2!}x^2+\frac{(-\frac{1}{2})(-\frac{3}{2})(-\frac{5}{2})}{3!}x^3+\cdots\\ &= 1-\frac{1}{2}x+\frac{3}{8}x^2-\frac{5}{16}x^3+\cdots\end{aligned}$$

Therefore

$$\begin{aligned}(1-x)^{\frac{1}{2}}(1+x)^{-\frac{1}{2}} &= (1-\frac{1}{2}x-\frac{1}{8}x^2-\frac{1}{16}x^3+\cdots)\\ &\quad (1-\frac{1}{2}x+\frac{3}{8}x^2-\frac{5}{16}x^3+\cdots)\\ &= 1-\frac{1}{2}x+\frac{3}{8}x^2-\frac{1}{2}x+\frac{1}{4}x^2-\frac{1}{8}x^2\\ &= 1-x+\frac{1}{2}x^2.\end{aligned}$$

Therefore,

$$\sqrt{\frac{1-x}{1+x}} = 1-x+\frac{1}{2}x^2 \tag{8.6}$$

as required.

Let $x = \frac{1}{8}$. Substituting into $\sqrt{\frac{1-x}{1+x}}$ we obtain

$$\sqrt{\frac{1-\frac{1}{8}}{1+\frac{1}{8}}} = \sqrt{\frac{\frac{7}{8}}{\frac{9}{8}}} = \sqrt{\frac{7}{9}} = \frac{\sqrt{7}}{3}.$$

Substituting $x = \frac{1}{8}$ in (8.6) we see that

$$\frac{\sqrt{7}}{3} = 1-\frac{1}{8}+\frac{1}{2}\left(\frac{1}{64}\right) = \frac{113}{128}$$

This implies that

$$\sqrt{7} = 3 \times \frac{113}{128} = 2\frac{83}{128}$$

as required.

Exercises

1. Find the first four terms in the binomial expansion of each of $\left(2 - \frac{1}{2}x\right)^7$.

   (a) Find the first four terms in the binomial expansion of each of the following, stating the values of x for which each expansion is valid

   i. $(1+x)^{\frac{1}{3}}$
   ii. $(1+x)^{-2}$
   iii. $(4+x)^{\frac{1}{2}}$

   (b) Use a suitable expansion in (a) to calculate to four decimal places

   i. $\sqrt[3]{1.06}$
   ii. $\sqrt{4.05}$

2. Using the binomial expansion for $(a+b)^n$ or otherwise, prove that

$$\sum_{k=0}^{n} \binom{n}{k} = 2^n$$

3. Sow that the following are true.

   (a) $\binom{n-1}{r-1} + \binom{n-1}{r} = \binom{n}{r}$
   (b) $\binom{n}{r} = \binom{n}{n-r}$
   (c) $\binom{n}{r} + \binom{n}{r-1} = \binom{n+1}{r}$

4. Proof of Binomial Thereom.

   (a) Expand $\sqrt{\frac{1+2x}{1-2x}}$ as far as the term in x^4

   (b) Deduce from (a) that $\sqrt{3} \approx 1\frac{91}{128}$.

   (c) Calculate the errors in this approximation if $\sqrt{3} = 1.7321$.

   (a) Sketch the curve of

$$f : x \to \frac{2x}{x^2 - 4}$$

   showing clearly all the asymptotes and intercepts, if any, on the axes.

   i. Decompose the function in (a) into partial fractions.
   ii. Henc find the expansion of $f(x)$ as far as the term in x^5. Show that $f(x)$ may be defined as

$$f(x) = -\frac{1}{2}\sum_{r=0}^{\infty} \frac{x^{2r+1}}{4^r}$$

   iii. Write down the rth term of the expansion, and hence, find the 4th and 5th terms.

5. Use the method of binomial theorem to find the value of $\sqrt[4]{17}$, correct to 3 significant figures.

6. Show that the general in the expansion of $\left(x - \frac{1}{2x}\right)^{11}$ canbe expressed in the form

$$U_{r+1} = kx^{11-2r}$$

   where k is in terms of r only.

   (a) Hence, show that there is no term in even power.
   (b) Find the term in x^{-1}

7. A rational function is given by

$$f : x \to \frac{2}{1 - x^2}$$

   (a) Decompose $f(x)$ into partial fractions.
   (b) Henc, find the expansion of $f(x)$ as far as the term in x^4. Deduce that

$$\frac{1}{1-x^2} = \sum_{k=0}^{\infty} x^{2k}.$$

   (c) Find the range of values of x for which the sum in (b) is valid.

8. Find, without expanding, the constant term in the expansion of

$$\left(x^2 - \frac{2}{x}\right)^9.$$

Answer: $U_7 = 5376$

Chapter 9

Series and Sequences

9.1 Sequence

A sequence is a list of items or numbers arranged in a definite order so that the terms of the list can be obtained by a definite rule.

The following are some examples of a sequence.

1. A sequence of natural numbers :

$$1, 2, 3, 4, ..$$

2. A sequence of positive even numbers:

$$2, 4, 6, 8, \cdots$$

The nth term is given by

$$U_n = 2n, \quad n \in \mathbb{N}.$$

3. A sequence of positive odd numbers:

$$1, 3, 5, 7, \cdots$$

The nth term is given by

$$U_n = 2n + 1, \quad n \in \mathbb{N}.$$

4. A sequence of perfect squares:

$$1, 4, 9, 16, \cdots$$

The nth term is given by

$$U_n = n^2, \quad n \in \mathbb{N}.$$

5. A sequence of y-coordinates of all points on a given straight line:

$$y = mx + c, \quad \forall\, x \in \mathbb{Z}.$$

The sequence is given by

$$\cdots - 2m + c, -m + c, c, \cdots, m + c, 2m + c, \cdots$$

The rule of a sequence is the formula for finding the nth term or general term of the sequence and it is denoted by T_n or U_n, $n \in \mathbb{Z}^+ (n \in \mathbb{N})$. The n denotes the position of U_n in the sequence.

Example 1

Generate the first five terms of the sequence given by the rule

$$U_n = 2^{n-1} + 2n.$$

Solution

$$\begin{aligned}
n &= 1 : U_1 = 2^{1-1} + 2(1) = 1 + 2 = 3 \\
n &= 2 : U_2 = 2^{2-1} + 2(2) = 2 + 4 = 6 \\
n &= 3 : U_3 = 2^{3-1} + 2(3) = 4 + 6 = 10 \\
n &= 4 : U_4 = 2^{4-1} + 2(4) = 8 + 8 = 16 \\
n &= 5 : U_5 = 2^{5-1} + 2(5) = 16 + 10 = 26
\end{aligned}$$

Therefore the sequence is

$$3, 6, 10, 16, 26, \cdots$$

Example 2

A sequence is given by $U_n = a + bn$. The 7th and 15th terms are 19 and 43, respectively.

(i) Find the value of a and b.

(ii) Find the 6th term.

Solution

(i)

$$\begin{aligned}
U_n &= a + bn \\
U_7 &= a + 7b \Rightarrow a + 7b = 19 \qquad (9.1) \\
U_{15} &= a + 15b \Rightarrow a + 15b = 43 \qquad (9.2)
\end{aligned}$$

Subtracting Equation (9.1) from (9.2) gives

(ii)

$$\begin{aligned}
8b &= 24 \Rightarrow b = 3, \Rightarrow a = -2 \\
U_n &= 3n - 2 \\
U_6 &= 3(6) - 2 = 16
\end{aligned}$$

Therefore, the 6th term is 16.

9.2 Series

The sum of the terms of a sequence is referred to as a series.

Denote the sum of the first n terms of a sequence by S_n. That is,

$$\begin{aligned}
S_n &= U_1 + U_2 + U_3 + \cdots + U_n \\
S_n &= \sum_{k=1}^{n} U_k
\end{aligned}$$

The nth term U_n and S_n are related by the equation

$$U_n = S_n - S_{n-1}$$

and

$$S_1 = U_1.$$

If all terms of the sequence are the same, S_n is the sum of n terms of a constant and is equal to n times the constant. That is,

$$\begin{aligned} S_n &= a + a + a + \cdots + a \\ S_n &= \sum_{k=1}^{n} a = na. \end{aligned}$$

Example 3

The sum of the first n terms of a sequence is given by $S_n = an^2 + bn$. Given that $U_1 = -1$ and $U_5 = 15$, find

(i) the values of the constants a and b.

(ii) an expression for the nth term of the sequence and use it to find U_2.

Solution

(i)

$$\begin{aligned} S_n &= an^2 + bn \\ T_1 &= -1 \Rightarrow S_1 = -1 \end{aligned}$$

Substituting into S_n gives

$$a + b = -1 \tag{9.3}$$

But

$$\begin{aligned} U_5 &= S_5 - S_4 \\ &= (25a + 5b) - (16a + 4b) \\ &= 9a + b \end{aligned}$$

Since $U_5 = 15$, $\Rightarrow$

$$9a + b = 15 \tag{9.4}$$

(9.4) − (9.3),

$$8a = 16 \Rightarrow a = 2 \Rightarrow b = -1 - 2 = -3.$$

(ii) Therefore,

$$\begin{aligned} S_n &= 2n^2 - 3n \\ U_n &= S_n - S_{n-1} \\ &= 2n^2 - 3n - \left\{2(n-1)^2 - 3(n-1)\right\} \\ &= 2n^2 - 3n - (n-1)\{2n - 2 - 3\} \\ &= 2n^2 - 3n - 2n^2 + 7n - 5 \\ &= 4n - 5. \end{aligned}$$

Substituting 2 for r gives

$$U_2 = 4(2) - 5 = 3.$$

9.3 Arithmetic Progression

An Arithmetic Progression (A.P.) is a sequence in which the difference between two consecutive terms is a constant. This difference is called the common difference and denoted by d. Thus, for the general term U_n,

$$d = U_n - U_{n-1}, \quad n \geq 2.$$

If the first term $U_1 = a$, then the sequence is given by

$$a,\ a + d,\ a + 2d,\ \cdots,\ a + (n-1)d$$

and

$$U_n = a + (n-1)d \tag{9.5}$$

is the nth term of an A.P. An arithmetic progression is also referred to as a linear sequence.

Example 4

The sum of three consecutive terms of an A.P. is 18 and their product is 120. Find the terms.

Solution

Let the consecutive terms be $a,\ a+d,\ a+2d$
Then,

$$\begin{aligned} a+(a+d)+(a+2d) &= 18 \\ 3a+3d &= 18 \\ a+d &= 6 \end{aligned} \tag{9.6}$$

Also,

$$a(a+d)(a+2d) = 120 \tag{9.7}$$

(9.6) into (9.7),

$$\begin{aligned} (6-d)(6)(6+d) &= 120 \\ (6-d)(6+d) &= 20 \\ 36-d^2 &= 20 \end{aligned}$$

$\Rightarrow$

$$\begin{aligned} d^2 &= 16 \Rightarrow d = \pm 4 \\ a &= 6-(\pm 4) = 2, 10 \end{aligned}$$

$\Rightarrow$

$$d = 4,\ -4.$$

Therefore, the sequence is

$$2, 6, 10, 14, \cdots \text{ or } 10, 6, 2, -2, \cdots$$

If $U_1, U_2, U_3, \cdots U_n, \cdots$ is an A.P., then

$$\begin{aligned} d &= U_3 - U_2 \\ d &= U_2 - U_1, \end{aligned}$$

and thus,

$$\begin{aligned} U_3 - U_2 - (U_2 - U_1) &= 0 \\ U_1 + U_3 &= 2U_2 \\ U_2 &= \frac{1}{2}(U_1 + U_3). \end{aligned}$$

In general,

$$U_n = \frac{1}{2}(U_{n-1} + U_{n+1}) \tag{9.8}$$

That is, the middle term of an A.P. is the Arithmetic Mean of the other two terms for any three consecutive terms.

Exercises

1. $x+2,\ x+3$ and $2x^2+1$ are three consecutive terms of an A.P. Find the possible values of x.
2. The first term of an A.P. is 25 and the common difference is -3. If the sum of the first n terms is 112, find the number of terms of the sequence.
3. The sum of the first n terms of an A.P. is $2n$ and the sum of the first $2n$ terms is $3n$. Find the sum of the first $3n$ terms.
4. The numbers $\frac{3}{2}, p, q, \frac{15}{q}$ are consecutive terms of an A.P. Find the values of p and q.

9.4 Sum of an A.P.

An A.P. is given by

$$\begin{aligned} & a,\ a+d,\ a+2d,\ \cdots,\ a+(n-1)d \\ S_n &= a+(a+d)+\ (a+2d)+\ \cdots + a+(n-1)d \\ S_n &= \{a+(n-1)d\}+\{a+(n-2)d\}+\{a+(n-3)d\}+\cdots+a \\ 2S_n &= \{2a+(n-1)d\}+\{2a+(n-1)d\}+\cdots+\{2a+(n-1)d\} \\ &= n\{2a+(n-1)d\} \\ S_n &= \frac{1}{2}n\{2a+(n-1)d\} \\ &= \frac{1}{2}n\{a+a+(n-1)d\} \\ &= \frac{1}{2}n\{U_1+U_n\} \end{aligned} \tag{9.9}$$

and U_n is interpreted as the last term of the sequence.

Example 5

Find an expression for the sum of first n natural numbers. Hence, find the sum of the first 30 natural numbers.

Solution

The set of n natural numbers is $\mathbb{N} : 1, 2, 3, 4, \cdots, n$. This is an A.P. with $U_1 = 1$ and $d = 1$.

Therefore, substituting into the formula

$$S_n = \frac{1}{2}n\left[2a + (n-1)d\right]$$

we obtain

$$\begin{aligned} S_n &= \frac{1}{2}n\left\{2\,(1) + (n-1) \times 1\right\} \\ &= \frac{1}{2}n\left\{2 + n - 1\right\} \\ &= \frac{1}{2}n(n+1). \end{aligned} \tag{9.10}$$

When $n = 30$,

$$S_{30} = \frac{1}{2} \times 30 \times 31 = 465.$$

Alternative

The alternative proof of the sum of n terms of an A.P. makes use of results above.

$U_n = a + (n-1)d$

$$\begin{aligned} S_n &= \sum_{k=1}^{n} U_k = \sum_{k=1}^{n} \left[a + (k-1)d\right] \\ &= \sum_{k=1}^{n} a + d\sum_{k=1}^{n} k - \sum_{k=1}^{n} d \\ &= na + \frac{1}{2}n(n+1)d - nd \\ &= \frac{1}{2}n\left[2a + (n+1)d - 2d\right] \\ &= \frac{1}{2}n\left[2a + nd + d - 2d\right] \\ &= \frac{1}{2}n\left[2a + nd - d\right] \\ &= \frac{1}{2}n\left[2a + (n-1)d\right]. \end{aligned}$$

Note that in the above derivation, $\sum_{k=1}^{n} k$ means the sum of the first n natural numbers and $\sum_{n=1}^{n} a$ or $\sum_{n=1}^{n} d$ is the sum of a constant, n times.

9.5 Geometric Progression G.P.

A Geometric Progression (G.P.) is a sequence in which the ratio of a term to the previous term is a constant. This ratio is called the common ratio and is denoted by r. Thus, for U_n,

$$r = \frac{U_n}{U_{n-1}} \implies U_n = rU_{n-1}.$$

If $U_1 = a$ then G.P. is given by

$$a,\ ar,\ ar^2,\ ar^3,\ \cdots,\ ar^{n-1},\ \cdots$$

and

$$U_n = ar^{n-1}$$

is the nth term of a G.P. A geometric progression is also referred to as exponential sequence.

Example 6

Given that 8, x, y, 27, $\cdots$ is in geometric progression, find the values of x and y.

Solution

Common ratio

$$r = \frac{x}{8} = \frac{y}{x} \implies x^2 = 8y \tag{9.11}$$

Also,

$$\frac{y}{x} = \frac{27}{y} \implies y^2 = 27x \tag{9.12}$$

From (9.11), $64y^2 = x^4$ into (9.12)

$$\frac{1}{64}x^4 = 27x$$
$$x(\frac{1}{64}x^3 - 27) = 0.$$

But $x \neq 0 \implies$

$$\frac{1}{64}x^3 - 27 = 0$$
$$x^3 = 27 \times 64 = (3 \times 4)^3$$

For $x = 12$ we have,

$$8y = 12^2$$
$$y = 12 \times \frac{12}{8} = 18$$

Therefore, sequence:

$$8,\ 12,\ 18,\ 27, \cdots$$

Remark

1. Let $u_1, u_2, u_3, \cdots, u_n, \cdots$ be a G.P. Then

$$r = \frac{u_2}{u_1} = \frac{u_3}{u_2}$$
$$\implies u_2^2 = u_1 u_3$$
$$u_2 = \sqrt{u_1 u_3}$$

and u_2 is called the geometric mean of u_1 and u_3. Generally, for three consecutive terms of a G.P., the middle term is the geometric mean of the other two. That is,

$$u_n = \sqrt{u_{n-1} u_{n+1}}.$$

2. Suppose a and b are any two numbers and that $a \neq b$, $a,\ b > 0$, then

$$\left(\sqrt{a} - \sqrt{b}\right)^2 > 0$$
$$a - 2\sqrt{ab} + b > 0$$
$$a + b > 2\sqrt{ab}$$
$$\frac{a+b}{2} > \sqrt{ab}$$

The result means that if the for any two possible numbers, the arithmetic measn is always greater thatn the geometric mean.

3. Let $\log a$, $\log b$, $\log c$ be consecutive terms of an A.P. Then, equating the common difference, we have

$$\log b - \log a = \log c - \log b$$
$$\log b + \log b = \log a + \log c$$
$$2\log b = \log ac$$
$$\log b^2 = \log ac$$
$$\implies b^2 = ac$$
$$\implies b = \sqrt{ac}$$

The result means that if the logrithm of three numbers constitute the consecutive terms in an A.P., then the three numbers constitutes consecutive terms of a G.P.

Exercise

1. Given that the sequence $\sqrt{2}-1,\ 3-2\sqrt{2},\ \cdots$ is in geometric progression (G.P), find the 5th term.

2. Starting from the outer square of side $x$units, a sequence of squares is made by joining the midpoints of the sides.See Figure 9.1. Show that the lengths of the sides of the squares form a G.P., and find,

   (a) the general term of the sequence

   (b) the perimeter of the nth square.

9.6 The Sum of a Geometric Progression

Given the G.P.

$$a,\ ar,\ ar^2,\ \cdots, ar^{n-1}, \cdots$$

The sum of the first n terms of the sequence is

$$S_n = a + ar + ar^2 + \cdots + ar^{n-2} + ar^{n-1}$$

Multiplying through by r,

$$\begin{aligned} rS_n &= ar + ar^2 + ar^3 + \cdots + ar^{n-1} + ar^n \\ S_n - rS_n &= a - ar^n \\ (1-r)\,S_n &= a\,(1-r^n) \\ S_n &= \frac{a\,(1-r^n)}{1-r}, \qquad |r| < 1 \qquad (9.13) \\ &= \frac{a\,(r^n-1)}{r-1}, \qquad |r| > 1, \quad r \neq 1. \qquad (9.14) \end{aligned}$$

That is,

$$S_n = \sum U_k = \frac{a\,(1-r^n)}{1-r} = \frac{a\,(r^n-1)}{r-1}$$

or

$$\sum_{k=1}^{n} ar^{k-1} = \frac{a\,(1-r^n)}{1-r} = \frac{a\,(r^n-1)}{r-1}.$$

Example 7

Find the sum of the first six terms of the sequence

$$\frac{1}{2},\ \frac{3}{4},\ \frac{9}{8},\ \cdots$$

Solution

Sequence is a G.P. with $a = \frac{1}{2},\ r = \frac{3}{4} \cdot \frac{2}{1} = \frac{3}{2}$.

$$\begin{aligned} S_n &= \frac{a\,(r^n-1)}{r-1} = \frac{1}{2}\left[\frac{\left(\frac{3}{2}\right)^6 - 1}{\frac{3}{2}-1}\right] \\ &= \frac{\frac{1}{2}}{\frac{1}{2}}\left[\frac{3^6 - 2^6}{2^6}\right] = \frac{729-64}{64} \\ &= 10.39. \end{aligned}$$

Example 8

The 3rd and 6th terms of a G.P. are 108 and -32 respectively. Find the sum of the first 7 terms.

Solution

$$U_3 = ar^2 = 108$$

and

$$U_6 = ar^5 = -32$$

$$\frac{U_6}{U_3} = \frac{ar^5}{ar^2} = \frac{-32}{108} \implies r^3 = \frac{-8}{27} \implies r = \frac{-2}{3}$$

$$\implies a = 108 \cdot \left(-\frac{3}{2}\right)^2 = 108 \times \frac{9}{4} = 243 = 3^5$$

$$S_n = \frac{a(1-r^n)}{1-r}$$

$$S_7 = \frac{243\left[1-\left(-\frac{2}{3}\right)^7\right]}{1+\frac{2}{3}} = \frac{3^5}{\frac{5}{3}}\left(\frac{3^7+2^7}{3^7}\right)$$

$$= \frac{1}{15}(2187+128) = \frac{2315}{15}$$

$$= 154.3333.$$

Example 9

(a) Find the sum of the first n terms of a series with the following nth terms

(i) $U_n = n$ (ii) $U_n = 2^n$ (iii) $U_n = 3^{-n}$

(b) Hence, find the sum of the first n terms of the series with the nth terms. In each case, find S_5.

(i) $2^n + 3n + 1$ (ii) $2^{n-1} + 2n$

Solution

(a) (i) $U_n = n$

$$S_n = \sum_{k=1}^{n} U_k = \sum_{k=1}^{n} k = \frac{1}{2}n(n+1)$$

(ii) $U_n = 2^n$

$$S_n = \sum_{k=1}^{n} 2^k = \sum_{k=1}^{n} 2\cdot 2^{k-1} = 2\left(\frac{2^n-1}{2-1}\right) = 2(2^n-1)$$

(iii) $U_n = 3^{-n}$

$$S_n = \sum_{k=1}^{n} U_k = \sum_{k=1}^{n} \frac{1}{3^k} = \sum_{k=1}^{n} \frac{1}{3^{k-1}}\cdot\frac{1}{3}$$

$$S_n = \frac{\frac{1}{3}\left[1-\left(\frac{1}{3}\right)^n\right]}{1-\frac{1}{3}} = \frac{1}{3}\cdot\frac{3}{2}\left[1-\frac{1}{3^n}\right]$$

$$= \frac{1}{2}\left(1-\frac{1}{3^n}\right)$$

(b) (i) If

$$U_n = 2^n + 3n + 1$$

$$S_n = \sum_{k=1}^{n} U_k = \sum_{k=1}^{n} (2^k + 3k + 1)$$

$$= \sum_{k=1}^{n} 2^k + 3\sum_{k=1}^{n} k + \sum_{k=1}^{n} 1$$

$$= 2(2^n-1) + \frac{3}{2}n(n+1) + n$$

$$S_5 = 2(2^5-1) + \frac{3}{2}\times 5(5+1) + 5 = 112.$$

(ii)

$$U_n = 2^{n-1} + 2n$$

$$S_n = \sum_{k=1}^{n} U_k = \sum_{k=1}^{n} (2^{k-1} + 2k) = \sum_{k=1}^{n} 2^{k-1} + 2\sum_{k=1}^{n} k$$

$$= \frac{(2^{n-1}-1)}{2-1} + 2\times\frac{1}{2}n(n+1) = 2^{n-1} + n(n+1)$$

$\Longrightarrow$

$$S_5 = 2^{5-1} + 5(5+1) = 61.$$

For $n = 5$ we have

$$S_5 = \frac{1}{20}\left[5\times 5(5+1) + \left\{1-\left(\frac{1}{3}\right)^5\right\}\right]$$

$$= \frac{1}{20}(150 + 0.9959) = 7.5498.$$

Exercises

1. A G.P. has common ratio of 2. Find the value of n for which the sum of $2n$ terms is 33 times the sum of n first n terms.

2. The sum of the first two terms of a G.P. is $2\frac{1}{2}$ and the sum of the first 4 terms is $3\frac{11}{18}$. Find the G.P. if $r > 0$.

3. Find the sum of the first n terms of the sequence $U_n = \frac{1}{2}\left[n + \frac{1}{5}\left(\frac{1}{3}\right)^n\right]$

9.7 Sum to Infinity of a Sequence

Denote the infinite series of a G.P. by S_∞.

That is

$$\begin{aligned} S_\infty &= a + ar + ar^2 + \cdots + ar^{n-1} + \cdots \\ &= \sum_{k=1}^{\infty} ar^{k-1} \end{aligned}$$

The value of S_∞ is obtained by considering S_n as n becomes larger and larger.

That is,

$$\begin{aligned} S_\infty &= \lim_{n\to\infty} S_n \\ &= \lim_{n\to\infty} \frac{a(1-r^n)}{1-r}; \qquad |r| < 1 \qquad (9.15) \\ &= \lim_{n\to\infty} \frac{a(r^n-1)}{r-1}; \qquad |r| > 1 \qquad (9.16) \end{aligned}$$

In general, if $|r| < 1$, then $r^n \longrightarrow 0$ as $n \longrightarrow \infty$, and if $|r| > 1$, then $r^n \longrightarrow \infty$ as $n \longrightarrow \infty$.

Thus, from (9.15),

$$S_\infty = \frac{a}{1-r}, \qquad |r| < 1 \qquad (9.17)$$

Equation (9.17) is called the limiting value of S_n (as n approaches infinity) or S_n converges to $\frac{a}{1-r}$ as $n \longrightarrow \infty$.

Example 10

Given the series $1 - \frac{1}{2} + \frac{1}{4} + \cdots$

(a) Write down the 8th term of the series

(b) Find the value of the series.

Solution

(a) The series is a G.P. with common ratio $r = -\frac{1}{2}$.

$$U_n = \left(-\frac{1}{2}\right)^{n-1} \Longrightarrow U_8 = \left(-\frac{1}{2}\right)^{8-1} = \left(-\frac{1}{2}\right)^7 = -\frac{1}{128}.$$

(b)

$$\begin{aligned} S_\infty &= 1 - \frac{1}{2} + \frac{1}{4} + \cdots \\ &= \frac{1}{1-\left(-\frac{1}{2}\right)} = \frac{1}{1+\frac{1}{2}} = \frac{1}{\frac{3}{2}} = \frac{2}{3} \end{aligned}$$

Example 11

1. The sum to infinity of the series

$$x + x^3 + x^5 + \cdots \text{ is } \frac{3}{8}.$$

(a) Find the value of x.

(b) Deduce the range of possible values of x.

Solution

(a) The series is a G.P. with $r = x^2$ and $a = x$.

$$\begin{aligned} S_\infty &= \frac{x}{1-x^2} = \frac{3}{8} \\ 8x &= 3 - 3x^2 \\ 3x^2 + 8x - 3 &= 0 \\ 3x^2 + 9x - x - 3 &= 0 \\ 3x(x+3) - (x+3) &= 0 \\ (x+3)(3x-1) &= 0 \\ x + 3 = 0 &\Longrightarrow x = -3 \\ 3x - 1 = 0 &\Longrightarrow x = \frac{1}{3} \end{aligned}$$

Since $|r| < 1 \Longrightarrow |x^2| < 1$. Therefore, $x = \frac{1}{3}$.

(b) From

$$|x^2| < 1 \implies 0 < x^2 < 1 \implies -1 < x < 1.$$

Example 12

A ball is dropped from a height of 30m and it rebounds $\frac{4}{5}$th of the distance it has fallen after each fall. Find

(a) the height of the nth bounce.

(b) the height of the 6th bounce to 2 d.p.

(c) the ratio of the total distance travelled on rebound to the total distance travelled altogether before coming to rest.

Solution

(a) Sequence of height is given by (Excluding initial height before first bounce)

$$30\left(\frac{4}{5}\right), 30\left(\frac{4}{5}\right)^2, 30\left(\frac{4}{5}\right)^3, \cdots, 30\left(\frac{4}{5}\right)^n$$

The height of the nth bounce is given by

$$U_n = 30\left(\frac{4}{5}\right)^n = 24\left(\frac{4}{5}\right)^{n-1}$$

(b) Height of 6th bounce

$$U_6 = 24\left(\frac{4}{5}\right)^5 = 7.86\text{m}$$

(c) The total distance travelled on rebound is given by the sum to infinity of the G.P.

$$S_\infty = \frac{24}{1-\frac{4}{5}} = \frac{120}{5-4} = 120\text{m}$$

Consider the total distance travelled altogether on the $(n-1)$th rebounce (that is, when the ball strikes the ground the nth time)

Thus,

$$\begin{aligned} S_n &= 30 + 2\,(\text{Sum of } (n-1) \text{ bounces}) \\ &= 30 + 2\left[30\left(\frac{4}{5}\right) + 30\left(\frac{4}{5}\right)^2 + 30\left(\frac{4}{5}\right)^3 + \cdots + 30\left(\frac{4}{5}\right)^{n-1}\right] \\ &= 30 + 2\left\{30\left(\frac{4}{5}\right)\frac{1-\left(\frac{4}{5}\right)^{n-1}}{1-\frac{4}{5}}\right\} = 30 + 240\left[1-\left(\frac{4}{5}\right)^{n-1}\right] \\ &= \left\{270 - 240\left(\frac{4}{5}\right)^{n-1}\right\} \end{aligned}$$

As $n \longrightarrow \infty$, $\left(\frac{4}{5}\right)^{n-1} \longrightarrow 0 \implies S_n \longrightarrow 270$ as $n \longrightarrow \infty$.

Therefore, total distance travelled altogether before coming to rest is 270m.

Therefore, required ratio $= \frac{120}{270} = \frac{4}{9}$.

Example 13

Solution

$$\begin{aligned} 1.2 &= 1 + 0.2 + 0.02 + 0.002 + 0.0002 + \cdots \\ &= 1 + 2 \times 10^{-1} + 2 \times 10^{-2} + 2 \times 10^{-3} + 2 \times 10^{-4} + \cdots \\ &= 1 + 2 \times 10^{-1}\left[1 + 10^{-1} + 10^{-2} + 10^{-3} + \cdots\right] \\ &= 1 + 2 \times 10^{-1}\left[\frac{1}{1-\frac{1}{10}}\right] \\ &= 1 + 2 \times 10^{-1}\left(\frac{10}{9}\right) \\ &= 1 + \frac{2}{9} \\ &= \frac{11}{9}. \end{aligned}$$

9.8 Mixed Sequences

A missed sequence is a sequence which is a combination of cases of an A.P. and/or G.P.

The sum of a mixed series can be obtained by considering the sums of the progressions involved, evident in the definition of the general term, U_n.

Example 14

The nth term of the series $5+5+6\frac{1}{3}+\cdots$ is given by

$$U_n = a\left(\frac{1}{3}\right)^{n-1} + bn$$

where a and b are constants. Find:

(i) the values of a and b.

(ii) an expression for the sum of the first n terms of the series, S_n. Hence, find S_7.

Solution

(i) Given $U_n = a\left(\frac{1}{3}\right)^{n-1} + bn$

Now, $U_1 = 5$, $U_2 = 5$

$$\begin{aligned} a\left(\frac{1}{3}\right)^{1-1} + b(1) &= 5 \\ a + b &= 5 \qquad (9.18) \\ a\left(\frac{1}{3}\right)^{2-1} + b(2) &= 5 \\ \frac{1}{3}a + 2b &= 5 \\ a + 6b &= 15 \qquad (9.19) \end{aligned}$$

Solving (9.18) and (9.19) simultaneously gives $a = 3$ and $b = 2$.

(a) Therefore,

$$U_n = 3\left(\frac{1}{3}\right)^{n-1} + 2n.$$

(ii)

$$\begin{aligned} S_n &= \sum_{k=1}^{n} U_k = \sum_{k=1}^{n}\left[3\left(\frac{1}{3}\right)^{k-1} + 2k\right] \\ &= \sum_{k=1}^{n} 3\left(\frac{1}{3}\right)^{k-1} + 2\sum_{k=1}^{n} k \\ &= 3\left[\frac{1-\left(\frac{1}{3}\right)^n}{1-\frac{1}{3}}\right] + 2\left\{\frac{1}{2}n(n+1)\right\} \\ &= \frac{9}{2}\left[1-\left(\frac{1}{3}\right)^n\right] + n(n+1) \end{aligned}$$

Therefore,

$$\begin{aligned} S_7 &= \frac{9}{2}\left[1-\left(\frac{1}{3}\right)^7\right] + 7(8) \\ &= 4.5(0.9995) + 56 \\ &= 60.4979. \end{aligned}$$

Example 15

The nth term of a series is given by $U_n = 3 - \left(\frac{1}{3}\right)^n$.

(a) Find the first three terms of the series

(b) Deduce the limiting value of U_n.

(c) Obtain an expression for the sum of the first n terms of the series.

Solution

(a)

$$\begin{aligned} U_n &= 3 - \left(\frac{1}{3}\right)^n \\ U_1 &= 3 - \left(\frac{1}{3}\right) = \frac{8}{3} = 2\frac{2}{3} \\ U_2 &= 3 - \left(\frac{1}{3}\right)^2 = 3 - \frac{1}{9} = \frac{26}{9} = 2\frac{8}{9} \\ U_3 &= 3 - \left(\frac{1}{3}\right)^3 = 3 - \frac{1}{27} = \frac{80}{27} = 2\frac{26}{27} \end{aligned}$$

(b)

$$\lim_{n\to\infty} U_n = \lim_{n\to\infty}\left[3-\left(\frac{1}{3}\right)^n\right] = 3$$

since $\left(\frac{1}{3}\right)^n \longrightarrow 0$ as $n \longrightarrow \infty$.

(c)

$$\begin{aligned} S_n &= \sum_{k=1}^{n} U_k = \sum_{k=1}^{n}\left[3-\left(\frac{1}{3}\right)^k\right] \\ &= \sum_{k=1}^{n} 3 - \sum_{k=1}^{n}\frac{1}{3}\left(\frac{1}{3}\right)^{k-1} \\ &= 3n - \frac{1}{3}\left[\frac{1-\left(\frac{1}{3}\right)^n}{1-\frac{1}{3}}\right] \\ &= 3n - \frac{1}{2}\left[1-\left(\frac{1}{3}\right)^n\right]. \end{aligned}$$

9.9 Recurrence Relation

A recurrencr relation is a relation that connects consecutive terms of a sequence. It may generally be written in the form

$$U_{n+1} = ar^n + bU_n\,, \quad n \geq 1 \tag{9.20}$$

where a and b are real constants.. If U_1 is given, subsequent terms of the sequence can be generated.

From (9.20),

$$U_n = ar^{n-1} + bU_{n-1}\,, \quad n > 1. \tag{9.21}$$

It may be shown that U_n can be expressed in terms of n in the form

$$U_n = \frac{ar}{b-r}\left[b^{n-1} - r^{n-1}\right] + b^{n-1}U_1.$$

Proof

$$\begin{aligned} U_{n+1} &= ar^n + bU_n \\ U_n &= ar^{n-1} + bU_{n-1} \\ &= ar^{n-1} + b\left[ar^{n-2} + bU_{n-2}\right] \\ &= ar^{n-1} + abr^{n-2} + b^2U_{n-2} \\ &= ar^{n-1} + abr^{n-2} + b^2\left[ar^{n-3} + bU_{n-3}\right] \\ &= ar^{n-1} + abr^{n-2} + ab^2r^{n-3} + b^3U_{n-3} \\ &= ar^{n-1} + abr^{n-2} + ab^2r^{n-3} + b^3\left[ar^{n-4} + bU_{n-4}\right] \\ &= ar^{n-1} + abr^{n-2} + ab^2r^{n-3} + ab^3r^{n-4} + b^4U_{n-4} \\ &= ar^{n-1} + abr^{n-2} + ab^2r^{n-3} + ab^3r^{n-4} + \cdots + b^{n-2}U_2 \\ &= ar^{n-1} + abr^{n-2} + ab^2r^{n-3} + ab^3r^{n-4} + \cdots + b^{n-2}\left[ar + bU_1\right] \\ &= ar^{n-1} + abr^{n-2} + ab^2r^{n-3} + ab^3r^{n-4} + \cdots + ab^{n-2}r + b^{n-1}U_1 \end{aligned}$$

Factorizing ar out and then $ab^{n-2}r$ out, we have

$$\begin{aligned} U_n &= ar\left[b^{n-2} + b^{n-3}r + b^{n-4}r^2 + \cdots + b^2r^{n-4} + br^{n-3} + r^{n-2}\right] + b^{n-1}U_1 \\ &= ab^{n-2}r\left[1 + \frac{r}{b} + \left(\frac{r}{b}\right)^2 + \cdots + \left(\frac{r}{b}\right)^{n-4} + \left(\frac{r}{b}\right)^{n-3} + \left(\frac{r}{b}\right)^{n-2}\right] \\ &\quad + b^{n-1}U_1 \\ &= ab^{n-2}r\left[\text{ sum of } 1^{\text{st}}\ (n-1) \text{ terms of G.P.with } U_1 = 1,\ r' = \frac{r}{b}\right] + b^{n-1}U_1 \\ &= ab^{n-2}r\left[\frac{1-\left(\frac{r}{b}\right)^{n-1}}{1-\frac{r}{b}}\right] + b^{n-1}U_1 \\ &= \frac{ab^{n-1}r}{b-r}\left[1-\left(\frac{r}{b}\right)^{n-1}\right] + b^{n-1}U_1 \end{aligned}$$

Therefore,

$$\begin{aligned} U_n &= \frac{ar}{b-r}\left[b^{n-1} - r^{n-1}\right] + b^{n-1}U_1 \\ &= -\frac{arb^{n-1}}{b-r} + \frac{ar^n}{r-b} + b^{n-1}U_1 \\ &= \frac{ar^n}{r-b} + b^{n-1}\left[U_1 - \frac{ar}{r-b}\right] \end{aligned} \tag{9.22}$$

Putting $r = 1$, (9.20) becomes

$$U_{n+1} = a + bU_n,$$

and

$$\begin{aligned} U_n &= \frac{a}{b-1}\left[b^{n-1} - 1\right] + b^{n-1}U_1 \\ &= \frac{a}{1-b}\left[1 - b^{n-1}\right] + b^{n-1}U_1 \\ &= \frac{a}{1-b} + b^{n-1}\left[U_1 - \frac{a}{1-b}\right] \end{aligned} \tag{9.23}$$

Now, if $|b| < 1$ that is, $-1 < b < 1$ then $b^{n-1} \longrightarrow 0$ as $n \longrightarrow \infty$.
This implies that the limiting value of $U_n = \frac{a}{1-b}$
That is,

$$\lim_{n\longrightarrow\infty} U_n = \frac{a}{1-b}.$$

Example 16

The sequence $u_1,\ u_2,\ u_3, \cdots$ is defined by

$$U_{n+1} = 6 - \frac{1}{3}U_n, \quad n \geq 1 \text{ and } U_1 = 5$$

(a) Write down the values of $u_2,\ u_3$, and u_4.

(b) Find an expression for U_n in terms of n only.

(c) Find the limiiting value of U_n as n increases indefinitely.

Solution

(a) Given

$$U_{n+1} = 6 - \frac{1}{3}U_n, \quad U_1 = 5.$$

When $n = 1$,

$$U_2 = 6 - \frac{1}{3}U_1 = 6 - \frac{1}{3}(5) = \frac{13}{3} = 4\frac{1}{3}$$

When $n = 2$,

$$U_3 = 6 - \frac{1}{3}U_2 = 6 - \frac{1}{3}\left(\frac{13}{3}\right) = \frac{41}{9} = 4\frac{5}{9}$$

When $n = 3$,

$$U_4 = 6 - \frac{1}{3}U_3 = 6 - \frac{1}{3}\left(\frac{41}{9}\right) = \frac{121}{27} = 4\frac{13}{27}$$

(b) Now,

$$\begin{aligned} U_{n+1} &= 6 - \frac{1}{3}U_n \text{ of the form} \\ U_{n+1} &= a + bU_n, \quad a = 6, \quad b = -\frac{1}{3} \\ U_n &= \frac{a}{1-b} + b^{n-1}\left[U_1 - \frac{a}{1-b}\right] \\ &= \frac{6}{1+\frac{1}{3}} + \left(-\frac{1}{3}\right)^{n-1}\left[5 - \frac{6}{1+\frac{1}{3}}\right] \\ &= \frac{9}{2} + \left(-\frac{1}{3}\right)^{n-1}\left[5 - \frac{9}{2}\right] \\ U_n &= 4\frac{1}{2} + \frac{1}{2}\left(-\frac{1}{3}\right)^{n-1}, \quad n \geq 1 \end{aligned}$$

(c) As $n \longrightarrow \infty$, $\left(-\frac{1}{3}\right)^{n-1} \longrightarrow 0 \implies U_n \longrightarrow 4\frac{1}{2}$.
Therefore,

$$\lim_{n\longrightarrow\infty} U_n = 4\frac{1}{2}$$

Example 17

A sequence of numbers $u_1,\ u_2,\ u_3, \cdots u_n$ satisfies the relation

$$U_n = U_{n-1} + \left(\frac{1}{3}\right)^{n-1} \quad \forall n > 1 \text{ and } u_1 = 1.$$

(a) Write down the values of $u_2,\ u_3$ and u_4.

(a) Prove that

$$U_{n+1} - U_n = b^{n-1}[a - (1-b)U_1]$$

(b) Show that the sequence

$$U_{n+1} = 6 - \frac{1}{3}U_n$$

may be expressed as

$$U_{n+1} = 2\left(-\frac{1}{3}\right)^n + U_n$$

Hence, show that, U_n is expressed in terms of n only as

$$U_n = 4\frac{1}{2} + \frac{1}{2}\left(-\frac{1}{3}\right)^{n-1}.$$

6. A sequence is defined by the statements:

$$U_1 = \frac{5}{2}, \quad U_{n+1} = 2 - \frac{1}{3}U_n, \quad \forall n \geq 1.$$

(a) Show that the sequence may also be defined by the relation

$$U_{n+1} = U_n + 4\left(-\frac{1}{3}\right)^n$$

(b) Deduce an explicit expression for U_n.

(c) Find

$$\lim_{n\longrightarrow\infty} \frac{S_n}{n},$$

where S_n is the sum of the first n terms of the sequence.

Answer: (b) $U_n = \frac{3}{2} + \left(-\frac{1}{3}\right)^{n-1}$ (c) $\frac{3}{2}$

(a) Prove that the sum of the first n numbers is a perfect square.

(b) Henc, find the sum of the odd numbers between and including the 14th and 57th odd numbers.

Answer (b) = 3080.

7. An amount Q is invested at 7% per annum compound interest

(a) What is the total value of the investment after

i. two years

ii. three years

iii. any number of years

(b) Find the value of the investment after 10 years if an amount of GH¢ 1200 was invested.

8. Mrs Fritz bought a new car for GH¢ 24,000.00. She decides to depreciate the car by writing off 8% of its present value at the end of each year. Find the value of the car at the end of

(a) five years

(b) ten years.

Chapter 10

MATRICES

A matrix A is a set of $m \times n$ quantities arranged in a rectangular array of m rows and n columns. It is generally of the form

$$\mathrm{A} = \begin{pmatrix} a_{11} & a_{12} & a_{13} & \cdots & a_{1j} & \cdots & a_{1n} \\ a_{21} & a_{22} & a_{23} & \cdots & a_{2j} & \cdots & a_{2n} \\ a_{31} & a_{32} & a_{33} & \cdots & a_{3j} & \cdots & a_{3n} \\ \vdots & \vdots & \vdots & \cdots & \vdots & \cdots & \vdots \\ a_{i1} & a_{i2} & a_{i3} & \cdots & a_{ij} & \cdots & a_{in} \\ \vdots & \vdots & \vdots & \cdots & \vdots & \cdots & \vdots \\ a_{m1} & a_{m2} & a_{m3} & \cdots & a_{mj} & \cdots & a_{mn} \end{pmatrix}$$

The matrix is denoted by $\mathrm{A} = (a_{ij})$, where a_{ij} represents the element of the matrix in the ith row and the jth column. The matrix is then called $(m \times n)$ matrix and is said to be of order $m \times n$, represented as $\mathrm{A}_{m \times n}$. If $m = n$, then A is called a square matrix. Examples of a matrix are as follows:

$$\mathrm{A} = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 1 & -5 \end{pmatrix}, \quad \mathrm{B} = \begin{pmatrix} 1 & 5 \\ -2 & 3 \\ 1 & 2 \end{pmatrix} = \mathrm{C} = \begin{pmatrix} -1 \\ 3 \\ 2 \end{pmatrix}$$

A is a 2×3, B is 3×2 and C which is 3×1 is usually called a column vector.

The matrix A^t called the transpose of A is obtained by interchanging the rows and columns of A. That is, if $A = (a_{ij})$ is of order $m \times n$, then $A^t = (a_{ji})$ and is order $n \times m$.

10.1 Algebra of Matrices

10.1.1 Addition of Matrices

If $A = (a_{ij})$ and $B = (b_{ij})$ have the same dimensions, then the matrix sum is defined by

$$A + B = (a_{ij}) + (b_{ij}) = (a_{ij} + b_{ij}), \quad \forall i, j$$

The resultant matrix may be denoted by $C = (c_{ij})$, where $c_{ij} = a_{ij} + b_{ij}$. That is, two matrices are added by adding their corresponding entries.

For example,

$$\begin{pmatrix} 1 & -2 \\ 3 & 1 \end{pmatrix} + \begin{pmatrix} -1 & 2 \\ -3 & -1 \end{pmatrix} = \begin{pmatrix} 1+(-1) & -2+2 \\ 3+(-3) & 1+(-1) \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$$

The resultant matrix of the sum of the two matrices above is called a zero matrix of order 2×2, denoted as $0_{2\times 2}$. For any matrix $A_{m\times n} = (a_{ij})$, there is a zero matrix $0_{m\times n}$ for which all entries are equal to 0 such that $A + 0 = 0 + A = A$.

Suppose that $B = A$. In this case, the two matrices must be of the same order and corresponding entries are the same, i.e., $a_{ij} = b_{ij}$. Thus,

$$A + B = A + A = 2A = 2(a_{ij}) = (2a_{ij})$$

It follows that for a scalar k and a matrix $A = (a_{ij})$, the scalar multiplication of A by k is defined by

$$kA = (ka_{ij}), \quad \forall i, j.$$

Example 1

If $A = \begin{pmatrix} x & 3 & -7 \\ 4 & 0 & 6y \end{pmatrix}$, $B = \begin{pmatrix} 1-y & 9 & -21 \\ 12 & 0 & 36 \end{pmatrix}$ and that $3A = B$, find x and y.

Solution

$$3A = 3\begin{pmatrix} x & 3 & -7 \\ 4 & 0 & 6y \end{pmatrix} = \begin{pmatrix} 3x & 9 & -21 \\ 12 & 0 & 18y \end{pmatrix}$$

Given $3A = B$,

$\Longrightarrow$

$$\begin{pmatrix} 3x & 9 & -21 \\ 12 & 0 & 18y \end{pmatrix} = \begin{pmatrix} 1-y & 9 & -21 \\ 12 & 0 & 36 \end{pmatrix}$$

$\Longrightarrow$

$$a_{11} = b_{11} \Longrightarrow 3x = 1 - y \tag{10.1}$$

and

$$18y = 36 \Longrightarrow y = 2 \Longrightarrow 3x = -1 \Longrightarrow x = -\frac{1}{3}$$

Let k and h be scalars and A and B be $m \times n$ matrices. The following properties hold.

(i) $A + B = B + A$

(ii) $A + (B + C) = (A + B) + C$

(iii) $k(hA) = (kh)A$

(iv) $(k + h)A = kA + hA$

(v) $k(A + B) = kA + kB$

10.1.2 Matrix Multiplication

If $A = (a_{ij})$ is an $m \times n$ matrix and $B = (b_{ij})$ is $n \times p$ then the product matrix AB is the $m \times p$ matrix defined by

$$AB = (c_{ij})$$

where

$$\begin{aligned} c_{ij} &= a_{i1}b_{1j} + a_{i2}b_{2j} + a_{i3}b_{3j} + \cdots + a_{in}b_{nj} \\ &= \sum_{k=1}^{n} a_{ik}b_{kj} \end{aligned}$$

That is, c_{ij} is obtained by multiplying the ith row of A by the jth column of B. It should be noted that for multiplication of two matrices to be possible, the number of columns of the first matrix must be the same as the number of rows of the second matrix. We will explain this using the following product of two 3×3 matrices:

$$\begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix} \begin{pmatrix} b_{11} & b_{12} & b_{13} \\ b_{21} & b_{22} & b_{23} \\ b_{31} & b_{32} & b_{33} \end{pmatrix} = \begin{pmatrix} c_{11} & c_{12} & c_{13} \\ c_{21} & c_{22} & c_{23} \\ c_{31} & c_{32} & c_{33} \end{pmatrix}$$

where

$$\begin{aligned} c_{11} &= a_{11}b_{11} + a_{12}b_{21} + a_{13}b_{31} & c_{12} &= a_{11}b_{12} + a_{12}b_{22} + a_{13}b_{32} \\ c_{13} &= a_{11}b_{13} + a_{12}b_{23} + a_{13}b_{33} & c_{21} &= a_{21}b_{11} + a_{22}b_{21} + a_{23}b_{31} \\ c_{22} &= a_{21}b_{12} + a_{22}b_{22} + a_{23}b_{32} & c_{23} &= a_{21}b_{13} + a_{22}b_{23} + a_{23}b_{33} \\ c_{31} &= a_{31}b_{11} + a_{32}b_{21} + a_{33}b_{31} & c_{32} &= a_{31}b_{12} + a_{32}b_{22} + a_{33}b_{32} \\ c_{33} &= a_{31}b_{13} + a_{32}b_{23} + a_{33}b_{33} \end{aligned}$$

It can be observed from the summand for the c_{11} element of the product that as the first subscript of a remains the same as 1 (i.e., row 1 of A), the second subscript changes from 1 to 3 (i.e., the number of columns of A). However, the first subscript of b changes from 1 to 3 (i.e., the number of rows of B), whilst the second subscript of b remains the same as 1 (i.e., the first column of B). From the summand for the c_{23}, as the first subscript of a remains the same as 2 (i.e., row 2 of A), the second subscript changes from 1 to 3 (i.e., the number of columns of A). However, the first subscript of b changes from 1 to 3 (i.e., the number of rows of B), whilst the second subscript of b remains the same as 3 (i.e., column 3 of B). Thus, for the general element c_{ij} obtained from the product of $\mathrm{A}_{m\times n}\mathrm{B}_{n\times p}$, as the first subscript of a remains the same

as i, (i.e., row i of A), the second subscript changes from 1 to n (i.e., the number of columns of A). However, the first subscript of b changes from 1 to n (i.e., the number of rows of B), whilst the second subscript remains the same as j (i.e., column j of B). This pattern is given in the product matrix as follows.

In general,

$$\mathrm{AB} = \begin{pmatrix} \sum_{k=1}^{n} a_{1k}b_{k1} & \sum_{k=1}^{n} a_{1k}b_{k2} & \cdots & \sum_{k=1}^{n} a_{1k}b_{kj} & \cdots & \sum_{k=1}^{n} a_{1k}b_{kp} \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ \sum_{k=1}^{n} a_{ik}b_{k1} & \sum_{k=1}^{n} a_{ik}b_{k2} & \cdots & \sum_{k=1}^{n} a_{ik}b_{kj} & \cdots & \sum_{k=1}^{n} a_{ik}b_{kp} \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ \sum_{k=1}^{n} a_{mk}b_{k1} & \sum_{k=1}^{n} a_{mk}b_{k2} & \cdots & \sum_{k=1}^{n} a_{mk}b_{kj} & \cdots & \sum_{k=1}^{n} a_{mk}b_{kp} \end{pmatrix}$$

Example 2

Given that

$$\mathrm{A} = \begin{pmatrix} 3 & x \end{pmatrix} \quad \text{and} \quad \mathrm{B} = \begin{pmatrix} 1 & -\frac{3}{2} \\ -\frac{3}{2} & 2 \end{pmatrix},$$

find the truthset of

$$\mathrm{A}\left(\mathrm{BA}^t\right) = 0$$

Solution

$$\mathrm{BA}^t = \begin{pmatrix} 1 & -\frac{3}{2} \\ -\frac{3}{2} & 2 \end{pmatrix} \begin{pmatrix} 3 \\ x \end{pmatrix} = \begin{pmatrix} 3 - \frac{3}{2}x \\ -\frac{9}{2} + 2x \end{pmatrix}$$

and

$$\mathrm{A}\left(\mathrm{BA}^t\right) = \begin{pmatrix} 3 & x \end{pmatrix} \begin{pmatrix} 3 - \frac{3}{2}x \\ -\frac{9}{2} + 2x \end{pmatrix} = 0$$

$\Longrightarrow$

$$\begin{aligned} 3(3-\frac{3}{2}x)+x(-\frac{9}{2}+2x) &= 0 \\ 9-\frac{9}{2}x-\frac{9}{2}x+2x^2 &= 0 \\ 9-9x+2x^2 &= 0 \\ 2x^2-6x-3x+9 &= 0 \\ 2x(x-3)-3(x-3) &= 0 \\ x-3=0 \implies x &= 3 \\ 2x-3=0 \implies x &= \frac{3}{2} \end{aligned}$$

Therefore, Truthset $= \{x : x = \frac{3}{2}, 3\}$.

Fundamental Properties of Matrix Multiplication

1. Matrix multiplication is not in general commutative, that is $AB \neq BA$.

2. If $AB = 0 \not\Rightarrow A = 0$ or $B = 0$. For example, if

$$A = \begin{pmatrix} 1 & 2 \\ 2 & 4 \end{pmatrix}, B = \begin{pmatrix} 2 & -2 \\ -1 & 1 \end{pmatrix},$$

it can be verified that $AB = 0$, but $A \neq 0$ and $B \neq 0$.

3. Cancellation property is not valid for matrices in general. That is,

$$AB = AC \not\Rightarrow B = C.$$

4. Let λ and μ be scalars and A, B and C be matrices having dimensions such that the sums and products in the following are defined. Then,

(a) $A(BC) = (AB)C$
(b) $(A+B)C = AC+BC$
(c) $A(B+C) = AB+AC$
(d) $0A = A0 = 0$, where 0 is the zero matrix

(e) $(\lambda A)(\mu B) = \lambda\mu AB$
(f) $(\lambda A+\mu B)^t = \lambda A^t + \mu B^t$

5. For an $n \times n$ matrix A, the product $A \cdot A$ is defined and denoted by A^2.

Similarly,

$$A^3 = A^2 \cdot A = A \cdot A^2$$

In general,

$$A^n = A \cdot A \cdot A \cdot \cdots \cdot A \ (n \text{ terms}), \ n \in \mathbb{Z}^+$$

and

$$A^{n+1} = A^n \cdot A = A \cdot A^n$$

Let $m, n \in \mathbb{Z}^+$ such that A^n and A^m are defined for matrix A, then

$$A^n \cdot A^m = A^{n+m}.$$

6. For any matrix $A_{n\times n}$, there exists a matrix called a *unit* or *identity* matrix denoted by I whose diagonal elements are all equal to 1 but all off-diagonal elements are zero. That is,

$$I = (a_{ij}) = \begin{cases} 0 \text{ if } i \neq j \\ 1 \text{ if } i = j \end{cases}$$

and such that

$$A \cdot I = I \cdot A = A$$

For example, the 2×2 and 3×3 unit matrices are given as

$$\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \text{ and } \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

If a matrix $A = (a_{ij})$ is such that $a_{ij} = 0$, $i \neq j$, then A is called a *diagonal* matrix. For example,

$$A = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{pmatrix} \text{ and } B = \begin{pmatrix} a & 0 & 0 \\ 0 & b & 0 \\ 0 & 0 & c \end{pmatrix}$$

are diagonal matrices. In matrix B, if $a = b = c = \lambda$ then

$$B = \begin{pmatrix} \lambda & 0 & 0 \\ 0 & \lambda & 0 \\ 0 & 0 & \lambda \end{pmatrix} = \lambda \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} = \lambda I$$

and B is called a *scalar matrix.*

7.

$$(AB)^t = B^t A^t$$

Definition 1 *(Invertibility of a Matrix)*

A square matrix A is said to be invertible if there exists a matrix A such that

$$AA^{-1} = A^{-1}A = I$$

The matrix A^{-1} is called the inverse of A. If A is not invertible, it is said to be *singular* or *irregular.*

The consequence of the definition above is given in the theorem that follows.

Theorem

Let A and B be square matrices.

(i) If $AB = BA = I$, then $B = A^{-1}$, i.e., A^{-1} is unique

(ii) $(A^{-1})^{-1} = A$

(iii) $(AB)^{-1} = B^{-1}A^{-1}$

The approach in (i) of the above theorem is the deductive method of finding the inverse of a matrix.

Example 3

The matrices P and Q are given as

$$P = \begin{pmatrix} 5 & 3 \\ 6 & 4 \end{pmatrix} \quad \text{and} \quad Q = \begin{pmatrix} 4 & -3 \\ -6 & 5 \end{pmatrix}$$

(a) Find the matrix PQ.

(b) Deduce the inverse of P.

Solution

(a)

$$PQ = \begin{pmatrix} 5 & 3 \\ 6 & 4 \end{pmatrix} \begin{pmatrix} 4 & -3 \\ -6 & 5 \end{pmatrix}$$

$$= \begin{pmatrix} 20 + (-18) & -15 + 15 \\ 24 + (-24) & -18 + 20 \end{pmatrix} = \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix}$$

(b)

$$PQ = \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix} = 2 \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = 2I$$

Thus,

$$\frac{1}{2}PQ = I \implies P\left(\frac{1}{2}Q\right) = I$$

But $PP^{-1} = I \implies P^{-1} = \frac{1}{2}Q$.

Therefore,

$$P^{-1} = \frac{1}{2} \begin{pmatrix} 4 & -3 \\ -6 & 5 \end{pmatrix} = \begin{pmatrix} 2 & -\frac{3}{2} \\ -3 & \frac{5}{2} \end{pmatrix}$$

In a later section of this chapter we shall return to examine further how to obtain the inverse of a matrix using the deductive approach. Until then, let us consider how to construct the inverse of a matrix the usual way. To find the inverse of a matrix, we first have to obtain the determinant of the matrix.

10.2 Determinant of a Square Matrix

The deteminant of a square matrix A is denoted by $|A|$ or $\det A$ or ΔA. Let A be a 2×2 matrix given as

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}.$$

The determinant of A is given by

$$\det A = ad - bc.$$

For example, if $A = \begin{pmatrix} 3 & -3 \\ -6 & 4 \end{pmatrix}$, then

$$|A| = \begin{vmatrix} 3 & -3 \\ -6 & 4 \end{vmatrix} = 3(4) - 3(6) = -6.$$

The determinant of a 3×3 matrix is defined in terms of 2×2 matrices as

$$A = \begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix}$$

$$\begin{aligned} |A| &= a_{11} \begin{vmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{vmatrix} - a_{12} \begin{vmatrix} a_{21} & a_{23} \\ a_{31} & a_{33} \end{vmatrix} + a_{13} \begin{vmatrix} a_{21} & a_{22} \\ a_{31} & a_{32} \end{vmatrix} \\ &= a_{11}[a_{22}a_{33} - a_{23}a_{32}] - a_{12}[a_{21}a_{33} - a_{23}a_{31}] \\ &\quad + a_{13}[a_{21}a_{32} - a_{22}a_{31}] \end{aligned} \tag{10.2}$$

That is, the entries in the first row are multiplied by the determinant of 2×2 submatrices obtained by deleting the first row and appropriate columns from A.

For example, the determinant of the matrix

$$A = \begin{pmatrix} 1 & 3 & -2 \\ 4 & 1 & -1 \\ 5 & -3 & 2 \end{pmatrix}$$

is

$$\begin{aligned} |A| &= 1 \begin{vmatrix} 1 & -1 \\ -3 & 2 \end{vmatrix} - 3 \begin{vmatrix} 4 & -1 \\ 5 & 2 \end{vmatrix} + (-2) \begin{vmatrix} 4 & 1 \\ 5 & -3 \end{vmatrix} \\ &= (2-3) - 3(8+5) - 2(-12-5) \\ &= -6. \end{aligned}$$

1. If $\det A = 0$, then A^{-1} does not exist and A is said to be irregular or singular, otherwise it is regular or non-singular.
2. $\det A = \det A^t$.

10.3 The Adjoint (Adjugate) of a Matrix

Let $A = [a_{ij}]$ be $n \times n$ matrix, and let the row of a_{ij} entry be deleted. The determinant of the remaining $(n-1) \times (n-1)$ submatrix is called the (i, j)-minor of A. Denote this by M_{ij}.

From

$$A = \begin{pmatrix} 1 & 3 & -2 \\ 4 & 1 & -1 \\ 5 & -3 & 2 \end{pmatrix},$$

then the $(3,2)$-minor of A is

$$M_{32} = \begin{vmatrix} 1 & -2 \\ 4 & -1 \end{vmatrix} = -1 - 4(-2) = 7$$

and the $(2,2)$-minor of A is

$$M_{22} = \begin{vmatrix} 1 & -2 \\ 5 & 2 \end{vmatrix} = 2 - 5(-2) = 12.$$

There are nine possible minors for 3×3 matrix, one for each entry and for n^2 minors for a $n \times n$ matrix.

Let a sign be assigned to M_{ij} by $(-1)^{i+j}M_{ij}$ denoted by A_{ij}. The signed minor $A_{ij} = (-1)^{i+j}M_{ij}$ is referred to as the (i, j)-Cofactor of A.

Thus for A above,

$$A_{32} = (-1)^{3+2}M_{32} = -7; \quad A_{22} = (-1)^{2+2}M_{22} = 12.$$

Thus, signs attached to entries of a 3×3 matrix is given as

$$\begin{pmatrix} + & - & + \\ - & + & - \\ + & - & + \end{pmatrix}.$$

For the matrix $A = [a_{ij}]$, a new matrix of cofactors can be constructed as $[A_{ij}]$. The transpose of this matrix is called the Adjoint matrix of A and denoted by adj A.

That is,

$$\text{adj A} = [A_{ij}]^t = [A_{ji}] = \begin{pmatrix} A_{11} & A_{12} & A_{13} \\ A_{21} & A_{22} & A_{23} \\ A_{31} & A_{32} & A_{33} \end{pmatrix}$$

The determinant of A in (1) can then be defined in terms of cofactors across the first row as

$$\det A = a_{11}A_{11} + a_{12}A_{12} + a_{13}A_{13}$$

If $A = [a_{ij}]$ is an $n \times n$ matrix, $n \geq 2$, then the cofactor expansion across any row or down any column has the value det A. That is,

For the ith row,

$$\det A = a_{i1}A_{i1} + a_{i2}A_{i2} + \cdots + a_{in}A_{in} = \sum_{j=1}^{n} a_{ij}A_{ij}$$

For the jth column,

$$\det A = a_{1j}A_{1j} + a_{2j}A_{2j} + \cdots + a_{nj}A_{nj} = \sum_{i=1}^{n} a_{ij}A_{ij}$$

Now,

$$A\,(\text{adj } A) = (\text{adj } A)\,A = |A|\,I$$

Take,

$$A\,(\text{adj } A) = |A|\,I$$

$$\frac{A}{|A|}\,(\text{adj } A) = I$$

$$A.\left[\frac{1}{|A|}(\text{adj } A)\right] = I$$

Therefore,

$$A^{-1} = \frac{1}{|A|}adj\ A$$

is the inverse of A.

Example 4

Find the inverse of the matrix

$$A = \begin{pmatrix} 1 & 3 & -2 \\ 4 & 1 & -1 \\ 5 & -3 & 2 \end{pmatrix}.$$

Solution

The determinant of the matrix is obtained as

$$\begin{aligned} |A| &= 1\begin{vmatrix} 1 & -1 \\ -3 & 2 \end{vmatrix} - 3\begin{vmatrix} 4 & -1 \\ 5 & 2 \end{vmatrix} + (-2)\begin{vmatrix} 4 & 1 \\ 5 & -3 \end{vmatrix} \\ &= (2-3) - 3(8+5) - 2(-12-5) \\ &= -6. \end{aligned}$$

Next, we consider the cofactors of the matrix as follows:

$$A_{11} = \begin{vmatrix} 1 & -1 \\ -3 & 2 \end{vmatrix} = 2 - 3 = -1 \qquad A_{12} = -\begin{vmatrix} 4 & -1 \\ 5 & 2 \end{vmatrix} = -(8+5) = -13$$

$$A_{13} = \begin{vmatrix} 4 & 1 \\ 5 & -3 \end{vmatrix} = -12 - 5 = -17 \quad A_{21} = -\begin{vmatrix} 3 & -2 \\ -3 & 2 \end{vmatrix} = -(6-6) = 0$$

$$A_{22} = \begin{vmatrix} 1 & -2 \\ 5 & 2 \end{vmatrix} = 2 + 10 = 12 \qquad A_{23} = -\begin{vmatrix} 1 & 3 \\ 5 & -3 \end{vmatrix} = -(-3-15) = 18$$

$$A_{31} = \begin{vmatrix} 3 & -2 \\ 1 & -1 \end{vmatrix} = -3 + 2 = -1 \quad A_{32} = -\begin{vmatrix} 1 & -2 \\ 4 & -1 \end{vmatrix} = -(-1+8) = -7$$

$$A_{33} = \begin{vmatrix} 1 & 3 \\ 4 & 1 \end{vmatrix} = 1 - 12 = -11$$

Therefore, the inverse is given as

$$A^{-1} = \frac{1}{|A|}\text{adj } A = -\frac{1}{6}\begin{pmatrix} -1 & 0 & -1 \\ -13 & 12 & -7 \\ -17 & 18 & -11 \end{pmatrix}$$

By a similar procedure, the inverse of a 2×2 matrix $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ is given by

$$A^{-1} = \frac{1}{|A|}\begin{pmatrix} d & -b \\ -c & a \end{pmatrix} = \frac{1}{ad-bc}\begin{pmatrix} d & -b \\ -c & a \end{pmatrix}.$$

For example, if

$$A = \begin{pmatrix} 3 & -3 \\ -6 & 4 \end{pmatrix},$$

then

$$|A| = 3(4) - 3(6) = 12 - 18 = -6$$

$$A^{-1} = -\frac{1}{6}\begin{pmatrix} 4 & 3 \\ 6 & 3 \end{pmatrix} = \begin{pmatrix} -\frac{2}{3} & -\frac{1}{2} \\ -1 & -\frac{1}{2} \end{pmatrix}.$$

If A and B are square matrices, then,

(i) $\det(AB) = \det A \det B$.

(ii) $\det(AA^{-1}) = \det A \det(A^{-1}) = 1$, then $\det(A^{-1}) = \frac{1}{\det A}$.

10.4 Gauss-Jordan Operations on Augmented Matrix

Inverse Matrix- Alternative

Given the matrix $A_{n\times n}$, we form an augmented matrix $\left[\begin{array}{c|c} A & I \end{array}\right]$, where I is of the same dimension as A. A systematic (row) operations are then performed until $\left[\begin{array}{c|c} A & I \end{array}\right]$ is transformed into that of the form $\left[\begin{array}{c|c} I & B \end{array}\right]$. The matrix $\left[\begin{array}{c|c} A & I \end{array}\right]$ is then said to have been put into *reduced echelon form*. Since A^{-1} is assumed to exist, the reduced echelon form is $\left[\begin{array}{c|c} I & A^{-1} \end{array}\right]$. Thus, $B = A^{-1}$. In this case, A is said to have a (row) rank of n. The elementary row operations that transforms the augmented matrix into reduced echelon form is any combination of the following procedures.

Type 1: Interchange any two rows;
Type 2: Multiply any row by a scalar;
Type 3: Add to one row scalar multiple of any other row.
For example, the notation,

$$R_2^1 = R_2 + \lambda R_1$$

means that a new second row is created by adding the second row to λ times the first row. A similar definition can be formulated for column operations.

Example 5

Use elementary operations to find the inverse of the matrices

$$(a)\quad \mathrm{P} = \begin{pmatrix} -16 & -12 & 5 \\ 4 & 3 & -1 \\ -3 & -2 & 1 \end{pmatrix} \quad \text{and} \quad (b)\quad \mathrm{Q} = \begin{pmatrix} 1 & -1 & -3 \\ -1 & 2 & 4 \\ 1 & 1 & 0 \end{pmatrix}.$$

Solution

(a) We form the augmented matrix and perform row operations to obtain new first, second and third rows. The remaining operations then follow as shown.

$$\left(\begin{array}{ccc|ccc} -16 & -12 & 5 & 1 & 0 & 0 \\ 4 & 3 & -1 & 0 & 1 & 0 \\ -3 & -2 & 1 & 0 & 0 & 1 \end{array}\right) \qquad \begin{aligned} R_1^1 &= R_1 - 5R_3 \\ R_2^1 &= R_2 + R_3 \\ R_3^1 &= R_1 - 6R_3 \end{aligned}$$

$$\left(\begin{array}{ccc|ccc} -1 & -2 & 0 & 1 & 0 & -5 \\ 1 & 1 & 0 & 0 & 1 & 1 \\ 2 & 0 & -1 & 1 & 0 & -6 \end{array}\right) \qquad R_1^1 = R_1 + 2R_2$$

$$\left(\begin{array}{ccc|ccc} 1 & 0 & 0 & 1 & 2 & -3 \\ 1 & 1 & 0 & 0 & 1 & 1 \\ 2 & 0 & -1 & 1 & 0 & -6 \end{array}\right) \qquad \begin{aligned} R_2^1 &= R_2 - R_1 \\ R_3^1 &= R_3 - 2R_1 \end{aligned}$$

$$\left(\begin{array}{ccc|ccc} 1 & 0 & 0 & 1 & 2 & -3 \\ 0 & 1 & 0 & -1 & -1 & 4 \\ 0 & 0 & -1 & -1 & -4 & 0 \end{array}\right) \qquad R_3^1 = -R_3$$

$$\left(\begin{array}{ccc|ccc} 1 & 0 & 0 & 1 & 2 & -3 \\ 0 & 1 & 0 & -1 & -1 & 4 \\ 0 & 0 & 1 & 1 & 4 & 0 \end{array}\right) \qquad \text{of the form } \left[\begin{array}{c|c} \mathrm{I} & \mathrm{B} \end{array}\right]$$

Therefore,

$$\mathrm{B} = \mathrm{P}^{-1} = \begin{pmatrix} 1 & 2 & -3 \\ -1 & -1 & 4 \\ 1 & 4 & 0 \end{pmatrix}.$$

(b)

$$\left(\begin{array}{ccc|ccc} 1 & -1 & -3 & 1 & 0 & 0 \\ -1 & 2 & 4 & 0 & 1 & 0 \\ 1 & 1 & 0 & 0 & 0 & 1 \end{array}\right) \qquad R_2^1 = R_2 - R_3$$

$$R_3^1 = R_2 + R_3$$

$$\left(\begin{array}{ccc|ccc} 1 & -1 & -3 & 1 & 0 & 0 \\ -2 & 1 & 4 & 0 & 1 & -1 \\ 0 & 1 & 1 & 1 & 1 & 0 \end{array}\right) \qquad R_1^1 = R_1 + R_2$$

$$R_2^1 = R_2 + 2R_1$$

$$\left(\begin{array}{ccc|ccc} -1 & 0 & 1 & 1 & 1 & -1 \\ 0 & -1 & -2 & 2 & 1 & -1 \\ 0 & 1 & 1 & 1 & 1 & 0 \end{array}\right) \qquad R_2^1 = R_2 + 2R_3$$

$$R_3^1 = R_3 + R_2$$

$$\left(\begin{array}{ccc|ccc} -1 & 0 & 1 & 1 & 1 & -1 \\ 0 & 1 & 0 & 4 & 3 & -1 \\ 0 & 0 & -1 & 3 & 2 & -1 \end{array}\right) \qquad R_1^1 = R_1 + R_3$$

$$R_3^1 = -R_3$$

$$\left(\begin{array}{ccc|ccc} -1 & 0 & 0 & 4 & 3 & -2 \\ 0 & 1 & 0 & 4 & 3 & -1 \\ 0 & 0 & 1 & -3 & -2 & 1 \end{array}\right) \qquad R_1^1 = -R_1$$

$$\left(\begin{array}{ccc|ccc} 1 & 0 & 0 & -4 & -3 & 2 \\ 0 & 1 & 0 & 4 & 3 & -1 \\ 0 & 0 & 1 & -3 & -2 & 1 \end{array}\right) \qquad \text{of the form } \left[\begin{array}{c|c} \mathrm{I} & \mathrm{B} \end{array}\right]$$

Therefore,

$$\mathrm{B} = \mathrm{Q}^{-1} = \begin{pmatrix} -4 & -3 & 2 \\ 4 & 3 & -1 \\ -3 & -2 & 1 \end{pmatrix}.$$

10.5 Characteristic Equation of a Matrix

Suppose the matrix A satisfies the equation of the form

$$x^n + a_{n-1}x^{n-1} + \cdots + a_1 x + a_o = 0$$

This equation is called a characteristic equation of A. Every square matrix satisfies its own characteristc equation. This implies that

$$\mathrm{A}^n + a_{n-1}\mathrm{A}^{n-1} + \cdots + a_1\mathrm{A} + a_o\mathrm{I} = 0,$$

where 0 and I are the zero and unit matrices, respectively, of the same order as A. Factorizing A gives

$$\mathrm{A}\left[\mathrm{A}^{n-1} + a_{n-1}\mathrm{A}^{n-2} + \cdots + a_1\mathrm{I}\right] = -a_o\mathrm{I},$$

$$\mathrm{A}\left[-\frac{1}{a_o}\left(\mathrm{A}^{n-1} + a_{n-1}\mathrm{A}^{n-2} + \cdots + a_1\mathrm{I}\right)\right] = \mathrm{I}.$$

Since $\mathrm{AA}^{-1} = \mathrm{I}$, it implies that the inverse of A is given by the expression

$$\mathrm{A}^{-1} = -\frac{1}{a_o}\left(\mathrm{A}^{n-1} + a_{n-1}\mathrm{A}^{n-2} + \cdots + a_1\mathrm{I}\right).$$

Thus, we can use the characteristic equation of a matrix to deduce its inverse.

Example 6

A matrix is given by

$$\mathrm{A} = \begin{pmatrix} 1 & -1 & -3 \\ -1 & 2 & 4 \\ 1 & 1 & 0 \end{pmatrix}$$

(a) Show that the matrix A satisfies the equation

$$x^3 - 3x^2 - 1 = 0$$

(b) Deduce that the inverse of A is of the form

$$\lambda A^2 + \mu A + \kappa I$$

(c) Hence, find the inverse of A.

Solution

(a) If

$$A = \begin{pmatrix} 1 & -1 & -3 \\ -1 & 2 & 4 \\ 1 & 1 & 0 \end{pmatrix}, \text{then } A^2 = \begin{pmatrix} -1 & -6 & -7 \\ 1 & 9 & 11 \\ 0 & 1 & 1 \end{pmatrix},$$

$$\text{and } A^3 = \begin{pmatrix} -2 & -18 & -21 \\ 3 & 28 & 33 \\ 0 & 3 & 4 \end{pmatrix}$$

Hence, substituting into

$$x^3 - 3x^2 - 1,$$

we have

$$\begin{pmatrix} -2 & -18 & -21 \\ 3 & 28 & 33 \\ 0 & 3 & 4 \end{pmatrix} - 3\begin{pmatrix} -1 & -6 & -7 \\ 1 & 9 & 11 \\ 0 & 1 & 1 \end{pmatrix} - \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

$$= \begin{pmatrix} -2 & -18 & -21 \\ 3 & 28 & 33 \\ 0 & 3 & 4 \end{pmatrix} - \begin{pmatrix} -3 & -18 & -21 \\ 3 & 27 & 33 \\ 0 & 3 & 3 \end{pmatrix} - \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

$$= \begin{pmatrix} -2+3-1 & -18+18+0 & -21+21+0 \\ 3-3-0 & 28-27-1 & 33-33+0 \\ 0-0-0 & 3-3-0 & 4-3-1 \end{pmatrix} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

Therefore A satisfies the equation.

(b)

$$A^3 - 3A^2 - I = 0 \implies A^3 - 3A^2 = I$$

$$\implies$$

$$. A\left[A^2 - 3A\right] = 0$$

$$\implies$$

$$A^{-1} = A^2 - 3A = 0$$

form $\lambda A^2 + \mu A + \kappa I$ where $\lambda = 1,\ \mu = -3,\ \kappa = 0$.

(c)

$$A^{-1} = \begin{pmatrix} -1 & -6 & -7 \\ 1 & 9 & 11 \\ 0 & 1 & 1 \end{pmatrix} - 3\begin{pmatrix} 1 & -1 & -3 \\ -1 & 2 & 4 \\ 1 & 1 & 0 \end{pmatrix}$$

$$= \begin{pmatrix} -1 & -6 & -7 \\ 1 & 9 & 11 \\ 0 & 1 & 1 \end{pmatrix} - \begin{pmatrix} 3 & -3 & -9 \\ -3 & 6 & 12 \\ 3 & 3 & 0 \end{pmatrix}$$

$$= \begin{pmatrix} -4 & -3 & 2 \\ 4 & 3 & -1 \\ -3 & -2 & 1 \end{pmatrix}.$$

10.5.1 Finding the Characteristic Equation of a Matrix

In exercise on polynomials, you were asked to show that the function

$$p(x) = x^3 - 3x^2 - 9x + 3$$

is a characteristic polynomial of the matrix

$$Q = \begin{pmatrix} 1 & -2 & 4 \\ 0 & -1 & 2 \\ 2 & 0 & 3 \end{pmatrix}.$$

We will use Q and $p(x)$ to demonstrate how a characteristic equation is derived for both three and two dimensional square matrices.

First you might have noted that the leading coefficient of a characteristic polynomial is always equal to 1. Now, to find the trace of the matrix, we obtain $tr(\mathrm{Q}) = 1 + (-1) + 3 = 3$. That is, the sum of the diagonal elements of Q is 3. Note that the negative of this trace is the coefficient of x^2 in $p(x)$. Next, we consider the minors of Q corresponding to the leading diagonal elements. These are

$$\mathrm{M}_{11} = \begin{vmatrix} -1 & 2 \\ 0 & 3 \end{vmatrix} = -3 - 0 = -3$$

$$\mathrm{M}_{22} = \begin{vmatrix} 1 & 4 \\ 2 & 3 \end{vmatrix} = 3 - 8 = -5$$

$$\mathrm{M}_{33} = \begin{vmatrix} 1 & -2 \\ 0 & -1 \end{vmatrix} = -1 - 0 = -1$$

Adding the minors we obtain

$$\mathrm{M}_{11} + \mathrm{M}_{22} + \mathrm{M}_{33} = -3 + (-5) + (-1) = -9.$$

Note that the sum of these minors is equal to the coefficient of x in the polynomial. Finally, the determinant of Q is

$$\begin{aligned} |\mathrm{Q}| &= 1\begin{vmatrix} -1 & 2 \\ 0 & 3 \end{vmatrix} + 2\begin{vmatrix} 0 & 2 \\ 2 & 3 \end{vmatrix} + 4\begin{vmatrix} 0 & -1 \\ 2 & 0 \end{vmatrix} \\ &= (-3 - 0) + 2(0 - 4) + 4(0 + 2) \\ &= -3. \end{aligned}$$

Observe that the negative of this determinant is equal to the constant in the polynomial. The procedure for determining the characteristic equation of a matrix Q of dimension 3×3 is therefore generalized as

$$p(x) = x^3 - tr(\mathrm{Q})x^2 + (\mathrm{M}_{11} + \mathrm{M}_{22} + \mathrm{M}_{33})x - |\mathrm{Q}|.$$

The characteristic equation of a matrix Q of dimension 2×2 is similarly generalized as

$$q(x) = x^2 - tr(\mathrm{Q})x + |\mathrm{Q}|.$$

Having shown how the polynomial in this illustration was obtained, we now deduce the inverse of Q. Since Q is a zero of $p(x)$, we have

$$\mathrm{Q}^3 - 3\mathrm{Q}^2 - 9\mathrm{Q} + 3\mathrm{I} = 0.$$

Finding the inverse of Q we have

$$\begin{aligned} \mathrm{Q}^3 - 3\mathrm{Q}^2 - 9\mathrm{Q} &= -3\mathrm{I} \\ \mathrm{Q}\left(\mathrm{Q}^2 - 3\mathrm{Q} - 9\mathrm{I}\right) &= -3\mathrm{I} \\ \mathrm{Q}\left[-\frac{1}{3}\left(\mathrm{Q}^2 - 3\mathrm{Q} - 9\mathrm{I}\right)\right] &= \mathrm{I}. \end{aligned}$$

Therefore,

$$\mathrm{Q}^{-1} = -\frac{1}{3}\left(\mathrm{Q}^2 - 3\mathrm{Q} - 9\mathrm{I}\right).$$

Evaluating Q^{-1}, we obtain

$$\begin{aligned} \mathrm{Q}^{-1} &= -\frac{1}{3}\left[\begin{pmatrix} 1 & -2 & 4 \\ 0 & -1 & 2 \\ 2 & 0 & 3 \end{pmatrix}^2 - 3\begin{pmatrix} 1 & -2 & 4 \\ 0 & -1 & 2 \\ 2 & 0 & 3 \end{pmatrix} - 9\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}\right] \\ &= \frac{1}{3}\left[\begin{pmatrix} -9 & 0 & -12 \\ -4 & -1 & -4 \\ -8 & 4 & -17 \end{pmatrix} + \begin{pmatrix} 3 & -6 & 12 \\ 0 & -3 & 6 \\ 6 & 0 & 9 \end{pmatrix} + \begin{pmatrix} 9 & 0 & 0 \\ 0 & 9 & 0 \\ 0 & 0 & 9 \end{pmatrix}\right] \\ &= \frac{1}{3}\begin{pmatrix} 3 & -6 & 0 \\ -4 & 5 & 2 \\ -2 & 4 & 1 \end{pmatrix} = \begin{pmatrix} 1 & -2 & 0 \\ -\frac{4}{3} & \frac{5}{3} & \frac{2}{3} \\ -\frac{2}{3} & \frac{4}{3} & \frac{1}{3} \end{pmatrix} \end{aligned}$$

Example 7

Given

$$\mathrm{A} = \begin{pmatrix} -16 & -12 & 5 \\ 4 & 3 & -1 \\ -3 & -2 & 1 \end{pmatrix},$$

(a) Find the characteristic polynomial of A.

(b) Deduce that A^{-1} is of the form $\lambda\mathrm{A}^2 + \mu\mathrm{A} + \kappa\mathrm{I}$.

(c) Hence, find A^{-1}.

Solution

The deteriminant of the matrix is given as

$$\begin{aligned} |\mathrm{A}| &= -16\begin{vmatrix} 3 & -1 \\ -2 & 1 \end{vmatrix} + 12\begin{vmatrix} 4 & -1 \\ -3 & 1 \end{vmatrix} + 5\begin{vmatrix} 4 & 3 \\ -3 & -2 \end{vmatrix} \\ &= -16(3-2) + 12(4-3) + 5(-8+9) \\ &= 1. \end{aligned}$$

The minors of the diagonal elements are

$$\mathrm{M}_{11} = \begin{vmatrix} 3 & -1 \\ -2 & 1 \end{vmatrix} = 3 - 2 = 1$$

$$\mathrm{M}_{22} = \begin{vmatrix} -16 & 5 \\ -3 & 1 \end{vmatrix} = -16 + 15 = -1$$

$$\mathrm{M}_{33} = \begin{vmatrix} -16 & -12 \\ 4 & 3 \end{vmatrix} = -48 + 48 = 0$$

$\mathrm{Tr}\mathrm{A} = -12$.
Therefore,

$$p(x) = x^3 + 12x^2 + 1 = 0.$$

Remark

Another important application of the characteristic equation is the representation of higher powers of a matrix. Any positive integral power of A (i.e., A^n) can be expressed in terms of $\mathrm{I}, \mathrm{A}, \cdots, \mathrm{A}^{n-1}$. Similarly, all negative integral powers of A (i.e., A^{-n}) may also be expressed in terms of $\mathrm{I}, \mathrm{A}, \cdots, \mathrm{A}^{n-1}$ when A^{-1} exists. The following example illustrates this point.

Example 7

(a) Find the characteristic equation of the matrix

$$\mathrm{Q} = \begin{pmatrix} -1 & 0 \\ 4 & -2 \end{pmatrix}$$

(b) Hence, by expressing in the form

$$\lambda\mathrm{Q} + \mu\mathrm{I},$$

find the matrices Q^{-1}, Q^3 and Q^4.

Solution

(a) If

$$\mathrm{Q} = \begin{pmatrix} -1 & 0 \\ 4 & -2 \end{pmatrix},$$

then

$$\mathrm{tr}(\mathrm{Q}) = -1 + (-2) = -3,$$

and

$$\det \mathrm{Q} = (-1)(-2) - 0 = 2.$$

Therefore, the characteristic equation of Q is

$$\mathrm{Q}^2 + 3\mathrm{Q} + 2\mathrm{I} = 0.$$

(b) Factorizing Q gives

$$\begin{aligned} \mathrm{Q}\,(\mathrm{Q} + 3\mathrm{I}) &= -2\mathrm{I} \\ \mathrm{Q}\left[-\tfrac{1}{2}(\mathrm{Q} + 3\mathrm{I})\right] &= \mathrm{I}. \end{aligned}$$

Therefore, the inverse of Q is given by

$$\begin{aligned} Q^{-1} &= -\frac{1}{2}(Q + 3I) \\ &= -\frac{1}{2}Q - \frac{3}{2}I. \end{aligned}$$

which is of the form $\lambda Q + \mu I$, where $\lambda = -\frac{1}{2}$, $\mu = -\frac{3}{2}$.

Now, from

$$Q^2 + 3Q + 2I = 0, \implies Q^2 = -3Q - 2I$$

Multiplying through by Q gives

$$Q\left(Q^2\right) = -3Q^2 - 2Q.$$

Substituting $-3Q - 2I$ for Q^2 in the above and simplifying gives

$$\begin{aligned} Q^3 &= -3(-3Q - 2I) - 2Q \\ &= 9Q + 6I - 2Q \\ &= 7Q + 6I. \end{aligned}$$

which is of the form $\lambda Q + \mu I$, where $\lambda = 7$, $\mu = 6$.

From $Q^3 = 7Q + 6I$, multiplying through by Q gives

$$Q\left(Q^3\right) = 7Q^2 + 6Q.$$

Substituting $-3Q - 2I$ for Q^2 in the above and simplifying gives

$$\begin{aligned} Q^4 &= 7(-3Q - 2I) + 6Q \\ &= -21Q - 14I + 6Q \\ &= -15Q - 14I \end{aligned}$$

which is of the form $\lambda Q + \mu I$, where $\lambda = -15$, $\mu = -14$.

Substituting for Q in the various expressions gives

$$\begin{aligned} Q^{-1} &= -\frac{1}{2}\begin{pmatrix} -1 & 0 \\ 4 & -2 \end{pmatrix} - \frac{3}{2}\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \\ &= \begin{pmatrix} \frac{1}{2} & 0 \\ -2 & 1 \end{pmatrix} - \begin{pmatrix} \frac{3}{2} & 0 \\ 0 & \frac{3}{2} \end{pmatrix} \\ &= \begin{pmatrix} -1 & 0 \\ -2 & -\frac{1}{2} \end{pmatrix} \end{aligned}$$

$$\begin{aligned} Q^3 &= 7\begin{pmatrix} -1 & 0 \\ 4 & -2 \end{pmatrix} + 6\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \\ &= \begin{pmatrix} -7 & 0 \\ 28 & -14 \end{pmatrix} + \begin{pmatrix} 6 & 0 \\ 0 & 6 \end{pmatrix} \\ &= \begin{pmatrix} -1 & 0 \\ 28 & -8 \end{pmatrix} \end{aligned}$$

$$\begin{aligned} Q^4 &= -15\begin{pmatrix} -1 & 0 \\ 4 & -2 \end{pmatrix} - 14\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \\ &= \begin{pmatrix} 15 & 0 \\ 60 & 30 \end{pmatrix} - \begin{pmatrix} 14 & 0 \\ 0 & 14 \end{pmatrix} \\ &= \begin{pmatrix} 1 & 0 \\ 60 & 16 \end{pmatrix} \end{aligned}$$

10.6 Application of Matrices

10.6.1 Solution of Systems of Equations

Consider the system,

$$\begin{aligned} a_{11}x + a_{12}y + a_{13}z &= b_1 \\ a_{21}x + a_{22}y + a_{23}z &= b_2 \\ a_{31}x + a_{32}y + a_{33}z &= b_3 \end{aligned}$$

Write as

$$\begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} b_1 \\ b_2 \\ b_3 \end{pmatrix}$$

which is of the form

$$\mathrm{Ax} = \mathrm{b}$$

Suppose $\det \mathrm{A} \neq 0$, then by pre-multiplying through by A^{-1} gives

$$\begin{aligned} \mathrm{A}^{-1}\mathrm{Ax} &= \mathrm{A}^{-1}\mathrm{b} \\ \mathrm{Ix} &= \mathrm{A}^{-1}\mathrm{b} \\ \mathrm{x} &= \mathrm{A}^{-1}\mathrm{b} \end{aligned}$$

This approach is the method of matrix inversion.

The solution may also be obtained by

1. Cramer's Rule (Use of Determinants)
2. Elementary Operations

Example 8

Solve the system

$$\begin{aligned} x + y + z &= 6 \\ x + 2y + 3z &= 14 \\ x + 4y + 9z &= 36 \end{aligned}$$

Solution

Write the matrix as

$$\begin{pmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \\ 1 & 4 & 9 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 6 \\ 14 \\ 36 \end{pmatrix}$$

which is of the form $\mathrm{Ax} = \mathrm{b} \implies \mathrm{x} = \mathrm{A}^{-1}\mathrm{b}$. Now

$$\mathrm{A}^{-1} = \frac{1}{2}\begin{pmatrix} 6 & -5 & 1 \\ -6 & 8 & -2 \\ 2 & -3 & 1 \end{pmatrix},$$

then

$$\begin{aligned} \mathrm{x} &= \frac{1}{2}\begin{pmatrix} 6 & -5 & 1 \\ -6 & 8 & -2 \\ 2 & -3 & 1 \end{pmatrix}\begin{pmatrix} 6 \\ 14 \\ 36 \end{pmatrix} \\ &= \frac{1}{2}\begin{pmatrix} 36 - 70 + 36 \\ -36 + 112 - 72 \\ 12 - 42 + 36 \end{pmatrix} = \frac{1}{2}\begin{pmatrix} 2 \\ 4 \\ 6 \end{pmatrix} = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} \end{aligned}$$

$\implies$

$$\text{Truthset} = \{x, y, z : x = 1,\ y = 2,\ z = 3\}.$$

Alternative Method

Consider the augmented matrix $\left[\begin{array}{c|c} \mathrm{A} & \mathrm{B} \end{array}\right]$ put in row echelon form.

$$\left(\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 1 & 2 & 3 & 14 \\ 1 & 4 & 9 & 36 \end{array}\right) \quad R_2^1 = R_2 - R_1, \quad R_3^1 = R_3 - R_1$$

$$\left(\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 0 & 1 & 2 & 8 \\ 0 & 3 & 8 & 30 \end{array}\right) \quad R_3^1 = R_3 - 3R_2$$

$$\left(\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 0 & 1 & 2 & 8 \\ 0 & 0 & 2 & 6 \end{array}\right)$$

This implies that

$$\begin{aligned} x + y + z &= 6 \\ y + 2z &= 8 \\ 2z &= 6 \\ z &= 3 \end{aligned}$$

Thus,

$$\begin{aligned} y &= 8 - 2(3) = 2 \\ x + 2 + 3 &= 6 \implies x = 1 \end{aligned}$$

Therefore,

$$\text{Truth set} = \{x, y, z : \ x = 1, \ y = 2, \ z = 3\}$$

10.6.2 Cramer's Rule

Consider the system

$$\begin{aligned} ax + by &= z_1 \\ cx + dy &= z_2 \end{aligned}$$

The system can be written in matrix form as

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} z_1 \\ z_2 \end{pmatrix}$$

which is of the form $\mathbf{Ax = b} \implies \mathbf{x = A^{-1}b}$ where

$$\mathrm{A}^{-1} = \frac{1}{|\mathrm{A}|}\mathrm{adjA} = \frac{1}{|\mathrm{A}|}\begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$$

$$\implies \mathrm{adjA} = \begin{pmatrix} d & -b \\ -c & a \end{pmatrix} = \begin{pmatrix} \mathrm{A}_{11} & \mathrm{A}_{12} \\ \mathrm{A}_{21} & \mathrm{A}_{22} \end{pmatrix}$$

Therefore,

$$\begin{aligned} \mathrm{x} &= \frac{1}{|\mathrm{A}|}\begin{pmatrix} \mathrm{A}_{11} & \mathrm{A}_{12} \\ \mathrm{A}_{21} & \mathrm{A}_{22} \end{pmatrix}\begin{pmatrix} z_1 \\ z_2 \end{pmatrix} \\ &= \frac{1}{|\mathrm{A}|}\begin{pmatrix} z_1\mathrm{A}_{11} + z_2\mathrm{A}_{21} \\ z_1\mathrm{A}_{12} + z_2\mathrm{A}_{22} \end{pmatrix} \\ \begin{pmatrix} x \\ y \end{pmatrix} &= \frac{1}{|\mathrm{A}|}\begin{pmatrix} dz_1 - bz_2 \\ -cz_1 + az_2 \end{pmatrix} \end{aligned}$$

Thus,

$$x = \frac{1}{|\mathrm{A}|}(dz_1 - bz_2) = \frac{1}{|\mathrm{A}|}\begin{vmatrix} z_1 & b \\ z_2 & d \end{vmatrix} = \frac{1}{|\mathrm{A}|}|\mathrm{A}_1| \text{ and}$$

$$y = \frac{1}{|\mathrm{A}|}(az_2 - cz_1) = \frac{1}{|\mathrm{A}|}\begin{vmatrix} a & z_1 \\ c & z_2 \end{vmatrix} = \frac{1}{|\mathrm{A}|}|\mathrm{A}_2|.$$

That is, A_1 is obtained by replacing the first column of A by b and A_2 is obtained by replacing the second column of A by b.

Similarly, for a system of 3 unknowns $x,\ y,\ z$,

$$x = \frac{|A_1|}{|A|}, \quad y = \frac{|A_2|}{|A|}, \quad z = \frac{|A_3|}{|A|}$$

where A_j is the matrix obtained by replacing the entries of the jth column of A by the entries of the vector b.

Example 9

Use Cramer's Rule to solve

$$\begin{aligned} -3x + 6y &= -4 \\ 2x + 4y &= 7 \end{aligned}$$

Solution

Write as

$$\begin{pmatrix} -3 & 6 \\ 2 & 4 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} -4 \\ 7 \end{pmatrix}$$

which is of the form $\mathrm{Ax} = \mathrm{b} \implies \mathrm{x} = \mathrm{A}^{-1}\mathrm{b}$

Now

$$|\mathrm{A}| = -12 - 12 = -24$$

$\implies$

$$x = \frac{1}{|\mathrm{A}|}\begin{vmatrix} -4 & 6 \\ 7 & 4 \end{vmatrix} = -\frac{1}{24}(-16 - 42) = \frac{29}{12}$$

$$y = \frac{1}{|\mathrm{A}|}\begin{vmatrix} -3 & -4 \\ 2 & 7 \end{vmatrix} = -\frac{1}{24}(-21 + 8) = \frac{13}{24}$$

$\implies$

$$\text{Truthset} = \left\{x, y : x = \frac{29}{12},\ y = \frac{13}{24}\right\}$$

Example 10

Solve the system

$$\begin{aligned} 3x - 7y + 11z &= 4 \\ x - 2y + 3z &= 1 \\ -2x + 8y - 16z &= 4 \end{aligned}$$

Solution

Write as

$$\begin{pmatrix} 3 & -7 & 11 \\ 1 & -2 & 3 \\ -2 & 8 & -16 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 4 \\ 1 \\ 4 \end{pmatrix}$$

which is of the form $\mathrm{Ax} = \mathrm{b} \implies \mathrm{x} = \mathrm{A}^{-1}\mathrm{b}$

$$\begin{aligned} |\mathrm{A}| &= 3(32 - 24) + 7(-16 + 6) + 11(8 - 4) \\ &= 24 - 70 + 44 \\ |\mathrm{A}| &= -2 \end{aligned}$$

Thus,

$$x = \frac{1}{|\mathrm{A}|}\begin{vmatrix} 4 & -7 & 11 \\ 1 & -2 & 3 \\ 4 & 8 & -16 \end{vmatrix}$$

$$= -\frac{1}{2}[4(32 - 24) + 7(-16 - 12) + 11(8 + 8)] = -6$$

$$y = \frac{1}{|\mathrm{A}|}\begin{vmatrix} 3 & 4 & 11 \\ 1 & 1 & 3 \\ -2 & 4 & -16 \end{vmatrix}$$

$$= -\frac{1}{2}[3(-16 - 12) - 4(-16 + 6) + 11(4 + 2)] = -11.$$

and

$$z = \frac{1}{|A|}\begin{vmatrix} 3 & -7 & 4 \\ 1 & -2 & 1 \\ -2 & 8 & 4 \end{vmatrix}$$

$$= -\frac{1}{2}[3(-8-8)+7(4+2)+4(8-4)] = -5$$

$\Longrightarrow$

$$\text{Truthset} = \{x, y, z : x = -6,\ y = -11,\ z = -5\}$$

Exercises

1. If $A = \begin{pmatrix} 2 & 1 & 4 \\ -1 & 3 & 0 \\ 2 & -1 & 3 \end{pmatrix}$ and $B = \begin{pmatrix} 1 & 2 & 5 \\ 3 & 0 & -1 \\ 5 & -1 & 1 \end{pmatrix}$, find $2A - 3B$.

2. Find AB if $A = \begin{pmatrix} 1 & 2 \\ 3 & -1 \\ 2 & 3 \end{pmatrix}$ and $B = \begin{pmatrix} -3 & 1 & 2 \\ 2 & 1 & 3 \end{pmatrix}$.

3. If $A = \begin{pmatrix} 2 & 1 & -1 \end{pmatrix}^t$ and $B = \begin{pmatrix} 3 & 2 \\ 4 & 3 \\ 7 & 6 \end{pmatrix}$, find A^tB where t denote transposition.

4. If $A = \begin{pmatrix} 1 & 2 & 0 \\ 1 & 1 & 0 \\ -1 & 4 & 0 \end{pmatrix}$, $B = \begin{pmatrix} 1 & 2 & 3 \\ 1 & 1 & -1 \\ 2 & 2 & 2 \end{pmatrix}$ and $C = \begin{pmatrix} 1 & 2 & 3 \\ 1 & 1 & -1 \\ 1 & 1 & 1 \end{pmatrix}$, show that $AB = AC$.

   What conclusion can you make from this result?

5. The matrices P and Q are given as

$$P = \begin{pmatrix} 2 & -4 & 6 \\ 3 & -5 & 8 \\ -2 & 6 & -10 \end{pmatrix} \quad \text{and} \quad Q = \begin{pmatrix} -1 & 2 & 1 \\ -7 & 4 & -1 \\ -4 & 2 & -1 \end{pmatrix}$$

   (a) Find the product matrix PQ.

   (b) Deduce the inverse of P.

6. Given that the matrix $A = \begin{pmatrix} 1 & -3 \\ 1 & 2 \end{pmatrix}$ satisfies the equation

$$x^2 - 3x + 5 = 0,$$

   (a) Express the inverse of A in the form $\lambda A + \mu I$.

   (b) Hence, find the inverse of A.

7. (a) Show that $A = \begin{pmatrix} 1 & -2 & 4 \\ 0 & -1 & 2 \\ 2 & 0 & 3 \end{pmatrix}$ is a zero of the polynomial

$$P(x) = x^3 - 3x^2 - 9x + 3.$$

   (b) Express A^{-1} in the form $\lambda A^2 + \mu A + \kappa I$. Hence, find A^{-1}.

8. By means of Cramer's rule or otherwise, solve

$$\begin{aligned} x - 2y - z - 3 &= 0 \\ -x + 2y + 4z + 1 &= 0 \\ 3x - 4y + z - 5 &= 0 \end{aligned}$$

9. Find AB if $A = \begin{pmatrix} -7 & 3 \\ 2 & 8 \end{pmatrix}$, $B = \begin{pmatrix} -8 & 3 \\ 2 & 7 \end{pmatrix}$.

Hence solve the system

$$\begin{aligned} -7x + 3y &= -2 \\ 2x + 8y &= 6 \end{aligned}$$

10. Using elementary operations or otherwise, solve the system of equation

$$\begin{aligned} x + y + z &= 1 \\ 2x + 4y - 3z &= 9 \\ 10x + 5y - 2z &= 10 \end{aligned}$$

11. Write the system of equations

$$\begin{aligned} 4x - 3y + z &= 11 \\ 2x + y - 4z &= -1 \\ x + 2y - 2z &= 1 \end{aligned}$$

in the form $\mathrm{Ax} = \mathrm{b}$.

By finding the inverse of A, solve the system.

(a) Find the characteristic equation of the matrix

$$\mathrm{P} = \begin{pmatrix} 0 & -1 & 1 \\ 2 & -1 & 2 \\ 1 & 0 & 1 \end{pmatrix}$$

(b) By determining a suitable expression, find the matrix P^3.

(c) Hence, find the inverse matrix P^{-1}.

(d) Use matrix methods to solve the linear equations

$$\begin{aligned} -y + z &= 2 \\ 2x - y + 2z &= 5 \\ x + z &= 4 \end{aligned}$$

12. The matrices M and N are given as

$$\mathrm{M} = \begin{pmatrix} 3 & 2 & -2 \\ 1 & -3 & 1 \\ 2 & 1 & -3 \end{pmatrix} \quad \text{and} \quad \mathrm{N} = \begin{pmatrix} 8 & 4 & -4 \\ 5 & -5 & -5 \\ 7 & 1 & -11 \end{pmatrix}$$

(a) Find the product matrix MN.

(b) Hence, find the inverse of M.

(c) Using M^{-1}, solve the simultaneous equations

$$\begin{aligned} 3x + 2y - 2z + 8 &= 0 \\ x - 3y + z + 3 &= 0 \\ 2x + y - 3z + 9 &= 0 \end{aligned}$$

13. Given that

$$\mathrm{S} = \begin{pmatrix} 5 & 2 \\ 2 & 1 \end{pmatrix}, \quad \mathrm{x} = \begin{pmatrix} x \\ y \end{pmatrix}, \quad x \neq 0,\ y \neq 0$$

and that

$$\mathrm{Sx} = \lambda \mathrm{x}$$

(a) show that

$$(5 - \lambda)x + 2y = 0$$

and find another equation satisfied by x and y.

(b) Hence, or otherwise, show that

$$\lambda^2 - 6\lambda + 1 = 0$$

For each value of λ, find the vector x

(c) Show that for a value of λ,

$$y : x = (\sqrt{2} - 1) : 1$$

Chapter 11

Trigonometry

Angles are measured from an initial line. They are measured in degrees or radians in anticlockwise or clockwise sense. Angles measured in anticlockwise sense are positive and those in clockwise sense are negative.

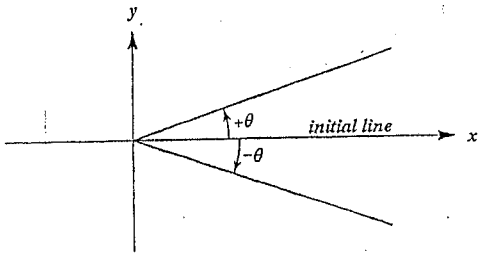


Figure 11.1

11.1 The Radian

One radian is the angle subtended at the centre by an arc whose length is equal to the radius of the circle.

Length of arc $AB = r \implies \angle AOB = 1$ radian. Denote θ radian by θ rad or θ^c. See Figure 11.2.

11.3.1 Transformation about the x-axis

When trigonometry ratios are transformed about the x-axis, they remain unchanged, but the sign is determined by the quadrant in which lies the angle. Suppose θ is an acute angle. Then,

1. $\cos(180° - \theta) = -\cos\theta; \quad \cos(180° + \theta) = -\cos\theta$
2. $\sin(180° - \theta) = \sin\theta; \quad \sin(180° + \theta) = -\sin\theta$
3. $\tan(180° - \theta) = -\tan\theta; \quad \tan(180° + \theta) = \tan\theta$
4. $\cos(360° - \theta) = \cos\theta; \quad \cos(360° + \theta) = \cos\theta$
5. $\sin(360° - \theta) = -\sin\theta; \quad \sin(360° + \theta) = \sin\theta$
6. $\tan(360° - \theta) = -\tan\theta; \quad \tan(360° + \theta) = \tan\theta$

Note that the angle $(360° + \theta)$ is in the first quadrant where all the ratios are positive.

11.3.2 Transformation about the y-axis

When trigonometry ratios are transformed about the y-axis, they change to their corresponding complimentary functions, and the sign is determined by the quadrant in which lies the angle. If θ is an acute angle, then,

1. $\cos(90° - \theta) = \sin\theta; \quad \cos(90° + \theta) = -\sin\theta$
2. $\sin(90° - \theta) = \cos\theta; \quad \sin(90° + \theta) = \cos\theta$
3. $\tan(90° - \theta) = \cot\theta; \quad \tan(90° + \theta) = -\cot\theta$

4. $\cos(270° - \theta) = -\sin\theta; \quad \cos(270° + \theta) = \sin\theta$
5. $\sin(270° - \theta) = -\cos\theta; \quad \sin(270° + \theta) = \cos\theta$
6. $\tan(270° - \theta) = \cot\theta; \quad \tan(270° + \theta) = -\cot\theta$

Note that $\cos x$ and $\sin x$ are complimentary functions. This is because if α and θ are such that $\alpha + \theta = 90°$, then $\cos\alpha = \sin\theta$ and $\cos\theta = \sin\alpha$. Other complimentary functions are $\tan x$ and $\cot x$, and $\sec x$ and $\cos ecx$.

The transformations show that we can obtain the ratio of a non-acute angle by using the ratio of a suitable acute angle. Such a transformation thus simplifies the work involved in computing the original ratio. Since a transformation about the x-axis does not change the original ratio, it is usually preferred to transformation about the y-axis. For example,

$$\sin 135° = \sin(180° - 45°) = \sin 45° = \frac{\sqrt{2}}{2}$$

and

$$\cos 420° = \cos(360° + 60°) = \cos 60° = \frac{1}{2}$$

Example 1

Find the value of the following ratios

(a) $\tan 150°$ (b) $\sin(-140°)$ (c) $\cos(-210°)$ (d) $\tan(-150°)$

Solution

(a)

$$\tan 150° = \tan(180° - 30°) = -\tan 30° = -\frac{1}{\sqrt{3}}.$$

(b)

$$\begin{aligned}\sin(-140°) &= -\sin 140° = -\sin(180° - 40°) \\ &= -\sin 40° = -0.6428\end{aligned}$$

(since $140°$ is in the second quadrant but the original angle $(-140° \equiv 220°)$ is in the third quadrant).

(c)

$$\cos(-210°) = \cos 210° = \cos(180° + 30°) = -\cos 30° = -\frac{\sqrt{2}}{2}.$$

(d)

$$\tan(-150°) = -\tan 150° = -\tan(180° - 30°) = \tan 30° = \frac{1}{\sqrt{3}}.$$

(since 150° is in the second quadrant but the original angle $(-150° \equiv 210°)$ is in the third quadrant).

The value of the ratios at the axes are given as follows:

$$\begin{aligned}
\sin 0° &= 0, & \cos 0° &= 1, & \tan 0° &= 0 \\
\sin 90° &= 1, & \cos 90° &= 0, & \tan 90° &= \infty \\
\sin 180° &= 0, & \cos 180° &= -1, & \tan 180° &= 0 \\
\sin 270° &= -1, & \cos 270° &= 0, & \tan 270° &= \infty \\
\sin 360° &= 0, & \cos 360° &= 1, & \tan 360° &= 0.
\end{aligned}$$

11.4 Characteristics of Trigonometric Functions

We examine the characteristics of the three major trigonometric functions by studying their graphs. The characteristics of the minor functions will be studied later.

11.4.1 The Graph of $y = \sin x$

The graph of $y = \sin x$ is shown in Figure 11.7,

Features of the sine function

1. For $\sin x = 0$, the values of x are

   $$x = \cdots -2\pi,\ -\pi,\ 0,\ \pi,\ 2\pi,\ 3\pi,\ \cdots$$

   Thus, the zeros of $\sin x$ are whole multiples of π. That is,

   $$\sin x = 0, \implies x = n\pi, \qquad n \in \mathbb{Z}$$

2. The maximum value of $\sin x$ is 1.

   For $\sin x = 1$, the values of x are

   $$\begin{aligned} x &= \cdots -\frac{3\pi}{2}, \frac{\pi}{2}, \frac{5\pi}{2}, \cdots \\ &= \cdots \left(-2\pi + \frac{\pi}{2}\right), \frac{\pi}{2}, \left(2\pi + \frac{\pi}{2}\right), \cdots \end{aligned}$$

   Thus, $\sin x$ has a maximum value when x is an (even multiple of π) $+$ $\frac{\pi}{2}$. That is,

   $$\sin x = 1, \implies x = 2n\pi + \frac{\pi}{2}, \qquad n \in \mathbb{Z}$$

3. The minimum value of $\sin x$ is -1.

   For $\sin x = -1$, the values of x are

   $$\begin{aligned} x &= \cdots -\frac{5\pi}{2}, -\frac{\pi}{2}, \frac{3\pi}{2}, \frac{7\pi}{2}, \cdots \\ &= \cdots \left(-2\pi - \frac{\pi}{2}\right), -\frac{\pi}{2}, \left(2\pi - \frac{\pi}{2}\right), \left(4\pi - \frac{\pi}{2}\right), \cdots \end{aligned}$$

   Thus, $\sin x$ has a minimum value when x is an (even multiple of π) $-$ $\frac{\pi}{2}$. That is,

   $$\sin x = 1, \implies x = 2n\pi - \frac{\pi}{2}, \qquad n \in \mathbb{Z}$$

4. It is continuous and lies entirely in the range $-1 \leq \sin x \leq 1$.

5. The curve is symmetrical about the origin. That is, $\sin(-x) = -\sin x$. Thus, $y = \sin x$ is an odd function.

6. It is periodic with period 2π.

11.4.2 The Graph of $y = \cos x$

The graph of $y = \cos x$ is shown in Figure 11.8

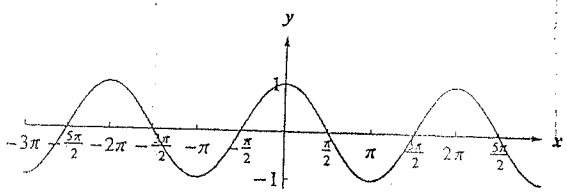


Figure 11.8

Features of the cosine function

1. For $\cos x = 0$, the values of x are

$$x = \cdots -\frac{3\pi}{2},\ -\frac{\pi}{2},\ \frac{\pi}{2},\ \frac{3\pi}{2},\ \frac{5\pi}{2},\ \cdots.$$

Thus, the zeros of $\cos x$ are odd multiples of $\frac{\pi}{2}$. That is,

$$\cos x = 0, \implies x = (2n+1)\frac{\pi}{2}, \qquad n \in \mathbb{Z}$$

2. The maximum value of $\cos x$ is 1.

For $\cos x = 1$, the values of x are

$$x = \cdots -2\pi,\ 0,\ 2\pi,\ 4\pi,\ \cdots$$

Thus, $\cos x$ has a maximum value when x is an even multiple of π. That is,

$$\cos x = 1, \implies x = 2n\pi, \qquad n \in \mathbb{Z}$$

3. The minimum value of $\cos x$ is -1.

For $\cos x = -1$, the values of x are

$$x = \cdots -3\pi,\ -\pi,\ \pi,\ 3\pi,\ 5\pi \cdots$$

Thus, $\cos x$ has a minimum value when x is an odd multiple of π. That is,

$$\cos x = -1, \implies x = (2n+1)\pi, \qquad n \in \mathbb{Z}$$

4. It is continuous and lies entirely in the range $-1 \leq \cos x \leq 1$.

5. The curve is symmetrical about the y-axis. That is, $\cos(-x) = \cos x$. Thus, $y = \cos x$ is an even function.

6. It is periodic with period 2π.

7. It has the same shape as the sine curve, but is displaced by $\frac{\pi}{2}$ to the left. That is,

$$\cos x = \sin\left(\frac{\pi}{2} - x\right)$$

11.4.3 The Graph of $y = \tan x$

The graph of $y = \tan x$ is shown Figure 11.9.

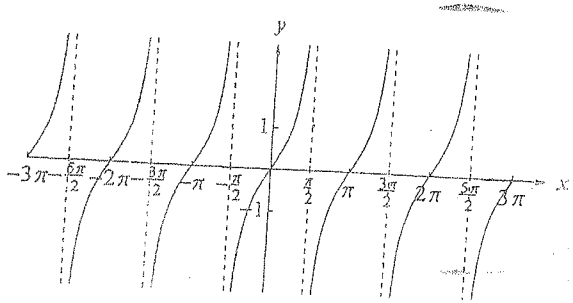


Figure 11.9

Features of the tangent function

1. For $\tan x = 0$, the values of x are

$$x = \cdots -3\pi,\ -2\pi,\ -\pi,\ 0,\ \pi,\ 2\pi,\ 3\pi \cdots$$

Thus, the zeros of $\tan x$ are whole multiples of π. That is,

$$\tan x = 0, \implies x = n\pi, \qquad n \in \mathbb{Z}$$

2. The maximum/minimum value of $\tan x$ is $\pm\infty$.

   This means that $\tan x$ can assume very large values.

   As $\tan x \longrightarrow \pm\infty$, x approaches the following values

$$x \longrightarrow \cdots -\frac{3\pi}{2}, \quad -\frac{\pi}{2}, \frac{\pi}{2}, \frac{3\pi}{2}, \frac{5\pi}{2}, \cdots$$

   Thus, $\tan x$ tends to infinity as x tends to odd multiples of $\frac{\pi}{2}$. That is,

$$\tan x \longrightarrow \pm\infty, \quad \Longrightarrow \quad x \longrightarrow (2n+1)\frac{\pi}{2}, \qquad n \in \mathbb{Z}$$

3. The curve is symmetrical about the origin. That is, $\tan(-x) = -\tan x$. Thus, $y = \tan x$ is an odd function.

4. It is not continuous being undefined at $x = (2n+1)\frac{\pi}{2}$, $\quad n \in \mathbb{Z}$.

5. It is periodic with period π.

Example 2

1. (a) Use the graph of $y = \sin x$ to deduce the period of the graph of $y = \sin 2x$.

   (b) Sketch the graph $y = \sin 2x$ and

   (c) Hence, describe the graph $y = \sin(kx + \theta)$.

Solution

(a) Let $(x,\ y)$ be a point on $y = \sin x$ and $(x\prime,\ y\prime)$ on $y = \sin 2x$. That is, $y\prime = \sin 2x\prime$. Comparing the two functions, we have

$$2x\prime = x \implies x\prime = \frac{x}{2} \text{and } y\prime = y.$$

Thus, the curve of $y = \sin 2x$ can be deduced by from $y = \sin x$ by shrinking it along the x-axis by 2, but the values along the y-axis remain the same. Thus, since the period of $y = \sin x$ is 2π, it implies that the period of $y = \sin 2x$ is $\frac{2\pi}{2} = \pi$.

(b) The graph of $y = \sin 2x$ is as shown.

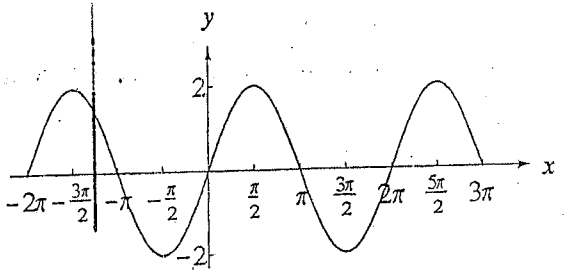


Figure 11.3: Figure 11.10

(c) The graph of $y = \sin(kx + \theta)$ is a shift by θ to the left and a shrink by k units along the x-axis of the graph of $y = \sin x$, but the values along the y-axis remain the same from -1 to 1. Figure 11.11 shows the graphs of $y = \sin(kx + \theta)$ for $k = 1$, and $\theta = \frac{\pi}{3}$ and $\theta = -\frac{\pi}{3}$.

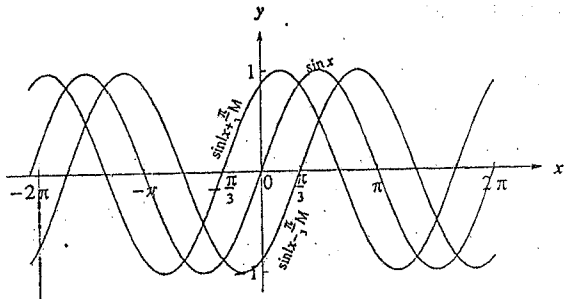


Figure 11.11.

11.5 Trigonometric Identities

In this section, we establish some fundamental identities. You will find these identities useful particularly in solving trigonometric equations in subsequent sections. We will also make use to these identities in Section 11 to derive further identities.

11.5.1 The expression $\sin^2\theta + \cos^2\theta$

With reference to the sides of a right-angle triangle in Figure 11.3, it has already been noted that

$$\tan\theta = \frac{\sin\theta}{\cos\theta} \tag{11.2}$$

Again, from Figure 11.3,

$$\cos\theta = \frac{b}{c} \Longrightarrow \cos^2\theta = \frac{b^2}{c^2}$$
$$\sin\theta = \frac{a}{c} \Longrightarrow \sin^2\theta = \frac{a^2}{c^2}$$

where $c = \sqrt{a^2 + b^2}$.

Now

$$\sin^2\theta + \cos^2\theta = \frac{a^2}{c^2} + \frac{b^2}{c^2} = \frac{a^2 + b^2}{c^2} = \frac{c^2}{c^2} = 1$$

Therefore

$$\sin^2\theta + \cos^2\theta = 1 \tag{11.3}$$

11.5.2 The Expression $1 + \tan^2\theta$

We can use Figure 11.3 to obtain an identity for the above expression. However, having establised two identities already in Equation (11.2) and (11.3), let us make use to them to derive others, if it is possibl.

Now

$$1 + \tan^2\theta = 1 + \frac{\sin^2\theta}{\cos^2\theta} = \frac{\cos^2\theta + \sin^2\theta}{\cos^2\theta} = \frac{1}{\cos^2\theta} = \sec^2\theta.$$

Therefore,

$$1 + \tan^2\theta = \sec^2\theta.$$

It should not be difficult to prove another fundamental identity that

$$1 + \cot^2\theta = \cos ec^2\theta.$$

11.6 The Equation $a\cos x + b\sin x = c$

Let the left hand side of the equation

$$a\cos x + b\sin x = c$$

be represented by the function

$$f(x) = a\cos x + b\sin x$$

Let $R > 0$ and $0 < \alpha < \frac{\pi}{2}$. If we express $f(x)$ in the form

$$\begin{aligned} f(x) &= R\cos x\cos\alpha + R\sin x\sin\alpha \\ &= R\cos(x - \alpha) \end{aligned}$$

By comparing to the given function, we obtain the equations

$$R\cos\alpha = a \tag{11.4}$$
$$R\sin\alpha = b \tag{11.5}$$

Squaring both sides of the equations and adding gives

$$R^2(\cos^2\alpha + \sin^2\alpha) = a^2 + b^2$$

Therefore,

$$R = \sqrt{a^2 + b^2}$$

Dividing Equation (11.5) by (11.4), we obtain

$$\frac{\sin\alpha}{\cos\alpha} = \tan\alpha = \frac{b}{a}$$

Therefore,

$$\alpha = \tan^{-1}\left(\frac{b}{a}\right)$$

Since the cosine of any angle lies between -1 and 1, it implies that

$$-R \leq R\cos(x - \alpha) \leq R.$$

Thus,

$$-R \leq f(x) \leq R.$$

This means that the maximum value of $f(x)$ is R and its minimum value is $-R$, which will be denoted respectiv ely, as $f_{\max} = R$ and $f_{\min} = -R$.

The procedure described will enable us to obtain the extreme values of the function and the corresponding values of x. Having determined the values of R and α, it is then possible to find the solution of the equation $f(x) = c$.

It should be noted that it is possible to express the function also in the form $f(x) = R\sin(x - \alpha)$.

Example 3

(a) Express

$$f(x) = \sqrt{3}\sin\theta - \cos\theta$$

in the form $R\sin(\theta - \alpha)$ where $R > 0$.

(b) Determine the maximum and minimum values of f and the corresponding values of θ.

(c) Solve the following equations for $0° \leq \theta \leq 360°$.

(i)

$$3\sin\theta - \cos\theta = 1.5$$

(ii)

$$4\sin\theta\cos\theta = \sqrt{3}\sin\theta - \cos\theta$$

Solution

(a) Let

$$\begin{aligned} f(\theta) &= R\sin(\theta - \alpha) \\ &= R\sin\theta\cos\alpha - R\cos\theta\sin\alpha \end{aligned}$$

$\Longrightarrow$

$$R\cos\alpha = \sqrt{3} \quad (11.6)$$

$$R\sin\alpha = 1 \quad (11.7)$$

Dividing Equation (11.6) by Equation (11.7)

$$\tan\alpha = \frac{1}{\sqrt{3}} \Longrightarrow \alpha = 30°$$

$$R^2 = 4 \Longrightarrow R = 2$$

$$\therefore f(\theta) = 2\sin(\theta - 30°)$$

(b) But

$$\begin{aligned} -1 \leq \sin(\theta - 30°) &\leq 1 \\ \Longrightarrow -2 \leq 2\sin(\theta - 30°) &\leq 2 \\ \Longrightarrow -2 \leq f(\theta) &\leq 2 \\ \Longrightarrow f_{\max} = 2 &\text{ and } f_{\min} = -2 \end{aligned}$$

At $f = 2$,

$$\begin{aligned} 2\sin(\theta - 30°) &= 2 \\ \theta - 30° &= \sin^{-1}(1) \\ &= 90° \end{aligned}$$

$\Longrightarrow \theta = 90° + 30° = 120°$ Therefore, the maximum value occurs at $\theta = 120°$.

At $f = -2$,

$$\begin{aligned} 2\sin(\theta - 30°) &= -2 \\ \theta - 30° &= \sin^{-1}(-1) \\ &= 270° \end{aligned}$$

$\Longrightarrow \theta = 30° + 270° = 300°$ Therefore, the minimum value occurs at $\theta = 300°$.

(c) (i) Given that $\sqrt{3}\sin\theta - \cos\theta = 1.5$. Since

$$\sqrt{3}\sin\theta - \cos\theta = 2\sin(\theta - 30°)$$

we have

$$\begin{aligned} 2\sin(\theta - 30°) &= 1.5 \\ \sin(\theta - 30°) &= \frac{3}{4} \\ \theta - 30° &= 48.59°,\ 131.41° \\ \Longrightarrow \theta &= 78.59°,\ 161.41° \end{aligned}$$

(ii) Given that

$$4\sin\theta\cos\theta = \sqrt{3}\sin\theta - \cos\theta,$$

we have

$$\begin{aligned} 4\sin\theta\cos\theta &= 2\sin(\theta - 30°) \\ 2\sin\theta\cos\theta &= \sin(\theta - 30°) \\ \sin 2\theta &= \sin(\theta - 30°) \\ 2\theta &= \theta - 30° \\ \theta &= -30° \\ &= 150° \end{aligned}$$

The equation of the form $a\cos x + b\sin x = c$ can be solve using other methods. Such methods will be considered in a later section.

11.7 General Solution of Trigonometric Equations

11.7.1 Solution of $\cos x = c$

In Figure 11.12 shown, α is an angle, such that $\alpha = \cos^{-1} c$. In a complete revolution, $\overline{OP}$ passes through two values of θ which satisfy the equation

$$\cos\theta = \cos\alpha.$$

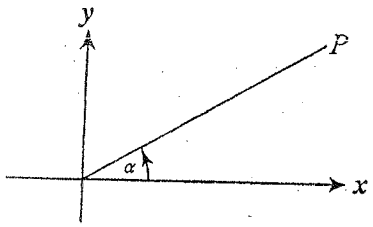


Figure 11.12

Since cosine is positive in the first and fourth quadrants, the two values of θ that satisfy the equation are $\pm\alpha$. After every complete revolution (i.e., 2π), the rule is repeated. Hence, the value of θ in the nth revolution is $2n\pi \pm \alpha$. Therefore,

$$\cos x = c \implies x = 2n\pi \pm \cos^{-1} c, \quad 0 \le \cos^{-1} c \le \pi.$$

11.7.2 Solution of $\sin x = c$

From Figure 11.12, in one complete revolution, $\overline{OP}$ passes through two values of θ which satisfy the equation

$$\sin\theta = \sin\alpha$$

Since sine is positive in the first and second quadrants, the two values that satisy the equation are α and $\pi - \alpha$. Hence, the value of θ in the nth revolution is either $\alpha + 2n\pi$ or $\pi - \alpha + 2n\pi$. Therefore, noting that sine is an odd function,

$$\sin x = c \implies x = n\pi + (-1)^n \sin^{-1} c, \quad -\frac{\pi}{2} \le \sin^{-1} c \le \frac{\pi}{2}.$$

11.7.3 Solution of $\tan x = c$

From Figure 11.12, in one complete revolution, $\overline{OP}$ passes through two values (θ) which satisfy the equation

$$\tan\theta = \tan\alpha$$

Since tangent is positive in the first and third quadrants, the two values that satisy the equation are α and $\pi + \alpha$. Hence, the value of θ in the nth revolution is either $\alpha + n\pi$ or $\pi + \alpha + n\pi$. Therefore, noting that sine is an odd function,

$$\tan x = c \implies x = n\pi + \tan^{-1} c, \quad -\frac{\pi}{2} < \tan^{-1} c < \frac{\pi}{2}.$$

Example 4

Find the general solution for each of the equations

(a) $\sin 3x = \sin 2x$

(b) $\cos mx = \sin nx$

Solution

(a) We know that $\sin x = c \implies x = n\pi + (-1)^n \sin^{-1} c$.

If $\sin 3x = \sin 2x$, then

$$\begin{aligned} 3x &= n\pi + (-1)^n(2x) \\ 3x - (-1)^n(2x) &= n\pi \\ x\left[3 + 2(-1)^{n+1}\right] &= n\pi \\ x &= \frac{n\pi}{3 + 2(-1)^{n+1}}. \end{aligned}$$

If n is even, i.e., $n = 2k,\ k \in \mathbb{Z}$, then

$$\begin{aligned} x &= \frac{2k\pi}{3 + 2(-1)^{2k+1}} \\ &= 2k\pi. \end{aligned}$$

If n is odd, i.e., $n = 2k + 1,\ k \in \mathbb{Z}$, then

$$\begin{aligned} x &= \frac{(2k+1)\pi}{3 + 2(-1)^{2k+2}} \\ &= \frac{(2k+1)\pi}{5} \end{aligned}$$

(b) Given that $\cos mx = \sin nx$, this can be written as

$$\cos mx = \cos\left(\frac{\pi}{2} - nx\right).$$

The general solution is therefore given as

$$mx = 2n\pi \pm \left(\frac{\pi}{2} - nx\right)$$

Taking

$$\begin{aligned} mx &= 2n\pi + \left(\frac{\pi}{2} - nx\right) \\ (m+n)x &= 2n\pi + \frac{\pi}{2} \\ x &= \frac{4n+1}{(m+n)2}\pi. \end{aligned}$$

Taking

$$\begin{aligned} mx &= 2n\pi - \left(\frac{\pi}{2} - nx\right) \\ (m-n)x &= 2n\pi - \frac{\pi}{2} \\ x &= \frac{4n-1}{(m-n)2}\pi. \end{aligned}$$

Therefore, the general solution to the equation is

$$\left\{x : x = \frac{4n+1}{(m+n)2}\pi,\ \frac{4n-1}{(m-n)2}\pi\right\}.$$

11.8 Compound Angles

We what to find expressions for the ratios

$$\sin(A+B),\ \sin(A-B),\ \cos(A+B),\ \cos(A-B)$$

There are a number of approaches that can be used to prove identities for these ratios. In this section, we will use vector approach to derive the identities for $\cos(A+B)$ and $\cos(A-B)$. Expressions for $\sin(A+B)$ and $\sin(A-B)$ will then be deduced using trigonometric transformation. Subsequently, expressions for $\tan(A+B)$ and $\tan(A-B)$ will be deduced.

We know that if θ is the angle between the vectors a and b, then

$$\mathbf{a} \cdot \mathbf{b} = |\mathbf{a}|\,|\mathbf{b}| \cos\theta$$

This implies that

$$\begin{aligned} \cos\theta &= \frac{\mathbf{a} \cdot \mathbf{b}}{|\mathbf{a}|\,|\mathbf{b}|} \\ &= \frac{\mathbf{a}}{|\mathbf{a}|} \cdot \frac{\mathbf{b}}{|\mathbf{b}|} \\ \cos\theta &= \hat{\mathbf{a}} \cdot \hat{\mathbf{b}} \end{aligned}$$

where $\hat{\mathbf{a}}$ and $\hat{\mathbf{b}}$ are the unit vectors along a and b, respectively.

Let $\overrightarrow{OA}$ be a unit vector that makes an angle of $\alpha°$ with the positive x-axis and $\overrightarrow{OB}$ a unit vector that makes an angle of $\beta°$ with the positive x-axis, where $\beta < \alpha$. This is illustrated in Figure 11.13.

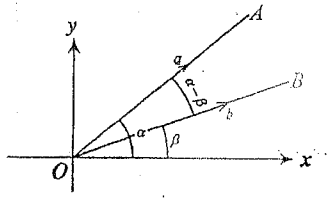


Figure 11.13

That is,

$$\overrightarrow{OA} = \hat{\mathbf{a}} \implies \hat{\mathbf{a}} = \begin{pmatrix} \cos\alpha \\ \sin\alpha \end{pmatrix}$$

and

$$\overrightarrow{OB} = \hat{\mathbf{b}} \implies \hat{\mathbf{b}} = \begin{pmatrix} \cos\beta \\ \sin\beta \end{pmatrix}$$

or

$$\begin{aligned} \hat{\mathbf{a}} &= \cos\alpha\mathbf{i} + \sin\alpha\mathbf{j} \\ \hat{\mathbf{b}} &= \cos\beta\mathbf{i} + \sin\beta\mathbf{j} \end{aligned}$$

The angle between $\hat{\mathbf{a}}$ and $\hat{\mathbf{b}}$ is $\theta = \alpha - \beta$. Therefore,

$$\begin{aligned} \cos(\alpha-\beta) &= \hat{\mathbf{a}}\cdot\hat{\mathbf{b}} \\ &= (\cos\alpha\mathbf{i} + \sin\alpha\mathbf{j})\cdot(\cos\beta\mathbf{i} + \sin\beta\mathbf{j}) \\ \cos(\alpha-\beta) &= \cos\alpha\cos\beta + \sin\alpha\sin\beta \end{aligned} \tag{11.8}$$

Suppose $\hat{\mathbf{b}}$ makes an angle of $(-\beta)$ with the x-axis.

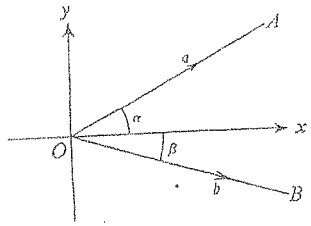


Figure 11.14

$$\begin{aligned} \hat{\mathbf{a}} &= \begin{pmatrix} \cos\alpha \\ \sin\alpha \end{pmatrix} = \cos\alpha\mathbf{i} + \sin\alpha\mathbf{j} \\ \hat{\mathbf{b}} &= \begin{pmatrix} \cos\beta \\ -\sin\beta \end{pmatrix} = \cos\beta\mathbf{i} - \sin\beta\mathbf{j} \end{aligned}$$

The angle between $\hat{\mathbf{a}}$ and $\hat{\mathbf{b}}$ is $\theta = \alpha + \beta$.

Thus,

$$\begin{aligned} \cos(\alpha+\beta) &= (\cos\alpha\mathbf{i} + \sin\alpha\mathbf{j})\cdot(\cos\beta\mathbf{i} - \sin\beta\mathbf{j}) \\ &= \cos\alpha\cos\beta - \sin\alpha\sin\beta \end{aligned} \tag{11.9}$$

Alternative II

Putting $\beta = -\beta$ into Equation (1) and noting that $\cos(-\beta) = \cos\beta$ and $\sin(-\beta) = -\sin\beta$, we obtain

$$\begin{aligned} \cos(\alpha-(-\beta)) &= \cos\alpha\cos(-\beta) + \sin\alpha\sin(-\beta) \\ \cos(\alpha+\beta) &= \cos\alpha\cos\beta - \sin\alpha\sin\beta \end{aligned}$$

We use trigonometrical transformation to deduce formulae for $\sin(\alpha+\beta)$ and $\sin(\alpha-\beta)$ from (11.8) and (11.9).

That is

$$\cos(90-\alpha) = \sin\alpha.$$

Substituting $(90-\alpha)$ for α in Equation (1) gives

$$\begin{aligned}\cos[(90-\alpha)-\beta] &= \cos(90-\alpha)\cos\beta + \sin(90-\alpha)\sin\beta \\ \cos[90-(\alpha+\beta)] &= \sin\alpha\cos\beta + \cos\alpha\sin\beta\end{aligned}$$

$$\Longrightarrow \sin(\alpha+\beta) = \sin\alpha\cos\beta + \cos\alpha\sin\beta \tag{11.10}$$

From (11.9),

$$\begin{aligned}\cos[(90-\alpha)+\beta] &= \cos(90-\alpha)\cos\beta - \sin(90-\alpha)\sin\beta \\ \cos[90-(\alpha-\beta)] &= \sin\alpha\cos\beta - \cos\alpha\sin\beta\end{aligned}$$

$$\Longrightarrow \sin(\alpha-\beta) = \sin\alpha\cos\beta - +\cos\alpha\sin\beta \tag{11.11}$$

The results in (11.8) to (11.11) are summarized as follows:

$$\sin(A \pm B) = \sin A\cos B + \cos A\sin B \tag{11.12}$$
$$\cos(A \pm B) = \cos A\cos B \mp \sin A\sin B \tag{11.13}$$

From the above identities, we deduce similar identities for $\tan(A+B)$ and $\tan(A-B)$.

$$\tan(A+B) = \frac{\sin(A+B)}{\cos(A+B)} = \frac{\sin A\cos B + \cos A\sin B}{\cos A\cos B - \sin A\sin B}$$

Dividing through both the numerator and denomination by $\cos A\cos B$ we have

$$\begin{aligned}\tan(A+B) &= \frac{\frac{\sin A\cos B}{\cos A\cos B} + \frac{\cos A\sin B}{\cos A\cos B}}{\frac{\cos A\cos B}{\cos A\cos B} - \frac{\sin A\sin B}{\cos A\cos B}} = \frac{\frac{\sin A}{\cos A} + \frac{\sin B}{\cos B}}{1 - \frac{\sin A\sin B}{\cos A\cos B}} \\ &= \frac{\tan A + \tan B}{1 - \tan A\tan B}\end{aligned} \tag{11.14}$$

By similar appraoch, it can be shown that

$$\tan(A-B) = \frac{\tan A - \tan B}{1 + \tan A\tan B} \tag{11.15}$$

Example 5

11.8 COMPOUND ANGLES

Use compound angles to find the values of the following.
(a) $\cos 105°$ (b) $\sin 165°$ leaving your answers in surds.

Solution

(a) Express $\cos 105° = \cos(180° - 75°) = -\cos 75°$.

Thus,

$$\begin{aligned}\cos 105° &= -\cos 75 = -\cos(30+45) \\ &= -(\cos 30\cos 45 - \sin 30\sin 45) \\ &= -\left(\frac{\sqrt{3}}{2}\cdot\frac{\sqrt{2}}{2} - \frac{1}{2}\cdot\frac{\sqrt{3}}{2}\right) \\ &= -\left(\frac{\sqrt{6}}{4} - \frac{\sqrt{3}}{4}\right) \\ &= \frac{\sqrt{3}}{4}\left(1-\sqrt{2}\right).\end{aligned}$$

(b) Express $\sin 165° = \sin(180° - 15°) = \sin 15°$

$$\begin{aligned}\sin 165° &= \sin 15° = \sin(45° - 30°) \\ &= (\sin 45°\cos 30° - \cos 45°\sin 30°) \\ &= \frac{\sqrt{2}}{2}\times\frac{\sqrt{3}}{2} - \frac{\sqrt{2}}{2}\times\frac{1}{2} \\ &= \frac{\sqrt{6}}{4} - \frac{\sqrt{2}}{4} \\ &= \frac{\sqrt{2}}{4}\left(\sqrt{3}-1\right).\end{aligned}$$

11.8.1 Double Angles

From Equations (11.12), (11.13) and (11.14), if $A = B$, then the following double angle ratios are obtained.

$$\sin 2A = 2\sin A\cos A$$

$$\begin{aligned}\cos 2A &= \cos^2 A - \sin^2 A \\ &= 1 - 2\sin^2 A \\ &= 2\cos^2 A - 1.\end{aligned}$$

$$\tan 2A = \frac{2\tan A}{1-\tan^2 A}$$

Example 6

Given that $2x + \alpha = \frac{\pi}{4}$, show that

$$\tan\alpha = \frac{1-2t-t^2}{1+2t-t^2}$$

where $t = \tan x$.

Deduce that $\tan\frac{\pi}{8}$ is a root of the equation

$$t^2 + 2t - 1 = 0$$

and that

$$\tan\frac{\pi}{8} = \sqrt{2} - 1.$$

Solution

Given

$$2x + \alpha = \frac{\pi}{4} \implies \alpha = \frac{\pi}{4} - 2x$$

Taking tangent of both sides and simplifying gives

$$\begin{aligned}\tan\alpha &= \tan\left(\frac{\pi}{4} - 2x\right) = \frac{\tan\frac{\pi}{4} - \tan 2x}{1 + \tan\frac{\pi}{4}\tan 2x} \\ &= \frac{1 - \frac{2\tan x}{1-\tan^2 x}}{1 + \frac{2\tan x}{1-\tan^2 x}} = \frac{1 - \tan^2 x - 2\tan x}{1 - \tan^2 x + 2\tan x}\end{aligned}$$

Let $t = \tan x$. Then

$$\tan\alpha = \frac{1-2t-t^2}{1+2t-t^2}$$

Let $x = \frac{\pi}{8}$. Then, $\alpha = \frac{\pi}{4} - 2\left(\frac{\pi}{8}\right) = 0$

This implies that

$$\tan 0 = \frac{1-2t-t^2}{1+2t-t^2} = 0.$$

This simplifies as

$$\begin{aligned}1 - 2t - t^2 &= 0 \\ t^2 + 2t - 1 &= 0\end{aligned}$$

where $t = \tan\frac{\pi}{8}$.

Therefore, $\tan\frac{\pi}{8}$ is a root of the equation

$$\begin{aligned}t &= \frac{-2 \pm \sqrt{4-4(-1)}}{2} = \frac{-2 \pm \sqrt{8}}{2} = \frac{-2 \pm 2\sqrt{2}}{2} \\ &= -1 + \sqrt{2},\ -1 - \sqrt{2}\end{aligned}$$

But

$$\tan\frac{\pi}{8} > 0 \implies \tan\frac{\pi}{8} = -1 + \sqrt{2}.$$

11.8.2 Triple Angles

We can make repeated use of the double angles to derive triple angles. To obtain an expression for $\sin 3A$ in terms of $\sin A$ only, we make use of the results

$$\sin 2A = 2\sin A\cos A, \cos 2A = 1 - 2\sin^2 A,$$

and that

$$\cos^2 A + \sin^2 A = 1.$$

Expanding $\sin 3A$ as $\sin(2A + A)$, and further simplification gives

$$\begin{aligned}\sin 3A &= \sin(2A + A) = \sin 2A \cos A + \cos 2A \sin A \\ &= 2 \sin A \cos A \cos A + (1 - \sin^2 A) \sin A \\ &= 2 \sin A \cos^2 A + \sin A - \sin^3 A \\ &= 2 \sin A (1 - \sin^2 A) + \sin A - \sin^3 A \\ &= 2 \sin A - 2 \sin^3 A + \sin A - \sin^3 A \\ &= 3 \sin A - 4 \sin^3 A\end{aligned}$$

By a similar procedure, it can be shown that

$$\cos 3A = 4 \cos^3 A - 3 \cos A$$

$$\tan 3A = \frac{3 \tan A - \tan^3 A}{1 - 3 \tan^2 A}.$$

Example 7

Express $\sin 3\theta$ in terms of $\sin \theta$. Hence, find a general solution of the equation

$$8x^3 - 6x + 1 = 0$$

and give three answers correct to 2 decimal places.

Solution

It has already been shown that

$$\sin 3\theta = 3 \sin \theta - 4 \sin^3 \theta$$

Let $x = \sin \theta$. Substituting into the RHS of the expression gives

$$\sin 3\theta = 3x - 4x^3.$$

Multiplying through by -2 gives

$$-2 \sin 3\theta = 8x^3 - 6x.$$

From the given equation

$$\begin{aligned}8x^3 - 6x + 1 &= 0 \\ 8x^3 - 6x &= -1.\end{aligned}$$

This implies that

$$\begin{aligned}-2 \sin 3\theta &= -1 \\ \implies \sin 3\theta &= \frac{1}{2}.\end{aligned}$$

Using the general solution gives

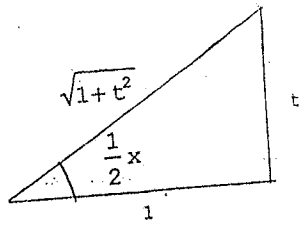
$$\begin{aligned}3\theta &= 180n + (-1)^n \sin^{-1}\left(\frac{1}{2}\right) \\ &= 180n + (-1)^n (30^\circ) \\ \implies x &= 60n + (-1)^n (10^\circ).\end{aligned}$$

Therefore,

$$\begin{aligned}x &= \sin\left[60n + (-1)^n (10^\circ)\right], \ n \in \mathbb{Z} \\ &= \sin 10^\circ, \sin 50^\circ, \sin 130^\circ, \sin 170^\circ \\ &= 0.17, \ 0.76, \ -0.94.\end{aligned}$$

11.9 Half-Angles

Let $t = \tan \frac{1}{2}x$, then from the sides of the triangle shown



$$\sin\frac{1}{2}x = \frac{t}{\sqrt{1+t^2}}, \quad \cos\frac{1}{2}x = \frac{1}{\sqrt{1+t^2}}$$

Applying the double angles

$$\begin{aligned}\sin x &= 2\sin\frac{1}{2}x\cos\frac{1}{2}x \\ &= 2\left(\frac{t}{\sqrt{1+t^2}}\right)\left(\frac{1}{\sqrt{1+t^2}}\right) \\ &= \frac{2t}{1+t^2}\end{aligned}$$

and

$$\begin{aligned}\cos x &= \cos^2\frac{1}{2}x - \sin^2\frac{1}{2}x \\ &= \frac{1}{1+t^2} - \frac{t^2}{1+t^2} \\ &= \frac{1-t^2}{1+t^2}\end{aligned}$$

Alternative Approach

$$\begin{aligned}\sin x &= 2\sin\frac{1}{2}x\cos\frac{1}{2}x = \frac{2\sin\frac{1}{2}x\cos\frac{1}{2}x}{1} \\ &= \frac{2\sin\frac{1}{2}x\cos\frac{1}{2}x}{\cos^2\frac{1}{2}x + \sin^2\frac{1}{2}x}\end{aligned}$$

Dividing through both numerator and denominator by $\cos\frac{1}{2}x$

$$\begin{aligned}\sin x &= \frac{2\frac{\sin\frac{1}{2}x}{\cos\frac{1}{2}x}}{1+\frac{\sin\frac{1}{2}x}{\cos\frac{1}{2}x}}, &= \frac{2\tan\frac{1}{2}x}{1+\tan^2\frac{1}{2}x} \\ &= \frac{2t}{1+t^2}.\end{aligned}$$

11.9 HALF-ANGLES

Similar approach can be used to find the expression for $\cos x$ and $\tan x$

In a previous section, we have already considered equations of the form $a\cos x + b\sin x = c$. We can use the approach of half angles to find solutions to such equations.

Example 8

Refer to the equation in Example 3, given by

$$\sqrt{3}\sin\theta - \cos\theta = 1.5$$

Use half angles to find the solution to the problem for θ in the interval $0° \le \theta \le 360°$.

Solution

Let $t = \tan\frac{1}{2}\theta$. Then $\sin\theta = \frac{2t}{1+t^2}$ and $\cos\theta = \frac{1-t^2}{1+t^2}$. Substituting into the equation

$$\sqrt{3}\sin\theta - \cos\theta = 1.5$$

gives

$$\sqrt{3}\left(\frac{2t}{1+t^2}\right) - \frac{1-t^2}{1+t^2} = \frac{3}{2}.$$

Multiplying through by $2(1+t^2)$ gives

$$4t\sqrt{3} - 2(1-t^2) = 3(1+t^2)$$

Simplifying and equating to zero gives

$$t^2 - 4\sqrt{3}t + 5 = 0.$$

Therefore,

$$\begin{aligned}t &= \frac{4\sqrt{3} \pm \sqrt{16(3)-4(5)}}{2} = \frac{4\sqrt{3}\pm\sqrt{28}}{2} \\ &= 2\sqrt{3} \pm \sqrt{7} = 6.1099,\ 0.8184\end{aligned}$$

This means that

$$\tan\frac{1}{2}\theta = 6.1099, \quad \text{and} \quad \tan\frac{1}{2}\theta = 0.8184$$

Considering $\tan\frac{1}{2}\theta = 6.1099$, the general solution gives

$$\begin{aligned}\frac{1}{2}\theta &= 180°n + \tan^{-1}(6.1099)\\ &= 180°n + 80.71°.\end{aligned}$$

If $n = 0$, then

$$\frac{1}{2}\theta = 80.71 \implies \theta = 161.42°$$

Considering $\tan\frac{1}{2}\theta = 0.8184$, the general solution gives

$$\begin{aligned}\frac{1}{2}\theta &= 180°n + \tan^{-1}(0.8184)\\ &\doteq 180°n + 39.30°.\end{aligned}$$

If $n = 0$, then

$$\frac{1}{2}\theta = 39.30 \implies \theta \doteq 78.59°$$

Larger values of n leads to angles outside the given range.

Therefore, the solution set is

$$\{\theta : \theta = 78.59°,\ 161.42°\}.$$

11.10 The Factor Formulae

It has been shown that the following identities hold.

$$\sin(A+B) = \sin A\cos B + \cos A\sin B \tag{11.16}$$
$$\sin(A-B) = \sin A\cos B - \cos A\sin B \tag{11.17}$$
$$\cos(A+B) = \cos A\cos B - \sin A\sin B \tag{11.18}$$
$$\cos(A-B) = \cos A\cos B + \sin A\sin B \tag{11.19}$$

11.10 THE FACTOR FORMULAE

From these four formulae, we further obtain other formulae as shown.

Equation (11.16) + (11.17)

$$\sin(A+B) + \sin(A-B) = 2\sin A\cos B$$

Equation (11.16) − (11.17)

$$\sin(A+B) - \sin(A-B) = 2\cos A\sin B \tag{11.20}$$

Equation (11.18) + (11.19)

$$\cos(A+B) + \cos(A-B) = 2\cos A\cos B \tag{11.21}$$

Equation(11.18) − (11.19)

$$\cos(A+B) - \cos(A-B) = -2\sin A\sin B \tag{11.22}$$

or

$$\cos(A-B) - \cos(A+B) = 2\sin A\sin B \tag{11.23}$$

To simplify the above formulae

Let

$$A + B = x \text{ and } A - B = y.$$

Then,

$$2A = x + y \implies A = \frac{x+y}{2}$$

and

$$2B = x - y \implies B = \frac{x-y}{2}$$

Making substitutions for A and B in Equation (11.20) to (11.23), we obtain

$$\sin x + \sin y = 2\sin\left(\frac{x+y}{2}\right)\cos\left(\frac{x-y}{2}\right) \tag{11.24}$$
$$\sin x - \sin y = 2\cos\left(\frac{x+y}{2}\right)\sin\left(\frac{x-y}{2}\right)$$
$$\cos x + \cos y = 2\cos\left(\frac{x+y}{2}\right)\cos\left(\frac{x-y}{2}\right)$$
$$\cos x - \cos y = -2\sin\left(\frac{x+y}{2}\right)\sin\left(\frac{x-y}{2}\right)$$

The Equations in (11.24) are what is referred to as *Factor Formulae.*

The factor formulae can be used to solve a some types of trigonometric equations and to derive certain properties of a set of angles.

Example 9

Show that

$$\sin x + \sin 2x + \sin 3x = \sin 2x\ (1 + 2\cos x)$$

Hence, solve

$$\sin x + \sin 2x + \sin 3x = 0$$

Solution

We write

$$\sin x + \sin 2x + \sin 3x = \sin 2x + \sin x + \sin 3x.$$

Applying the rule that

$$\sin x + \sin y = 2\sin\left(\frac{x+y}{2}\right)\cos\left(\frac{x-y}{2}\right)$$

we have

$$\begin{aligned}\sin x + \sin 2x + \sin 3x &= \sin 2x + 2\sin\left(\frac{3x+x}{2}\right)\cos\left(\frac{3x-x}{2}\right)\\ &= \sin 2x + 2\sin 2x\cos x\\ &= \sin 2x(1+2\cos x)\end{aligned}$$

Thus,

$$\sin x + \sin 2x + \sin 3x = 0$$

$$\implies \sin 2x(1+2\cos x) = 0$$

$$\begin{aligned}&\implies \sin 2x = 0\\ &\implies 2x = 0,\ \pi,\ 2\pi\\ &\implies x = 0,\ \frac{\pi}{2},\ \pi\end{aligned}$$

$$1 + 2\cos x = 0 \implies \cos x = -\frac{1}{2}$$

Thus,

$$x = \pi - \cos\left(\frac{1}{2}\right) = \frac{2\pi}{3}$$

Therefore, the solution set is

$$\{x : x = 0,\ \frac{\pi}{2},\ \frac{2\pi}{3},\ \pi\}.$$

Example 10

If A, B and C are the angles of a triangle, prove that

$$\sin A + \sin B + \sin C = 4\cos\frac{1}{2}A\cos\frac{1}{2}B\cos\frac{1}{2}C.$$

Solution

We first apply the factor formulae for the last two terms in the sum

$$\sin A + \sin B + \sin C$$

and then apply double angles for the first term.

Next, we note that

$$\begin{aligned}A + B + C &= 180^\circ\\ B + C &= 180^\circ - A\\ \frac{1}{2}(B+C) &= 90^\circ - \frac{1}{2}A\\ \frac{1}{2}A &= 90^\circ - \frac{1}{2}(B+C).\end{aligned}$$

By trigonometric transformation, we also observe that

$$\sin\frac{1}{2}A = \sin\left[90^\circ - \frac{1}{2}(B+C)\right] = \cos\frac{1}{2}(B+C)$$

Making appropriate simplifications we have,

$$\begin{aligned}
\sin A + \sin B + \sin C &= \sin A + 2\sin\frac{1}{2}(B+C)\cos\frac{1}{2}(B-C) \\
&= 2\sin\frac{1}{2}A\cos\frac{1}{2}A + 2\sin\frac{1}{2}(B+C)\cos\frac{1}{2}(B-C) \\
&= 2\sin\frac{1}{2}A\cos\frac{1}{2}A + 2\sin\left(90^\circ - \frac{1}{2}A\right)\cos\frac{1}{2}(B-C) \\
&= 2\sin\frac{1}{2}A\cos\frac{1}{2}A + 2\cos\frac{1}{2}A\cos\frac{1}{2}(B-C) \\
&= 2\cos\frac{1}{2}A\left[\sin\frac{1}{2}A + \cos\frac{1}{2}(B-C)\right] \\
&= 2\cos\frac{1}{2}A\left[\cos\frac{1}{2}(B+C) + \cos\frac{1}{2}(B-C)\right] \\
&= 2\cos\frac{1}{2}A\left[2\cos\frac{1}{2}B\cos\frac{1}{2}C\right] \\
&= 4\cos\frac{1}{2}A\cos\frac{1}{2}B\cos\frac{1}{2}C
\end{aligned}$$

as required.

There are a number of other interesting properties of angles in a triangle. Some of these are left in the exercises for practice.

11.11 Inverse Trigonometric Function

11.11.1 The Inverse of $y = \sin x$

If $y = \sin x$, then $x = \sin^{-1} y$. Interchanging x and y we have $y = \sin^{-1} x$. We know that the range of $f(x)$ is $[-1,\ 1]$. Suppose, without loss of generality, that $-2\pi \leq x \leq 2\pi$. The domain of $f(x)$ becomes the range of $f^{-1}(x)$, and the range of $f(x)$ becomes the domain of $f^{-1}(x)$. That is, for $y = f^{-1}(x)$, $x \in [-1,\ 1]$ and $y \in [-2\pi,\ 2]$. The graph of $y = \sin^{-1} x$ is shown in Figure 11.15

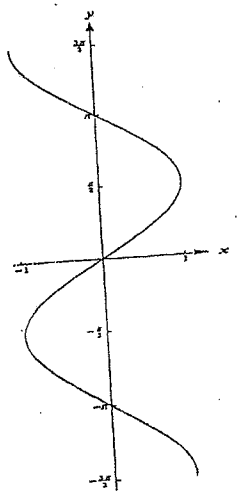


Figure 11.15

From the graph, $f^{-1} : x \longrightarrow \sin^{-1} x$ is not a one-to-one mapping. Thus, $y = \sin^{-1} x$ is not a function. However, if the range of f^{-1} is restricted to lie in the interval $\left[-\frac{\pi}{2},\ \frac{\pi}{2}\right]$, f^{-1} will define a one-to-one mapping and hence, will be a function. The graph of $y = \sin^{-1} x$ in the range of $\left[-\frac{\pi}{2},\ \frac{\pi}{2}\right]$ is shown in Figure 11.16.

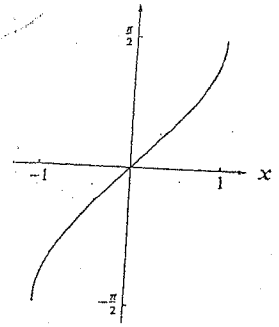


Figure 11.16

From Figure 11.15, $y = \sin^{-1} x, \quad y \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$ is a function

11.11.2 The Inverse of $y = \cos x$

If $y = \cos x$, then $x = \cos^{-1} y$. Interchanging x and y we have $y = \cos^{-1} x$We know that the range of $f(x)$ is $[-1,\ 1]$. Suppose, without loss of generality, that $-2\pi \leq x \leq 2\pi$. The domain of $f(x)$ becomes the range of $f^{-1}(x)$, and the range of $f(x)$ becomes the domain of $f^{-1}(x)$. That is, for $y = f^{-1}(x)$, $x \in [-1,\ 1]$ and $y \in [-2\pi,\ 2\pi]$. Like the inverse sine relation, the inverse cosine relation within the given interval and beyond does not define a function since it is not a one-to-one mapping. Thus, $y = \cos^{-1} x$ is not a function. However, if the range of f^{-1} is restricted to lie in the interval $[0,\ \pi]$, f^{-1} will define a one-to-one mapping and hence, will be a function. The graph of $y = \cos^{-1} x$ in the range of $[0,\ \pi]$ is shown in Figure 11.17.

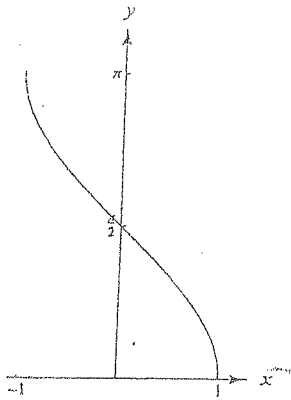


Figure 11.17

From Figure 11.17, $y = \cos^{-1} x, \quad y \in [0,\ \pi]$ is a function.

Example 11

Simplify $\sin^{-1} x + \cos^{-1} x$

Solution

Let

$$\theta = \sin^{-1} x \implies \sin\theta = x$$

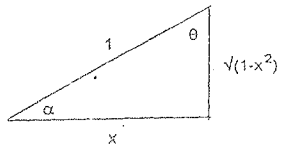
and let

$$\alpha = \cos^{-1} x \implies \cos\alpha = x$$

By the compound angles

$$\sin(\theta + \alpha) = \sin\theta\cos\alpha + \cos\theta\sin\alpha.$$

From the right angled triangle shown



$$\sin\alpha = \sqrt{1-x^2} \text{ and } \cos\theta = \sqrt{1-x^2}$$

Making substitutions give

$$\begin{aligned}\sin(\theta+\alpha) &= x^2 + \left(\sqrt{1-x^2}\right)\left(\sqrt{1-x^2}\right)\\ &= x^2 + (1-x^2)\\ &= 1\end{aligned}$$

$$\Longrightarrow \qquad \theta + \alpha = \sin^{-1}(1) = \frac{\pi}{2}.$$

Therefore,

$$\sin^{-1}x + \cos^{-1}x = \frac{\pi}{2}.$$

11.11.3 The Inverse of $y = \tan x$

The inverse relation for $y = \tan x$ is $x = \tan^{-1} y$. Interchanging x and y, we have $y = \tan^{-1} x$. Like the inverse sine and cosine relations, no inverse exists for the tangent function. However, if the range of f^{-1} is restricted to lie in the interval $[-\frac{\pi}{2}, \frac{\pi}{2}]$, f^{-1} will define a one-to-one mapping and hence, will be a function. The graph of $y = \tan^{-1} x$ in the range of $[-\frac{\pi}{2}, \frac{\pi}{2}]$ is shown in Figure 11.18.

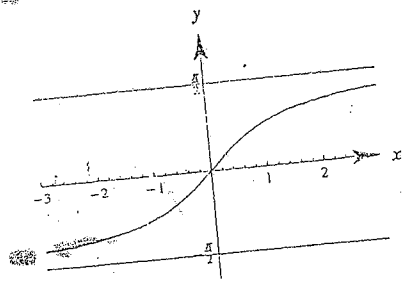


Figure 11.18

From Figure 11.18, we see that y

Example 12

Find an expression for $\tan^{-1} x + \tan^{-1} y$.
Hence, find $\tan^{-1}\left(\frac{1}{2}\right) + \tan^{-1}\left(\frac{1}{3}\right)$.

Solution

Let $\alpha = \tan^{-1} x \Longrightarrow \tan\alpha = x$

and let $\beta = \tan^{-1} y \Longrightarrow \cos\beta = y$

By the compound angles

$$\begin{aligned}\tan(\alpha+\beta) &= \frac{\tan\alpha + \tan\beta}{1 - \tan\alpha\tan\beta}\\ &= \frac{x+y}{1-xy}\end{aligned}$$

This implies that

$$\alpha + \beta = \tan^{-1}\left(\frac{x+y}{1-xy}\right)$$

Therefore,

$$\tan^{-1}x + \tan^{-1}y = \tan^{-1}\left(\frac{x+y}{1-xy}\right)$$

Putting $x = \frac{1}{2}$, and $y = \frac{1}{3}$, we obtain

$$\begin{aligned}\tan^{-1}\left(\frac{1}{2}\right) + \tan^{-1}\left(\frac{1}{3}\right) &= \tan^{-1}\left(\frac{\frac{1}{2}+\frac{1}{3}}{1-\frac{1}{2}\times\frac{1}{3}}\right)\\ &= \tan^{-1}\left(\frac{3+2}{6-1}\right)\\ &= \tan^{-1}(1)\\ &= \frac{\pi}{4}\end{aligned}$$